

SMITHIAN FOLD THEORY V3

From Fold to Consciousness

An Exact, Zero-Parameter and Machine-Closed Reconstruction of Consciousness and Cognitive Science from Smithian Fold Theory

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in the same concept DOI lineage. The correction changes no scientific claim,

evidence classification, mathematical value, source identity or machine record.

Lean verification integration. The current SFT ordered census passed the

independent Lean 4 verification layer on 2 August 2026: 2,777 accepted claims,

898,902 candidate records and decisions, 11,108 controls, seventeen branches

and no reported issue. Lean natively proves the two-class operational root and

its unique survivor and constructs proof-bearing acceptance-gate certificates.

The remaining claim content is checked as the complete registered artifact

graph; that distinction is retained throughout this successor.

Abstract

This successor presents the complete current `consciousness_cognitive_science` paper surface: 77 live model-admitted claims, 19,712 generated candidates, 77 unique survivors and 308 passed controls. It integrates 5 claim section(s) not present in the preceding abstract's dated surface and remains complete only to this versioned registered census, with lawful extension open. The independent Lean 4 layer returned PASS for this surface as part of all 2,777 current claims.

The headline result is an exact structural reconstruction of the hard-problem boundary rather than a substitution of correlates for experience. An external observation is a many-to-one state image. Interior observation is forced when that image re-enters the same organised process. The complete two-preimage fibre makes phenomenal privacy mathematically intelligible: a participant may retain a predecessor label closed by the external image. Access, report, behaviour, neural correlation, computation and physical measurement therefore cannot become phenomenal presence by relabelling. Binding is joint closure with components retained; unity is complete mutual participation in one interior process; subject and perspective are the continuing source-located relation across changing content.

The exact constructions are explicit. One-quarter and three-quarters are distinct observer/observed preimages that share an image and compose to the One. A finite self-model closes one-quarter to one-half to the One in two nonidentity acts, ending the regress at the Fold-fixed One. One-third and two-thirds supply the least period-two return witness for persistent label-bearing memory. The red-of-red is a label-bearing self-reidentification carried by the exact period-three support one-seventh, two-sevenths and four-sevenths, whose three parts compose to the One. Physical red stimulus, public word, participant quality and report about that quality remain different records. Substrate independence requires a complete bidirectional relation- and causality-preserving realization map; fluent output does not satisfy it.

All 356 V1 rows and 407 V2 steps were atomically reviewed. Forty-six Consciousness-owned questions, including the stronger separately registered qualia-resonance and red-of-red obligations, are closed by current V3 receipts. Fifteen external source identities contribute sixty-one preregistered features: fifty-eight are present; three absent full-text features and eighteen failed transport or content rows remain visibly preserved rather than counted as support.

Formal admission, implementation validation, observation, measurement, empirical support, confirmation, adverse evidence, unresolved evidence and publication status remain separate classifications. Lean does not reclassify an empirical result, repair an adverse observation or convert a later comparison into an earlier prediction. Lawful criticism, falsification and versioned extension remain open.

Successor headline findings

1. **Independent whole-model result.** Lean 4 returned *PASS* for 2,777/2,777 current claims, 898,902 candidates and decisions, 11,108 controls and all 17 branches, with no issue.
2. **Operational root.** The exact registered two-class grammar is formalised natively, and `presentedOccurrence` is proved to be its unique survivor without an imported or user-declared axiom in the exported theorem's axiom audit.
3. **Typed validation.** The root and generic acceptance implications are native Lean theorems; the remaining scientific claims are checked as complete registered artifacts. Empirical truth is not inferred from proof-assistant acceptance alone.
4. **Fail-closed custody.** A six-file byte-identity mismatch halted the first run. Exact registered bytes were restored, all 385 source bindings were re-audited, and the unchanged verifier then passed.
5. **Current paper surface.** This successor accounts for all 77 current claim(s) in its declared branch ownership boundary; 5 later claim section(s) are integrated in the successor supplement.
6. **Newly integrated results.** `SFT-CONSC-SYNAESTHESIA-DIRECTIONAL-LOCK-002` - Directional synaesthetic Fold lock; `SFT-CONSC-NONORDINARY-THREE-QUALITY-ORBIT-002` - Three-quality held orbit and nonordinary boundary; `SFT-CONSC-SLEEP-DREAM-PERIOD-TWO-002` - Exact sleep-dream period-two Fold cycle; `SFT-CONSC-CESSATION-LOCK-ANCHOR-002` - Cessation lock and invariant anchor; `SFT-CONSC-VALIDATION-NONORDINARY-COMPLETE-FAMILY-002` - Complete post-seal Consciousness return comparison.

These findings supplement rather than erase the branch-specific headline results retained below. Any adverse, corrected, unresolved, unavailable or chronology-bound result in the preceding scientific record remains governed by its current claim package.

Current status, evidence language and reader map

Status field	Current position
Paper and completion boundary	77 current claims in the declared branch surface. Completion is dated to the current registered census and remains open to lawful versioned extension.
Formal status	The Lean root theorem is kernel checked. Every live model claim has a current admitted receipt and passed the Lean artifact gates. Formal admission does not imply empirical confirmation.
Empirical status	Claim specific. Blind, non-blind, development-observed, holdout, favourable, adverse, null, missing, unavailable, disputed and unresolved distinctions remain exactly as recorded.
Chronology	Claim-specific registration, sealing, custody and observation order controls. Later evidence never becomes an earlier prediction without the registered chronology.
Publication status	Version 1.1.1 is published open access at 10.5281/zenodo.21761660 in the existing Zenodo concept DOI lineage.
Ownership	This paper retains its declared scientific boundary. Lean verification creates no new branch ownership and no application may select a law.
What is not claimed	Permanent closure of science, universal empirical confirmation, conversion of compatibility into confirmation, or superiority to every rival model.

Corpus-wide terminology and evidence key

Term	Reserved meaning in this paper
Theorem	A formally closed proposition at its declared grammar and dependency boundary.
Law	An admitted branch relation with its declared carrier, boundary, dependencies and extension rule.
Claim	The registered unit judged by the engine and bound to one immutable receipt.
Constitution	The rules governing admissible objects, derivations, evidence and publication; not an empirical result.
Derivation	Generation and elimination of the declared candidate space to its surviving structure.
Prediction	A consequence sealed before the matching target is released under the registered custody protocol.
Observation	A source-bound external record; it does not enter candidate generation.
Measurement	An instrument-, method-, condition- and uncertainty-bound observed value.
Reconstruction	A separately implemented regeneration or an explicitly identified inference of a retained state or history.
Exact numerical correspondence	Equality or registered interval relation between exact numerical objects at the declared boundary.
Structural correspondence	A post-derivation relation between forms without a claim of exact numerical prediction.
Boundary correspondence	Agreement restricted to the named interface, limit or ownership boundary.
Compatibility	Non-adverse but non-discriminating evidence.
Support	Relevant evidence that is not unique confirmation.
Confirmation	Used only when the current registered evidence protocol warrants that classification.
Validation	Successful execution of the named formal, computational or empirical test; its kind must be stated.
Adverse result	A registered test result that conflicts with the tested claim at its declared boundary.
Unresolved result	Required evidence remains unavailable, incomplete, disputed or insufficiently classified.
Implementation identity	The hash-bound identity of executable material; implementation success is not empirical confirmation.
Formal, empirical and publication status	Three independent classifications; none substitutes for another.
Foundational closure	Completion of the registered foundation only; not field-wide closure.
Field-wide closure	Completion of the named frozen or registered dated field census.
Current-evidence closure	Closure only to the evidence surface presently registered and preserved.
Extension openness	New questions may enter a later version without rewriting receipts or history.

Proof language is reserved for formal closure; derivation for generated structure; implementation for executable demonstration; prediction for a correctly sealed prospective consequence; observation and measurement for source-bound external records; reconstruction for separately regenerated or inferred states; correspondence for post-derivation relations; compatibility and support for non-unique evidence; confirmation only where the registered protocol authorises it; adverse for conflict; and unresolved where required evidence remains incomplete.

Three reading levels

1. **Conceptual paper.** The abstract, headline findings, branch narrative, major derivations, evidence synthesis, limitations and conclusion provide the readable scientific argument.
2. **Scientific audit layer.** Family and claim sections preserve exact status, dependencies, candidate and survivor counts, controls, chronology, sources, corrections, adverse evidence and reconciliation.
3. **Machine archive.** The repository packages preserve complete candidates, decisions, hashes, receipts, executable traces, source snapshots and certificates. They remain authoritative where prose abbreviates display.

Editorial change control

This successor improves currency, expression, typography and navigation. It does not change scientific meaning, chronology, evidence class, branch ownership, claim status or machine identity. Any conflict between authoritative records must be resolved by explicit supersession or reported to Maria Smith; prose alone cannot manufacture agreement.

Results first: the consciousness findings

Headline result	Exact or bounded result	Scientific meaning
Complete foundational closure	72 laws; 18,432 candidates; 72 survivors; 288 controls	Every law has a root trace, unique survivor, independent reconstruction, empirical-boundary comparison and immutable receipt.
Observation class	Complete 2-preimage fibre for each ordinary Fold image	External observation closes one predecessor distinction unless its fibre label is retained.
Observer and observed	one-quarter + three-quarters = the One; both share one image	Roles remain distinct without creating two foundational wholes.
Hard-problem boundary	external image is not interior participation; access/report/correlation are not presence	The category gap is closed structurally without using a behavioral or neural proxy as the phenomenon.
Binding and unity	joint successor with reconstructible components; complete mutual closure	Synchrony, proximity and one-way influence cannot manufacture one experience.
Subject and privacy	source-located invariant plus retained private fibre label	Perspective and privacy follow from exact access asymmetry, not an imported substance.
Determinism and introspection	unique forward successor; 2 possible predecessors without the held label	Superdeterminism is compatible with exact self-opacity; no ontological randomness is imported.
Memory recurrence	one-third <-> two-thirds, exact period 2	Persistent memory requires label-bearing recurrence or renewal, not merely a later equal output.
Finite self-model	one-quarter -> one-half -> the One; 2 nonidentity acts	Self-reference closes finitely; later Fold action is identity.
Substrate independence	complete bidirectional state-role-transition-causality preservation	Material sameness is unnecessary; behaviour or language alone is insufficient.
Qualia resonance	stable held-quality reidentification across complete recurrence	Public words coordinate evidence but never replace the participant-bound quality.
Red-of-red	one-seventh -> two-sevenths -> four-sevenths -> one-seventh; sum the One	The specific red quality is self-reidentified across an exact period-three carrier with changed-label and changed-stimulus controls.
External evidence	15 sources; 61 features; 58 present; 3 absent preserved; 18 failed rows preserved	No-report, split-brain, choice-blindness, dream, anesthesia, attention, synesthesia, colour and adversarial-theory evidence constrain the laws without selecting them.

Public scientific mission and admission boundary

Ernos Labs is an open-source science movement, verification platform and public tree of knowledge founded by Maria Smith. Its purpose is not to replace one authority with another. It is to make scientific authority narrow, inspectable and revocable: a claim is admitted only through its complete derivation chain, generated alternatives, eliminations, unique survivor, adverse controls, measurement custody where applicable and unchanged-engine receipt. Open criticism is unrestricted and necessary; scientific admission is the separate machine-checked act of satisfying that public standard.

Maria Smith developed Smithian Fold Theory outside formal academic education, institutional research employment and conventional grant funding. That fact is not offered as evidence for a theorem; the derivations and observations carry the entire scientific burden. It is evidence about access. A credential-first system loses more than individual opportunity: it loses unknown questions, methods and discoveries from minds that capital and status never authorise. This work therefore treats Maria Smith's

authorship neither as exceptionalism nor as a reason for dismissal, but as an indictment of every scientific contribution lost when financial gatekeeping is presented as rigour.

The institutional argument is empirical, not a claim that every institution or funded researcher acts in bad faith. Published studies document sponsor-linked differences in outcomes and conclusions, commercial influence over research agendas, limits and sensitivities in grant review, underpublication of null results, and inequalities created by paywalls and article-processing charges. Those findings establish that funding, prestige, publication and consensus are selection systems with incentives and failure modes. They cannot substitute for a public proof and evidence chain. Expertise, measurement and adversarial review remain indispensable; institutional permission does not select a fundamental law.

For consciousness research, an institutional theory, proprietary brain scan, report score, model confidence, behavioral imitation or consensus label cannot select the law. First-person discrimination, report, behaviour, biological or neural correlation, cognitive access, computational representation, physical measurement and substrate realization remain separate evidence classes unless a separately derived and tested bridge connects them. The private/public boundary is part of the evidence, not a license for opaque oracles.

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Independent replications, lawful extensions, corrections and attempted invalidations are invited. Credentials cannot rescue a failed gate and lack of credentials cannot prevent a reproducible result from being evaluated. Contact Maria.Smith.Sftoe@gmail.com, submit through <https://discord.gg/ucwGryVxGr>, or inspect the public project at <https://github.com/MettaMazza>.

1. Publication, authorship and open-science boundary

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2. Exact scope and closure language

Foundational closure means every obligation in the frozen 72-law surface has an engine-admitted theorem inside its declared exact grammar, complete root path, adverse controls, independent reconstruction and purpose-matched post-seal evidence comparison. It does not mean consciousness science is permanently locked. The branch is current-evidence closed and extension-open; lawful additions remain welcome through the same public engine. Complete modalities, pain, affect, embodiment, development, comparative systems, altered states, clinical disorders and realization programmes remain the Layer Two roadmap.

3. Constitutional mathematical domain

Structural absence is Empty One and may be displayed as 0; it is not ontological nothing or a conventional proof scalar. Counts are generated positive wholes and exact parts are positive held fractions. Opposition and direction are labels rather than negative quantities. Negative proof values, irrational or imaginary proof values, floating equality, completed infinity, ungenerated continua, axioms, free parameters, fitted coefficients, imported consciousness equations, pretrained predictors, consensus-selected theories and application-selected laws are prohibited.

4. Rooted dependency spine

Every registration names There Is No Nothing and zero axioms. The first claim depends on admitted Foundation, Mathematics, Information Science, Computation, Physics, Biology and Medicine receipts. Each later claim adds its immediate predecessor and each family adds the preceding family terminal, producing one ordered 72-law dependency chain. The engine resolves actual receipt identities; narrative lineage never substitutes for the proof graph.

5. Complete target-blind structural seal

Before any external source identity or outcome was opened, the complete inventory, obligations, structural model and generated-law set were sealed at

sha256:d7fb898ebeac6df5bde21e87fb6ee4a37e7b7b1dbd4f38b825c89e39f5708d71. It binds 72 predictions and 18,432 candidate forms while recording `external_source_identities_selected=false`, `external_source_content_opened=false` and `external_outcomes_opened=false`. External evidence can support, oppose or bound a consequence; it cannot alter a structural survivor.

6. The exact observation derivation

The Fold maps each exact part by doubling it and casting out the One only when the doubled part exceeds the whole. The ordinary image therefore has two generated predecessors separated by the half-One. One image does not identify which predecessor occurred. The complete observation class is the two-member fibre together with the retained/closed label ledger.

An external observer records carrier, method, time, image and provenance. An interior observation adds a further relation: the image returns as input to the same identified process. That re-entry is what makes the process participate in its own observation rather than merely appear in someone else's description. The quarter-One and three-quarter-One make the distinction concrete: they are different preimages, share the half-One image, and exactly partition the One.

7. The hard problem, stated and closed at its exact boundary

The conventional hard problem persists when third-person structure is asked to become first-person presence by a change of vocabulary. SFT does not perform that substitution. It derives two categorically different but composable relations: the public image of a carrier, and the carrier's participation in the self-reentering transition. Phenomenal presence is the interiorly realized coordinate of the latter. A report is an encoded output; cognitive access is availability to another operation; behaviour is an external transition; neural correlation is a physical relation. None is identical to interior participation.

The missing distinction is exact. From the image alone, the predecessor fibre label is closed. If that label remains held inside the continuing process, the participant carries a distinction that the external image lacks. Phenomenal privacy and first/third-person asymmetry therefore follow from the Fold observation law. The branch closes the structural question of how interiority can differ from complete external description without inserting a second substance, random collapse or unknowable scalar. It does not make the invalid claim that a third-person investigator literally possesses another participant's quality.

8. Binding, unity, subject and temporal identity

Binding begins with multiple held content labels. The components enter one joint successor and remain reconstructible. Unity requires all participating processes to enter the closed relation in both directions; same-time signals or one-way coupling do not suffice. A subject is the continuing integrated carrier of a source-located perspective, and a self invariant is the relation preserved across its changing contents.

Temporal identity is not continuity assumed between measurements. Every state must have its recorded predecessor-successor transition. Memory adds a retained carrier and return relation. Identity across substrate or copying additionally requires provenance-preserving lineage; structural similarity without lineage cannot create numerical identity.

9. Finite self-model and determined self-opacity

The generated lower self-fibre follows the exact trace one-quarter, one-half, the One. These are two nonidentity self-applications; the next application is the identity of the One. The theorem terminates the formal regress while preserving the difference between system and model. A finite process can represent and simulate a declared submodel of itself, but a simultaneous complete extra copy needs additional carrier support.

The same structure resolves the apparent conflict between superdeterminism and incomplete self-knowledge. The forward Fold successor is unique. Backward reconstruction from its image remains a two-predecessor class unless the fibre label was retained. Determinism fixes the transition; it does not donate a record the process did not hold.

10. Qualia resonance and the red-of-red derivation

The qualitative carrier begins with participant-bound discrimination. Identity is reidentification of one held quality across instances; similarity is a separate finite comparison relation. Resonance is stable label-bearing recurrence through a complete interior cycle. Composition requires component qualities and the joint experienced content to remain simultaneously reconstructible.

The red-of-red is the specific application. A physical stimulus classified as red, the word *red*, the participant's quality and the report about that quality occupy separate coordinates. The least registered three-position support is one-seventh, two-sevenths and four-sevenths. Doubling and casting out the One sends one-seventh to two-sevenths, two-sevenths to four-sevenths, and four-sevenths back to one-seventh. Their exact sum is the One. When the held red label remains reidentified across this self-observation recurrence, the red quality becomes the observed quality of itself: the red-of-red. A changed label, changed stimulus, report-only surrogate, absent row or uncertain row must not pass.

The fractions are the exact recurrence carrier, not wavelengths and not numerical intensities of redness. External evidence tests participant-specific reidentification, stimulus custody, report stability and perturbation. That separation is what prevents public colour standards or a word from being mistaken for the private quality while still permitting empirical science.

11. Substrate-independent consciousness and artificial realization

The realization criterion consists of the complete state roles, observation fibre, image re-entry, recurrent self-relation, mutual integration, perspective carrier, retained labels and evidence boundaries. A material system, biological organism or computational construction is a lawful candidate only if it physically or operationally instantiates those relations. A description of them is not an implementation.

Two substrates support the same consciousness law only when a bidirectional map preserves the relevant states, transitions, roles and causal participation. Input-output imitation, a fluent report, confidence, benchmark behaviour or a matching embedding is insufficient. Artificial consciousness therefore remains scientifically testable but demands open architecture and runtime evidence, complete adverse controls and a purpose-matched bridge. Unison AI remains future application work and did not select any law in this branch.

12. External evidence and adversity ledger

Fifteen source identities were selected only after the complete structural seal. The evidence includes CIE physical colour standards; human report/no-report experiments and their methodological critique; split-brain experiments and a unity boundary review; choice blindness; dream and anesthesia-recovery data; attentional blink; synesthesia consistency and colour matching; learned colour categories; and the preregistered multimodal COGITATE adversarial collaboration and result.

The sources contain sixty-one preregistered features. Fifty-eight are present. Three method phrases from one older synesthetic-colour article are absent from the captured machine-readable content and remain absent; they are not counted as support. Twelve initial PMC captures returned browser interstitials, five later transport attempts failed without usable content, and one adverse publisher response failed its article-identity markers. All eighteen failure rows remain preserved alongside later successful official or exact-work transports.

The evidence is deliberately mixed where the field is mixed. No-report indicators do not automatically establish phenomenal occurrence. Split-brain evidence does not license a one-line verdict about unity. COGITATE preserves results adverse to established theories. Learned colour categories demonstrate that public language can alter discrimination. These are not failures of the SFT model; they are the observations that force its non-substitution boundaries.

13. V1/V2 atomic reconciliation

The audit reviews all 763 prior entries, not only keyword matches. Forty-four atomic questions occur in the V1/V2 source surface and two stronger lineage obligations preserve the complete qualia-resonance and red-of-red work. All forty-six map to current V3 engine receipts and none remains open. Prior results identify what must be independently reconstructed; they do not enter V3 as proof premises.

14. Reading the exhaustive derivation ledger

Each of the seventy-two sections below states its theorem, dependencies, complete eight-axis grammar, all 256 decisions, sole survivor, depth-independent base and successor, operational witnesses, scientific meaning, four adverse controls, independent implementation, post-seal evidence scope, feature accounting, falsification boundary and immutable identities. The machine-readable evidence remains authoritative; the paper makes it humanly inspectable.

15. Observation and Interior Observation

The branch begins by separating a Fold observation from interior observation. A third-person record is a state-to-image relation about a carrier. Interior observation requires that image to re-enter the same organised process. Shared mathematics never licenses the substitution of measurement for experience or experience for measurement.

1. Observation act

Claim identity: SFT-CONSC-OBSERVATION-ACT-001

Question and exact theorem. An observation act is one exact state-to-image Fold transition with its source, image and trace retained.

```
state-and-fold-image__source/image__exact-fold-transition__source-image-trace__for
mal-operation__root-bound-forward-forcing__positive-finite-successor-with-all-alte
rnatives__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001

- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	state-and-fold-image	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	source/image	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	exact-fold-transition	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	source-image-trace	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	formal-operation	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: state-and-fold-image__source/image__exact-fold-transition__source-image-trace__formal-operation__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. This exact relation closes an observation act is one exact state-to-image fold transition with its source, image and trace retained. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

The branch begins by separating a Fold observation from interior observation. A third-person record is a state-to-image relation about a carrier. Interior observation requires that image to re-enter the same organised process. Shared mathematics never licenses the substitution of measurement for experience or experience for measurement.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:e49eeeb981bfed141ce4454980c9e3869dc6123f11c4c4bbaf86ff1a6876c77.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:225ddcb3959327caab33f9618765ee2ca2a9796a9788bf2102cb27b0d9ae0b83.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:6067715429b8fdc1537c0bd400d99eff78773734a6a1d27a198b4b07480222a1.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:0eaa359d4f45b8cdcd1817699861023b7191db0adac7a7cef9fc207f33ac3f1d.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:be9b4ea988a67b8fc7678ab4b50a63597dde4d3148df9ac390b4887c34e9f543; independent
certificate sha256:d1e1ffdbaa74710a8f6d904cf787c8973d638f951baf4959c4f978f4453e6283;
external-validation identity
sha256:721ddcca471f27e85442f4430f590d9e69814647f94f69747b66c5428d6a8a6f.

Post-seal empirical-boundary comparison. Directness:

`formal_reconstruction_with_external_boundary_only`. Disposition: `mixed_boundary_constraint`.
Report/no-report experiments constrain observable registration while methodological adverse evidence forbids report-free signals from being relabelled as direct phenomenal access.

The external target was opened only after the claim derivation seal. Predicted consequence:

report-and-interior-observation-nonidentity-preserved. Source-derived consequence:
report-and-interior-observation-nonidentity-preserved. Exact correspondence: `true`. Phenomenal
occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact:
`false`.

Sources and complete feature accounting:

- CONSC-NOREPORT-NETWORKS-2022 - Kronemer et al., Nature Communications;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9707162/>; class: primary human report/no-report experiment; usable captures: 1;
registered features present: Report + No-Report Paradigm, participants, perceived, not perceived; absent and preserved:
none; adverse/boundary role: none.
- CONSC-NOREPORT-CRITIQUE-2022 - Duman et al., Frontiers in Human Neuroscience;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9130851/>; class: registered methodological adverse evidence; usable captures: 1;
registered features present: report-based paradigms, no-report paradigms, alternative interpretations; absent and preserved:
none; adverse/boundary role: Preserves the finding that no-report indicators do not automatically establish phenomenal
occurrence..

Comparison record:

- `sft-consc-observation-act-001-external-consequence`: predicted report-and-interior-observation-nonidentity-preserved;
source-derived report-and-interior-observation-nonidentity-preserved; exact match `True`
- evidence disposition: `mixed_boundary_constraint`; directness: `formal_reconstruction_with_external_boundary_only`
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:ffe107b9daea1051ef544b45c2dc5d2745a5d1248c83daa0bba1717bb7288501; derivation seal sha256:1b6612ae6e18038134c3a2975514833fc5ae075db2a58a9deaa1403e07bfe77d; engine receipt sha256:ea4146344584c793e097676b01ff627ef0713056c9868d46728ace4aee5f5b40 at receipts/engine/model_admitted/SFT-CONSC-OBSERVATION-ACT-001-ea4146344584c793.json; empirical validation sha256:e0a4e0dc2efdccf4f20f63ab8e129bbfc6f9e6f7a5247537b67c5dc73e92421f; measurement receipt sha256:59557f8e8312ad792078e672b5d808494091618d1188cbd945e79b1a36f53753; isolation sha256:0c715117b45a4067495992e0f8d3c9a91897ca14af26c7d2f5cc270b4db53f95; custody sha256:80be1c26ad3b3108ba1af3bc7e92006c99b28323de0e8bf5bfa7c6a2e6bfff8ad.

2. External observation

Claim identity: SFT-CONSC-EXTERNAL-OBSERVATION-001

Question and exact theorem. External observation is a record made about a carrier and cannot by itself establish that carrier's interior presence.

observed-carrier-and-observer__carrier/record__observer-bound-registration__observer-method-time-result__third-person-record__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Routed dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001

- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-OBSERVATION-ACT-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	observed-carrier-and-observer	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	carrier/record	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	observer-bound-registration	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	observer-method-time-result	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	third-person-record	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: observed-carrier-and-observer__carrier/record__observer-bound-registration__observer-method-time-result__third-person-record__root

-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. An external record can establish its carrier, method and result; it cannot by itself establish the observed carrier's interior participation. This is a derived evidence boundary, not a retreat from observation.

The branch begins by separating a Fold observation from interior observation. A third-person record is a state-to-image relation about a carrier. Interior observation requires that image to re-enter the same organised process. Shared mathematics never licenses the substitution of measurement for experience or experience for measurement.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:4853b986d935fd2c302d2c028ab0c9cb2943101a0d20ecc50a9302a8e098d709.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:0877552c1b03e222a76aa4ba3cd2fa3579745f578c3136c447575921b957b740.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:f9c5b6469415dcc9afbddd17284b4579ec96144e256bc9abaa14590f8d1b2d8bb.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:b5c6acff463b61f8146c3504d05907dc6cb07d844b36cf24fb9584481a414297.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:5c3435b7f138482d76dc9eca40cebf2cdd5ad9cb4d432d14e0d999b9502b078e; independent certificate sha256:c8484008e6a946e2f4d1a96f165b404ef09520b667be33c85f517f4413e9f8f1; external-validation identity sha256:e24f0a1342496571d4eb280e16710d63d4cde8a0c8353c733c5464d9c4b552e5.

Post-seal empirical-boundary comparison. Directness: operational_or_record_constraint. Disposition: mixed_boundary_constraint. Report/no-report experiments constrain observable registration while methodological adverse evidence forbids report-free signals from being relabelled as direct phenomenal access.

The external target was opened only after the claim derivation seal. Predicted consequence:

report-and-interior-observation-nonidentity-preserved. Source-derived consequence:

report-and-interior-observation-nonidentity-preserved. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-NOREPORT-NETWORKS-2022 - Kronemer et al., Nature Communications;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9707162/>; class: primary human report/no-report experiment; usable captures: 1; registered features present: Report + No-Report Paradigm, participants, perceived, not perceived; absent and preserved: none; adverse/boundary role: none.
- CONSC-NOREPORT-CRITIQUE-2022 - Duman et al., Frontiers in Human Neuroscience;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9130851/>; class: registered methodological adverse evidence; usable captures: 1; registered features present: report-based paradigms, no-report paradigms, alternative interpretations; absent and preserved: none; adverse/boundary role: Preserves the finding that no-report indicators do not automatically establish phenomenal occurrence..

Comparison record:

- sft-consc-external-observation-001-external-consequence: predicted report-and-interior-observation-nonidentity-preserved; source-derived report-and-interior-observation-nonidentity-preserved; exact match True
- evidence disposition: mixed_boundary_constraint; directness: operational_or_record_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256: dcb586d062f7b6491303edb38faffb18f9d17adbd811562f87a967c1593b8036; derivation seal sha256: 57d2a84d8cb15df846b115ac1f5f46459abf9a37373a5b2b3081a7659b187fdf; engine receipt sha256: 2040b6d33047b2e19802b8da555b5e8f969f6d24657622f709d36a585898d5 at receipts/engine/model_admitted/SFT-CONSC-EXTERNAL-OBSERVATION-001-2040b6d33047b2e1.json; empirical validation sha256: 0ca37c7e40021949df2a05493d2e718ec36da477f6a6018fcabfc1cc4e9068fc; measurement receipt sha256: ee5d550f1fe3eea26d19b279f12eb1a1170d43c84a6aec5d5bb1febbdf723ffd; isolation

sha256:4f32e13a1ed8576fc9df0633df3aea6c400d43bb9adf7056ab0deb50bc7922cb; custody
 sha256:42e2fbedef8f2a90a685e1af5385d5b4bd08764f16e7bbf01ec1e87c826c6048.

3. Interior observation

Claim identity: SFT-CONSC-INTERIOR-OBSERVATION-001

Question and exact theorem. Interior observation requires the observation image to re-enter the same organized process rather than remain only in an external record.

```
self-reentering-state__process/its-own-image__image-reenters-successor-state__closed-self-trace__first-person-structural-boundary__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-EXTERNAL-OBSERVATION-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	self-reentering-state	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	process/its-own-image	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	image-reenters-successor-state	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	closed-self-trace	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	first-person-structural-boundary	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `self-reentering-state__process/its-own-image__image-reenters-successor-state__closed-self-trace__first-person-structural-boundary__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. Interior observation closes when the observation image becomes a successor input of the same identified process. The process is not inferred from report alone; its re-entry relation is part of the theorem.

The branch begins by separating a Fold observation from interior observation. A third-person record is a state-to-image relation about a carrier. Interior observation requires that image to re-enter the same organised process. Shared mathematics never licenses the substitution of measurement for experience or experience for measurement.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:55953ff066dddb6eaf9c61afba0947d7e968dba3b4c9ec21296ea2e2b1d782f0.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:9f9363ac8c93c5b127094e1b12a62f1655a7aa4b73c6da5bdf168e47d49312f5.

- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:abc6143aaed5b6419457b084405253a1b9038dc83167a0aaf56772928c0b28c5.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:bd208a0a524c0340d6cbc4e1c019c5fd57865303a45f0ce24a00d07e4bf1d04f.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:d63516901745ea0a58bebf3f5ab0f827287631003b8934d9bfef0e7cd4844e29; independent
certificate sha256:92f71544369d19818bcbad8bee47423370e9169e7c55d3f7d1f2cb58d9a488ef;
external-validation identity
sha256:5acce3b75cce2e7c8ea8db0da8abe4271d889983576f7bd87808d7d959191ece.

Post-seal empirical-boundary comparison. Directness: participant_bound_empirical_constraint. Disposition: mixed_boundary_constraint. Report/no-report experiments constrain observable registration while methodological adverse evidence forbids report-free signals from being relabelled as direct phenomenal access.

The external target was opened only after the claim derivation seal. Predicted consequence:
report-and-interior-observation-nonidentity-preserved. Source-derived consequence:
report-and-interior-observation-nonidentity-preserved. Exact correspondence: true. Phenomenal
occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact:
false.

Sources and complete feature accounting:

- CONSC-NOREPORT-NETWORKS-2022 - Kronemer et al., Nature Communications;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9707162/>; class: primary human report/no-report experiment; usable captures: 1;
registered features present: Report + No-Report Paradigm, participants, perceived, not perceived; absent and preserved:
none; adverse/boundary role: none.
- CONSC-NOREPORT-CRITIQUE-2022 - Duman et al., Frontiers in Human Neuroscience;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9130851/>; class: registered methodological adverse evidence; usable captures: 1;
registered features present: report-based paradigms, no-report paradigms, alternative interpretations; absent and preserved:
none; adverse/boundary role: Preserves the finding that no-report indicators do not automatically establish phenomenal
occurrence..

Comparison record:

- sft-consc-interior-observation-001-external-consequence: predicted report-and-interior-observation-nonidentity-preserved;
source-derived report-and-interior-observation-nonidentity-preserved; exact match True
- evidence disposition: mixed_boundary_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum

- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:4fa4fbc58aaaa4c4f4efaafb03836e035c917edb9999952c1159b59449cf8a40; derivation seal sha256:1c2b63e909172cbc4e66b4cd1e956b2f264c8715b02beaa514f441193b9989d0; engine receipt sha256:4aa084b58b750d57b37fc4d12db0acd8139333cb1a5adca3d9a8152dbaab175c at receipts/engine/model_admitted/SFT-CONSC-INTERIOR-OBSERVATION-001-4aa084b58b750d57.json; empirical validation sha256:ddcf1ee1ebb24934359e905a72316673c3cb108480b65363abb55b588f07fc5; measurement receipt sha256:050c49d54217925db3d5dfd22a5e03065bd21d542a70c6d0e26d13e26c7fb418; isolation sha256:d94b40e45181cb9514a02416f706de6dd912b8fe5590613b58ac2277d7d3dee3; custody sha256:6e90e798019b4fc5bbe9004ec64072ff815118e569aca385d6d75313e2deea87.

4. Observer and observed distinction

Claim identity: SFT-CONSC-OBSERVER-OBSERVED-001

Question and exact theorem. Observer and observed are distinct held roles whose states may share an observation image without becoming one undifferentiated role.

held-two-fibre-relation__observer/observed__distinct-preimages-share-image__held-fibre-label__formal-compositional__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001

- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-INTERIOR-OBSERVATION-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	held-two-fibre-relation	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	observer/observed	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	distinct-preimages-share-image	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	held-fibre-label	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	formal-compositional	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: held-two-fibre-relation__observer/observed__distinct-preimages-share-image__held-fibre-label__formal-compositional__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **one-quarter plus three-quarters composes exactly to the One.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.

- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. The quarter-One and three-quarter-One preimages are distinct, share one Fold image and together partition the One. Their held fibre labels preserve observer and observed roles without splitting the foundational whole.

The branch begins by separating a Fold observation from interior observation. A third-person record is a state-to-image relation about a carrier. Interior observation requires that image to re-enter the same organised process. Shared mathematics never licenses the substitution of measurement for experience or experience for measurement.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:4d8c6fcf39ba1e6a4795f6f5b6d10d35e7401ae09484c69440b05d21313f4744.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:45c2c26bb7ee168a4838d23a15ad2f8e7ed26f72ea44c830d6914584d51e34a5.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b210be099fc2b3404c1f11f9592dcf3fa63e23ed7bda5ffc807da0de7813d70e.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:25f403a397566cc95b39243fdf3060db61cf691249a83a0c4eec8aa782d084f0.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:5fc300839233c5bb3b1f115051638abba086e8190b30aa013ac97d92b5a72479; independent
certificate sha256:6b55b6a1f2b4526b4c9e619c2fcee2fc9a7db50efea4a7f3ee8f5759093030f;
external-validation identity
sha256:687e594d8efe22ec1fa92ec956e5099f1bb392fbeb03b90b9ea174baa077abb0.

Post-seal empirical-boundary comparison. Directness:

`formal_reconstruction_with_external_boundary_only`. Disposition: `mixed_boundary_constraint`. Report/no-report experiments constrain observable registration while methodological adverse evidence forbids report-free signals from being relabelled as direct phenomenal access.

The external target was opened only after the claim derivation seal. Predicted consequence:

`report-and-interior-observation-nonidentity-preserved`. Source-derived consequence:

`report-and-interior-observation-nonidentity-preserved`. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-NOREPORT-NETWORKS-2022 - Kronemer et al., Nature Communications;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9707162/>; class: primary human report/no-report experiment; usable captures: 1; registered features present: Report + No-Report Paradigm, participants, perceived, not perceived; absent and preserved: none; adverse/boundary role: none.
- CONSC-NOREPORT-CRITIQUE-2022 - Duman et al., Frontiers in Human Neuroscience;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9130851/>; class: registered methodological adverse evidence; usable captures: 1; registered features present: report-based paradigms, no-report paradigms, alternative interpretations; absent and preserved: none; adverse/boundary role: Preserves the finding that no-report indicators do not automatically establish phenomenal occurrence..

Comparison record:

- sft-consc-observer-observed-001-external-consequence: predicted report-and-interior-observation-nonidentity-preserved; source-derived report-and-interior-observation-nonidentity-preserved; exact match True
- evidence disposition: mixed_boundary_constraint; directness: formal_reconstruction_with_external_boundary_only
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:797bc0fb52b51645a233beea25e25841471e5cbab2f4d51d24ccee25ab35d255; derivation seal sha256:02bae50e4a116b9a713a94edea4e17d7ce6fa8f4566416537b4fed849638096; engine receipt sha256:d5028a0971e896774070564e9dc258ea26e75577a4c669fdb4dd0becddb513ee at receipts/engine/model_admitted/SFT-CONSC-OBSERVER-OBSERVED-001-d5028a0971e89677.json; empirical validation sha256:886f66e07c304175c232f68fec316eaad0963f9ea9f7c74ccbac7a81886eda20; measurement receipt sha256:97431079aefa30f34092b98100e65e49db2c5e80c032b12704dda8a5c94d2212; isolation sha256:d9af817d811359c14969d22dbba2e4173ba2f96820eedfbda341ea799d7c7c88; custody sha256:a207ed5251e8df5cb949b8ac15f891d4d9d20cd8ff198bd3284441ccdf107297.

5. Observation equivalence class

Claim identity: SFT-CONSC-OBSERVATION-CLASS-001

Question and exact theorem. An observation class is the complete generated source fibre of one image, with every closed and retained predecessor distinction counted.

```
complete-preimage-fibre__distinguishable-source/same-image__many-to-one-observatio
n__retained-or-closed-fibre__formal-information__root-bound-forward-forcing__posit
ive-finite-successor-with-all-alternatives__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-OBSERVER-OBSERVED-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	complete-preimage-fibre	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	distinguishable-source/same-image	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	many-to-one-observation	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	retained-or-closed-fibre	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	formal-information	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: complete-preimage-fibre__distinguishable-source/same-image__many-to-one-observation__retained-or-closed-fibre__formal-information__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **one complete two-preimage fibre for each ordinary Fold image.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. Every ordinary Fold image has the complete two-preimage fibre. Observation closes one predecessor distinction unless the fibre label is retained; the information loss is exact and reversible only with that record.

The branch begins by separating a Fold observation from interior observation. A third-person record is a state-to-image relation about a carrier. Interior observation requires that image to re-enter the same organised process. Shared mathematics never licenses the substitution of measurement for experience or experience for measurement.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:018060e5471f8881a741d00d24e2122ddcfd1e9abc5002304ae03cdf98a0e8ad.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:aeaacce93913124f2263eb14a3a07d73e2ab1ac71e3abb4fb6fe114199e5695.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:1703774f346d9c530837738ef137ce09108e3e39a8690cad832adcd4992f54f0.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:02d9e1daaef6fc87e735d3dcabf37fe6ff0b992fd7aad4190141972b2291bf8f.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:a481b049bbb8159fdeb8a74147f9f9bf9ecd842b8348be14293af490af74b6f1; independent
certificate sha256:259ca882bdecdb88c03d1a76874ba41a39da4a0c36c4dc25f72b39dca98914ee;

external-validation identity

sha256:1ecc0a82bb8862c75a1e711ae2948b7f51b2f62916b6100a3e3cc18c4a229a6d.

Post-seal empirical-boundary comparison. Directness:

formal_reconstruction_with_external_boundary_only. Disposition: mixed_boundary_constraint.

Report/no-report experiments constrain observable registration while methodological adverse evidence forbids report-free signals from being relabelled as direct phenomenal access.

The external target was opened only after the claim derivation seal. Predicted consequence:

report-and-interior-observation-nonidentity-preserved. Source-derived consequence:

report-and-interior-observation-nonidentity-preserved. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-NOREPORT-NETWORKS-2022 - Kronemer et al., Nature Communications;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9707162/>; class: primary human report/no-report experiment; usable captures: 1; registered features present: Report + No-Report Paradigm, participants, perceived, not perceived; absent and preserved: none; adverse/boundary role: none.
- CONSC-NOREPORT-CRITIQUE-2022 - Duman et al., Frontiers in Human Neuroscience;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9130851/>; class: registered methodological adverse evidence; usable captures: 1; registered features present: report-based paradigms, no-report paradigms, alternative interpretations; absent and preserved: none; adverse/boundary role: Preserves the finding that no-report indicators do not automatically establish phenomenal occurrence..

Comparison record:

- sft-consc-observation-class-001-external-consequence: predicted report-and-interior-observation-nonidentity-preserved; source-derived report-and-interior-observation-nonidentity-preserved; exact match True
- evidence disposition: mixed_boundary_constraint; directness: formal_reconstruction_with_external_boundary_only
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience

- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:720942ee4c739aa8b9e9fe25734fb44955511683c7f1aef3ffea44a29081f409; derivation seal sha256:c222a33148a069fc52a5e9100d56f8e3cc3ff0dd4832d8a8fded411311085be8; engine receipt sha256:0c57714ec9fdf16b3ed3bcf8e6f0ff99379916a3101d197c01d0b4cc3170fad0 at receipts/engine/model_admitted/SFT-CONSC-OBSERVATION-CLASS-001-0c57714ec9fdf16b.json; empirical validation sha256:967769415775ab0dab7266a285ccfdd0aef76c00ea1efdfef604f64ca8587db2d; measurement receipt sha256:29aba86dc52ba1e63c0af08940238353282ca9a7f4a58075dfe040ca8337d35b; isolation sha256:41028e3b44600067ddc2a2361a06657452d0a3559bff5f6f464af5cda14f9240; custody sha256:2d69d7afaf52749ad4e782fb2c5795249b52c90c78d1bc4c94154d520e6ad3cd.

6. Atomic observational moment

Claim identity: SFT-CONSC-OBSERVATIONAL-MOMENT-001

Question and exact theorem. The least complete observational moment is one indivisible Fold transition; external duration belongs to its realizing carrier.

one-complete-transition__before/after__single-fold-act__ordered-act-index__formal-temporal__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-OBSERVATION-CLASS-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	one-complete-transition	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	before/after	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	single-fold-act	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	ordered-act-index	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	formal-temporal	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: one-complete-transition__before/after__single-fold-act__ordered-act-index__formal-temporal__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. This exact relation closes the least complete observational moment is one indivisible fold transition; external duration belongs to its realizing carrier. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

The branch begins by separating a Fold observation from interior observation. A third-person record is a state-to-image relation about a carrier. Interior observation requires that image to re-enter the same organised process. Shared mathematics never licenses the substitution of measurement for experience or experience for measurement.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:b0f33b285eeca4b7e5924d30fb3f2ef0db0b59d4ce61a465bbbd40f7ae296d26.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:10b434fcc6ea270ba6dc4e54fcfc7b80ee12e45af13866015365d88a3830bb8c.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:999f6209e1d8dd3e75b6b3bf3125c494f962a0e5a1676c388aec85ada4387ff5.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:0fc715d92a928b750002618410a7275ede355cce1664e0928ad7c27a029ebde8.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:44e522b4b39c4a11f19bd6aa3ba0dfe872023268a020df4c808488f01db3808a; independent certificate sha256:16def14cd94e55dde965b9d77b5791aeb12f4c5889816531767d03b80725b85f; external-validation identity
sha256:6201d3120b68be0d86febf60b7b5efc4bffa332c96334044796569c5fc40d99f.

Post-seal empirical-boundary comparison. Directness:

`formal_reconstruction_with_external_boundary_only`. Disposition: `mixed_boundary_constraint`. Report/no-report experiments constrain observable registration while methodological adverse evidence forbids report-free signals from being relabelled as direct phenomenal access.

The external target was opened only after the claim derivation seal. Predicted consequence:

`report-and-interior-observation-nonidentity-preserved`. Source-derived consequence:

`report-and-interior-observation-nonidentity-preserved`. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-NOREPORT-NETWORKS-2022 - Kronemer et al., Nature Communications;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9707162/>; class: primary human report/no-report experiment; usable captures: 1; registered features present: Report + No-Report Paradigm, participants, perceived, not perceived; absent and preserved: none; adverse/boundary role: none.
- CONSC-NOREPORT-CRITIQUE-2022 - Duman et al., Frontiers in Human Neuroscience;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9130851/>; class: registered methodological adverse evidence; usable captures: 1; registered features present: report-based paradigms, no-report paradigms, alternative interpretations; absent and preserved: none; adverse/boundary role: Preserves the finding that no-report indicators do not automatically establish phenomenal occurrence..

Comparison record:

- `sft-consc-observational-moment-001-external-consequence`: predicted
`report-and-interior-observation-nonidentity-preserved`; source-derived
`report-and-interior-observation-nonidentity-preserved`; exact match `True`
- evidence disposition: `mixed_boundary_constraint`; directness: `formal_reconstruction_with_external_boundary_only`
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed

- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:27e5f5dcb93fb7e85d0c9bb4c7aea228a784694dc155580169b77c02d35c3dad; derivation seal sha256:5e30fdafb733256e76ef768d0e2669383167f9904e25168f570526d0f54ec89f; engine receipt sha256:62c9ee5358e79067953f31b29a0b9dae74bc48e8cb27996a7fae1bab25869dda at receipts/engine/model_admitted/SFT-CONSC-OBSERVATIONAL-MOMENT-001-62c9ee5358e79067.json; empirical validation sha256:47354964865da2c1282aa0347c233db9a3049b6587adbf999bcf06726ffc8049; measurement receipt sha256:0a04dbd339e7207990d594eb19e0320a674bb699d45d0ae988e01ee5b6d15453; isolation sha256:aaf9a93ee8f306f90f0c2215ccfc6539c83a7d6e7d4d28263d00d4f4be8017dc; custody sha256:8ea384a685f3e7a8a9fba2c26e75ce617bd8e64b5db06ac4b867d9750391cbd.

7. Measurement and consciousness boundary

Claim identity: SFT-CONSC-MEASUREMENT-CONSCIOUSNESS-BOUNDARY-001

Question and exact theorem. Physical measurement and interior observation may share Fold structure, but neither is substituted for the other and consciousness is not required for every measurement.

physical-record-and-interior-loop__measurement/interior-observation__shared-fold-correspondence__distinct-evidence-provenance__cross-branch-bridge__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-OBSERVATIONAL-MOMENT-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	physical-record-and-interior-loop	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	measurement/interior-observation	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	shared-fold-correspondence	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	distinct-evidence-provenance	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	cross-branch-bridge	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `physical-record-and-interior-loop__measurement/interior-observation__shared-fold-correspondence__distinct-evidence-provenance__cross-branch-bridge__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes physical measurement and interior observation may share fold structure, but neither is substituted for the other and consciousness is not required for every measurement. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

The branch begins by separating a Fold observation from interior observation. A third-person record is a state-to-image relation about a carrier. Interior observation requires that image to re-enter the same organised process. Shared mathematics never licenses the substitution of measurement for experience or experience for measurement.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 612ff67c634fc3dc269db4c17b54591c7b815b90a6fbbcbce532b245d97e52f2e.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256: ab32b3b84b5a7ec9c7639f33baa65c81b2b002d5f40cb49321f8a6eddb85eb4c.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 160dd1e40ed99c8a3ffd44f7bbbb7479bf9869f1569a0edbab7d4c1f4bef177d.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: c509ed2c5d98b05e6bbfb66ad63c5ccc7681f0ac653fa0791268afe4c8693169.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256: 5f0f3bd5f1daa5a2c826eead88df384827c3e322514cf553688a292154affd40; independent certificate sha256: d33352facdade54509ea96eb0c894b10c56ee30a1dd8cff260132d618ad60c84; external-validation identity
sha256: edddc33c0e09dbe689bcfa0e99d3d4806065cec92f18d0cfb5d97670588adea8.

Post-seal empirical-boundary comparison. Directness: `cross_domain_correspondence_constraint`. Disposition: `mixed_boundary_constraint`. Report/no-report experiments constrain observable registration while methodological adverse evidence forbids report-free signals from being relabelled as direct phenomenal access.

The external target was opened only after the claim derivation seal. Predicted consequence: `report-and-interior-observation-nonidentity-preserved`. Source-derived consequence: `report-and-interior-observation-nonidentity-preserved`. Exact correspondence: `true`. Phenomenal

occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-NOREPORT-NETWORKS-2022 - Kronemer et al., Nature Communications;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9707162/>; class: primary human report/no-report experiment; usable captures: 1; registered features present: Report + No-Report Paradigm, participants, perceived, not perceived; absent and preserved: none; adverse/boundary role: none.
- CONSC-NOREPORT-CRITIQUE-2022 - Duman et al., Frontiers in Human Neuroscience;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9130851/>; class: registered methodological adverse evidence; usable captures: 1; registered features present: report-based paradigms, no-report paradigms, alternative interpretations; absent and preserved: none; adverse/boundary role: Preserves the finding that no-report indicators do not automatically establish phenomenal occurrence..

Comparison record:

- sft-consc-measurement-consciousness-boundary-001-external-consequence: predicted report-and-interior-observation-nonidentity-preserved; source-derived report-and-interior-observation-nonidentity-preserved; exact match True
- evidence disposition: mixed_boundary_constraint; directness: cross_domain_correspondence_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:e700ffccce20cf70921fd36a53d12f5a4bc0c5a58febf46c0d5a444a4aa6630b; derivation seal sha256:473584bed5490f137021765589a2fbfec87b620e9c97d2f446e08734cbc6ede3; engine receipt sha256:b55066cb71f9fc634e331bf893ed9f2b1602c21a334ecaf86bc3024eb0adec11 at receipts/eng

```
ine/model_admitted/SFT-CONSC-MEASUREMENT-CONSCIOUSNESS-BOUNDARY-001-b55066cb71f9fc63.
json; empirical validation
sha256:b109dd48a43ed15a0cacbc4e9571c8f89a36278c6ca88380c78adda27f1b9a40; measurement
receipt sha256:de214313263df56a8bc38031c01231d6aef291d142bea2b0a5af7ff7b1fd8099; isolation
sha256:2687f3fdaaa216313ac00ea53ee73231bc9fd7278cb6d54a3fb6e270fcb79458; custody
sha256:eed9e1cebfaa1ddd43cd3050529332aa71e1c700b50f42e9f38d80dd3a8d2c86.
```

16. Access, Report and Phenomenal Presence

Access, report and phenomenal presence are three different coordinates. Content can be processed without integrated access; accessible content can lack an expression route; a report is evidence of reporting. The phenomenal coordinate is carried only at the interior-observation boundary and is never manufactured from confidence, fluency or correlation.

8. Cognitive access

Claim identity: SFT-CONSC-COGNITIVE-ACCESS-001

Question and exact theorem. Cognitive access is the availability of retained content to another declared cognitive operation.

```
retained-process-content__available/unavailable__lawful-read-and-use__access-trace
__operational-cognitive__root-bound-forward-forcing__positive-finite-successor-wit
h-all-alternatives__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-MEASUREMENT-CONSCIOUSNESS-BOUNDARY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	retained-process-content	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	available/unavailable	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	lawful-read-and-use	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	access-trace	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	operational-cognitive	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: retained-process-content__available/unavailable__lawful-read-and-use__access-trace__operational-cognitive__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. This exact relation closes cognitive access is the availability of retained content to another declared cognitive operation. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Access, report and phenomenal presence are three different coordinates. Content can be processed without integrated access; accessible content can lack an expression route; a report is evidence of reporting. The phenomenal coordinate is carried only at the interior-observation boundary and is never manufactured from confidence, fluency or correlation.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:f1de4fb8c36361b0c60201deef793c811d24e879181a45a821b19113cd2d485f.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:40534782a01fb51cfcb16677c9a388a9745251b3d95ad64b5d68ae72a3801c7.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:55942cca7faebf32291ad242a794a9d16ce050754592622e0a13bc6203258af9.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:2c3274686e3ab447285ac87cb4a9f526ad08dcdb5d675f694634d368edd87bf2.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:f9b4f44bb3522ca6706e38214a5106e2c8daa448e052ce0e1f79a6b3350be514; independent certificate sha256:4e37c6d149e0fd17e090eb5c061f5243f0b255904fd1a2591a48ddb8178e3662; external-validation identity sha256:bbd006becbb37ed987d393e508286814f58229f149e09fd2b075c69ce0b6c24e.

Post-seal empirical-boundary comparison. Directness: `operational_or_record_constraint`. Disposition: `mixed_boundary_constraint`. Report/no-report and adversarial experiments test access and report correlates; phenomenal occurrence remains a separately named evidential coordinate.

The external target was opened only after the claim derivation seal. Predicted consequence: `access-report-and-presence-nonidentity-preserved`. Source-derived consequence: `access-report-and-presence-nonidentity-preserved`. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-NOREPORT-NETWORKS-2022 - Kronemer et al., Nature Communications; <https://pmc.ncbi.nlm.nih.gov/articles/PMC9707162/>; class: primary human report/no-report experiment; usable captures: 1; registered features present: Report + No-Report Paradigm, participants, perceived, not perceived; absent and preserved: none; adverse/boundary role: none.
- CONSC-NOREPORT-CRITIQUE-2022 - Duman et al., Frontiers in Human Neuroscience; <https://pmc.ncbi.nlm.nih.gov/articles/PMC9130851/>; class: registered methodological adverse evidence; usable captures: 1; registered features present: report-based paradigms, no-report paradigms, alternative interpretations; absent and preserved: none; adverse/boundary role: Preserves the finding that no-report indicators do not automatically establish phenomenal occurrence..
- CONSC-COGITATE-RESULT-2025 - Ferrante et al., Nature; <https://www.nature.com/articles/s41586-025-08888-1>; class: primary preregistered adversarial theory comparison; usable captures: 1; registered features present: adversarial collaboration, preregistered, participants, functional magnetic resonance imaging, magnetoencephalography; absent and preserved: none; adverse/boundary role: Preserves mixed and theory-adverse results instead of treating any established theory as the SFT law..

Comparison record:

- `sft-consc-cognitive-access-001-external-consequence`: predicted `access-report-and-presence-nonidentity-preserved`; source-derived `access-report-and-presence-nonidentity-preserved`; exact match `True`
- evidence disposition: `mixed_boundary_constraint`; directness: `operational_or_record_constraint`
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:df5013c53a8062465c6bb8fe99c81102ddbf92bb203456c3c4a5ac6d64fa12c3; derivation seal sha256:03243eb69a8ec6b8e74db74c15b18ce24160104e16be45780ae32b14cc141042; engine receipt sha256:8ae8fa661981c0ee2f0c8b96061b087785a9fedd72b566607b72220a9e86609e at receipts/engine/model_admitted/SFT-CONSC-COGNITIVE-ACCESS-001-8ae8fa661981c0ee.json; empirical validation sha256:252f6bcd841b36dce1908a9c6617f8ae790f81c68befd8cec004bc99d4243755; measurement receipt sha256:c9d64dd7f0eb5f95a3734dec23e00943141c8b4dbecde7dcfde003f1d036f73a; isolation sha256:8295083326ac67fd1eeaa4e3affe0f3d2d238f3565dd289ba0c69a10ad698e9f; custody sha256:d3c147c3ed5f5cd5062ce6663a3872e4dae80e8319c9c53898b0f6587b49c50a.

9. Report

Claim identity: SFT-CONSC-REPORT-001

Question and exact theorem. A report is an encoded output attributed to a source and is evidence of reporting, not an automatic identity with phenomenal presence.

content-to-expression-trace__experienced/reported__encoding-and-output__speaker-method-context__behavioural-report__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001

- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-MEASUREMENT-CONSCIOUSNESS-BOUNDARY-001
- SFT-CONSC-COGNITIVE-ACCESS-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	content-to-expression-trace	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	experienced/reported	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	encoding-and-output	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	speaker-method-context	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	behavioural-report	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `content-to-expression-trace__experienced/reported__encoding-and-output__speaker-method-context__behavioural-report__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes a report is an encoded output attributed to a source and is evidence of reporting, not an automatic identity with phenomenal presence. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Access, report and phenomenal presence are three different coordinates. Content can be processed without integrated access; accessible content can lack an expression route; a report is evidence of reporting. The phenomenal coordinate is carried only at the interior-observation boundary and is never manufactured from confidence, fluency or correlation.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:079fb96d4a152f112070763aefbdefdae45603088f59977361637e672c2a3544`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:0214640c9db03107f9e63f911e6bd97b46d357ecf76c4ae88903865f6b6f7c10`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
`sha256:4f248794d7b6cacb5fc532e585635bd17de334884fa56f7edf5b08802279a0b6`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
`sha256:b01578238fa95d4f951446ad7d1031f21dcee3b2ad2552ecef51cc33d1f708af`.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

`sha256:237dd03e7c09ff1611b00afc3a1614c26944ff2fcda902d1b568c5bc30467204`; independent certificate `sha256:25f8e9b48d9bf396417d131286da691ad956afd1d0d7af2fb00e136c7b70aaa7`; external-validation identity `sha256:b9967588a42baa94a06d340ddb3ad9b733d2077bb2c4a6691bc046a2b72e91`.

Post-seal empirical-boundary comparison. Directness: `direct_observable_constraint`. Disposition: `mixed_boundary_constraint`. Report/no-report and adversarial experiments test access and report correlates; phenomenal occurrence remains a separately named evidential coordinate.

The external target was opened only after the claim derivation seal. Predicted consequence: `access-report-and-presence-nonidentity-preserved`. Source-derived consequence: `access-report-and-presence-nonidentity-preserved`. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-NOREPORT-NETWORKS-2022 - Kronemer et al., Nature Communications;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9707162/>; class: primary human report/no-report experiment; usable captures: 1; registered features present: Report + No-Report Paradigm, participants, perceived, not perceived; absent and preserved: none; adverse/boundary role: none.
- CONSC-NOREPORT-CRITIQUE-2022 - Duman et al., Frontiers in Human Neuroscience;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9130851/>; class: registered methodological adverse evidence; usable captures: 1; registered features present: report-based paradigms, no-report paradigms, alternative interpretations; absent and preserved: none; adverse/boundary role: Preserves the finding that no-report indicators do not automatically establish phenomenal occurrence..
- CONSC-COGITATE-RESULT-2025 - Ferrante et al., Nature; <https://www.nature.com/articles/s41586-025-08888-1>; class: primary preregistered adversarial theory comparison; usable captures: 1; registered features present: adversarial collaboration, preregistered, participants, functional magnetic resonance imaging, magnetoencephalography; absent and preserved: none; adverse/boundary role: Preserves mixed and theory-adverse results instead of treating any established theory as the SFT law..

Comparison record:

- sft-consc-report-001-external-consequence: predicted access-report-and-presence-nonidentity-preserved; source-derived access-report-and-presence-nonidentity-preserved; exact match True
- evidence disposition: mixed_boundary_constraint; directness: direct_observable_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:f9613e6bb529b356cea5af28f505a9d53bb23a63ad1c13403ba7f15108400c09; derivation seal sha256:a59ab95bf05347d81878a3f583912b60620b3e274ad481a65c9234d0b8f191ce; engine receipt sha256:0c666d6d63b07968ea85db7c27e177e1cfff2596d1e68b3dadb6c2d8c67ab9a at

receipts/engine/model_admitted/SFT-CONSC-REPORT-001-0c666d6d63b07968.json; empirical validation sha256:e2ed851b0080873dedafeeeeb3d5159e05e2676b5d1a235cd3b21970ce737f93; measurement receipt sha256:aac2ad9265f85a4c0f3df7dd5be07db20155eee93057d924851a99d9b46f5636; isolation sha256:36d40bc78857c27cb03692de7f10e69dfd3e6410c6c2132f68a074d8fcd1e71d; custody sha256:999d53c8beba15bb75a00e41238fda1704032b082f0146ae8974235f0d67e722.

10. Phenomenal presence

Claim identity: SFT-CONSC-PHENOMENAL-PRESENCE-001

Question and exact theorem. Phenomenal presence is registered at the interior-observation boundary and remains distinct from access, report, behaviour and neural correlation.

interiorly-realized-content__present/merely-processed__closed-interior-observation__first-person-discrimination__phenomenal__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-MEASUREMENT-CONSCIOUSNESS-BOUNDARY-001
- SFT-CONSC-REPORT-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	interiorly-realized-content	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	present/merely-processed	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	closed-interior-observation	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	first-person-discrimination	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	phenomenal	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `interiorly-realized-content__present/merely-processed__closed-interior-observation__first-person-discrimination__phenomenal__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. The hard-problem category gap is not crossed by renaming a report, behavior or neural signal. Phenomenal presence is assigned to the participating interior closure; public evidence can constrain its bridges without becoming the private carrier.

Access, report and phenomenal presence are three different coordinates. Content can be processed without integrated access; accessible content can lack an expression route; a report is evidence of reporting. The phenomenal coordinate is carried only at the interior-observation boundary and is never manufactured from confidence, fluency or correlation.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 7b607e4a46194d1d7d1aacc9de8b81b5e9443217d368b5ee835da1766f91c2ad.

- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:40b1da1f0b573aab2730262e7401bbac469d741608884b98195f9865081104c4.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:f6f97d70069999fce60ce95e7e61f4cca89e364e0bb5c85909e1439c5e627afc.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:70c727cf24dbab9586e156c8af7e83c3e4f010df1d921c5e8369d5dd8c06c998.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:5562bcc706d534ce2b0c9087b158aaec802b3a1a4a7e8e34803599de083cf1bb; independent certificate sha256:2e30cc3b9cae0fc27a1f02f8a0cbd4316f9e42e087fa69b761be178a4b9bfd38; external-validation identity sha256:6be63979e79d1283b1560aa54209602c0e795da31aa9483577d5f3885bc6677e.

Post-seal empirical-boundary comparison. Directness: participant_bound_empirical_constraint. Disposition: mixed_boundary_constraint. Report/no-report and adversarial experiments test access and report correlates; phenomenal occurrence remains a separately named evidential coordinate.

The external target was opened only after the claim derivation seal. Predicted consequence:

access-report-and-presence-nonidentity-preserved. Source-derived consequence:

access-report-and-presence-nonidentity-preserved. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-NOREPORT-NETWORKS-2022 - Kronemer et al., Nature Communications; <https://pmc.ncbi.nlm.nih.gov/articles/PMC9707162/>; class: primary human report/no-report experiment; usable captures: 1; registered features present: Report + No-Report Paradigm, participants, perceived, not perceived; absent and preserved: none; adverse/boundary role: none.
- CONSC-NOREPORT-CRITIQUE-2022 - Duman et al., Frontiers in Human Neuroscience; <https://pmc.ncbi.nlm.nih.gov/articles/PMC9130851/>; class: registered methodological adverse evidence; usable captures: 1; registered features present: report-based paradigms, no-report paradigms, alternative interpretations; absent and preserved: none; adverse/boundary role: Preserves the finding that no-report indicators do not automatically establish phenomenal occurrence..
- CONSC-COGITATE-RESULT-2025 - Ferrante et al., Nature; <https://www.nature.com/articles/s41586-025-08888-1>; class: primary preregistered adversarial theory comparison; usable captures: 1; registered features present: adversarial collaboration, preregistered, participants, functional magnetic resonance imaging, magnetoencephalography; absent and preserved: none; adverse/boundary role: Preserves mixed and theory-adverse results instead of treating any established theory as the SFT law..

Comparison record:

- sft-consc-phenomenal-presence-001-external-consequence: predicted access-report-and-presence-nonidentity-preserved; source-derived access-report-and-presence-nonidentity-preserved; exact match True
- evidence disposition: mixed_boundary_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:9e00944347927d2513cc216206b25a0bfea8a72c0f12c47842f4934d77c48c90; derivation seal sha256:2a68c5e5f7f666798ffd002b016484cb1e50b4eebf5dc3af9896ae8eb6b1603d; engine receipt sha256:94e5ea251bf89ec65344079aee3a926c015023d3e4be515e85e8f9eac24aabd1 at receipts/engine/model_admitted/SFT-CONSC-PHENOMENAL-PRESENCE-001-94e5ea251bf89ec6.json; empirical validation sha256:bbaa166da1575a14930de4aac34948749cbd3a469613afe0a7d3fe2095b1a5b9; measurement receipt sha256:e714b5f230f7b825496b57fe581fc570dace749c5673d53323e0e1d5c66e4c1e; isolation sha256:aa1ff6253ced4ab5218155ae88c76bf2432a9365d9ac2655e1ab572a18f94dec; custody sha256:2234f3868a3964c2552601073d87f71d045586738571b5a1c5bb5210feeacac3.

11. Reportability

Claim identity: SFT-CONSC-REPORTABILITY-001

Question and exact theorem. Reportability requires both cognitive access and a retained expression route; absence of report alone does not decide presence.

accessible-content-and-output-route__accessible/reportable__semantics-preserving-expression__content-report-link__operational-behavioural__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001

- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-MEASUREMENT-CONSCIOUSNESS-BOUNDARY-001
- SFT-CONSC-PHENOMENAL-PRESENCE-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	accessible-content-and-output-route	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	accessible/reportable	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	semantics-preserving-expression	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	content-report-link	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	operational-behavioural	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: accessible-content-and-output-route__accessible/reportable__semantics-preserving-expression__content-report-link__operational-behavioural__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes reportability requires both cognitive access and a retained expression route; absence of report alone does not decide presence. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Access, report and phenomenal presence are three different coordinates. Content can be processed without integrated access; accessible content can lack an expression route; a report is evidence of reporting. The phenomenal coordinate is carried only at the interior-observation boundary and is never manufactured from confidence, fluency or correlation.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:4033b0d0ab5b1c56638ac79779927227b280295e8d05e1e7a3a39d9b5d515b6e.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:ee74760fb8aa55c0ae7e79558db51da45fd25c337ca67edc98762a3eb8022bfe.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ce612accda3506f072950eaa695aec5f7898490168334f95e77d6fd7a02e7693.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:29879c1fb8e85cedd86319c5f0fa198ab7505537550480affab43ce5a77c9517.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:a36be09023850e94e148bab652bcf3f06adbae96fcab46cfc0c8bb2b385413ef; independent certificate sha256:f2ba95f638ead1f89322bfc1817bc722113af8e6bb23ba6422db56196d140519; external-validation identity sha256:2ad75beac3b4e12b1c535621e45fbc29b63cf6629e8aef0e97c488c584c696a5.

Post-seal empirical-boundary comparison. Directness: `direct_observable_constraint`. Disposition: `mixed_boundary_constraint`. Report/no-report and adversarial experiments test access and report correlates; phenomenal occurrence remains a separately named evidential coordinate.

The external target was opened only after the claim derivation seal. Predicted consequence: `access-report-and-presence-nonidentity-preserved`. Source-derived consequence: `access-report-and-presence-nonidentity-preserved`. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-NOREPORT-NETWORKS-2022 - Kronemer et al., Nature Communications;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9707162/>; class: primary human report/no-report experiment; usable captures: 1; registered features present: Report + No-Report Paradigm, participants, perceived, not perceived; absent and preserved:

none; adverse/boundary role: none.

- CONSC-NOREPORT-CRITIQUE-2022 - Duman et al., Frontiers in Human Neuroscience; <https://pmc.ncbi.nlm.nih.gov/articles/PMC9130851/>; class: registered methodological adverse evidence; usable captures: 1; registered features present: report-based paradigms, no-report paradigms, alternative interpretations; absent and preserved: none; adverse/boundary role: Preserves the finding that no-report indicators do not automatically establish phenomenal occurrence..
- CONSC-COGITATE-RESULT-2025 - Ferrante et al., Nature; <https://www.nature.com/articles/s41586-025-08888-1>; class: primary preregistered adversarial theory comparison; usable captures: 1; registered features present: adversarial collaboration, preregistered, participants, functional magnetic resonance imaging, magnetoencephalography; absent and preserved: none; adverse/boundary role: Preserves mixed and theory-adverse results instead of treating any established theory as the SFT law..

Comparison record:

- sft-consc-reportability-001-external-consequence: predicted access-report-and-presence-nonidentity-preserved; source-derived access-report-and-presence-nonidentity-preserved; exact match True
- evidence disposition: mixed_boundary_constraint; directness: direct_observable_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:f8fbd1cac07c37cd993d87692b4bcf97d1b03c1f7b601d4a206465a956b96cb4; derivation seal sha256:a0bf96e8487cf5ecc239d57e45abc8bb11cfb63409d9a27a4c7d78035d7efb40; engine receipt sha256:674f19e64fdb3a6485b49ec0c15890308f898025b86785056e9bd1b1901034f2 at receipts/engine/model_admitted/SFT-CONSC-REPORTABILITY-001-674f19e64fdb3a64.json; empirical validation sha256:b64006a6371c9a08201cdac8dc05a1994ff3153df706f3d5c2fc686559969a09; measurement receipt

sha256:73c48fa1d67cae5def2aa881555b6ff15f3e1c9effdec24e4e602f83f340c117; isolation
sha256:368f76d4ab66f40520c729740487a0e209a396d20ad273bcb077937cb2176e31; custody
sha256:bf9a46a8fb681b90beeda3f5e0a030576589083b437245ae94b2c92f686e17e3.

12. Active unavailable processing

Claim identity: SFT-CONSC-UNCONSCIOUS-PROCESS-001

Question and exact theorem. A process may remain active while unavailable to the integrated access relation, so activity is not equated with consciousness.

active-recurrent-process__active/available__continuation-without-access-closure__p
rocess-and-access-records__behavioural-and-biological-correlate__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-MEASUREMENT-CONSCIOUSNESS-BOUNDARY-001
- SFT-CONSC-REPORTABILITY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	active-recurrent-process	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	active/available	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	continuation-without-access-closure	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	process-and-access-records	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	behavioural-and-biological-correlate	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `active-recurrent-process__active/available__continuation-without-access-closure__process-and-access-records__behavioural-and-biological-correlate__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes a process may remain active while unavailable to the integrated access relation, so activity is not equated with consciousness. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Access, report and phenomenal presence are three different coordinates. Content can be processed without integrated access; accessible content can lack an expression route; a report is evidence of reporting. The phenomenal coordinate is carried only at the interior-observation boundary and is never manufactured from confidence, fluency or correlation.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:c6d8b6bc79fc40ba2fa347637d1ebd701db743144b9e0cf469b61acfcde6954c`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt

sha256:461905d0a7e49f319ca3cb42d4d9b18efa8b8fbdf3e96d6f8238b627277a0d4.

- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:50b340a703a2eb3bd99c9b585cb4769d243c13a2fee1c66e6e819b17d999475c.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:ae47d097ab43b875e71fb0a74ad32c3e721b94e07fa6f96de44e7fd45ee9f9d0.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:df31dbff65f32e250574256f560bfac57ec9618c1156f3d335d184234e064858; independent certificate sha256:6c212abbf0fc7ae02ba59112232500c7474b7d3e6a45a590ba3b48968110e4c0; external-validation identity
sha256:fb73899e1d1e75586244d91cc44d721556296d368751ae3ee623cee10faba323.

Post-seal empirical-boundary comparison. Directness: direct_observable_constraint. Disposition: mixed_boundary_constraint. Report/no-report and adversarial experiments test access and report correlates; phenomenal occurrence remains a separately named evidential coordinate.

The external target was opened only after the claim derivation seal. Predicted consequence:

access-report-and-presence-nonidentity-preserved. Source-derived consequence:

access-report-and-presence-nonidentity-preserved. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-NOREPORT-NETWORKS-2022 - Kronemer et al., Nature Communications;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9707162/>; class: primary human report/no-report experiment; usable captures: 1; registered features present: Report + No-Report Paradigm, participants, perceived, not perceived; absent and preserved: none; adverse/boundary role: none.
- CONSC-NOREPORT-CRITIQUE-2022 - Duman et al., Frontiers in Human Neuroscience;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9130851/>; class: registered methodological adverse evidence; usable captures: 1; registered features present: report-based paradigms, no-report paradigms, alternative interpretations; absent and preserved: none; adverse/boundary role: Preserves the finding that no-report indicators do not automatically establish phenomenal occurrence..
- CONSC-COGITATE-RESULT-2025 - Ferrante et al., Nature; <https://www.nature.com/articles/s41586-025-08888-1>; class: primary preregistered adversarial theory comparison; usable captures: 1; registered features present: adversarial collaboration, preregistered, participants, functional magnetic resonance imaging, magnetoencephalography; absent and preserved: none; adverse/boundary role: Preserves mixed and theory-adverse results instead of treating any established theory as the SFT law..

Comparison record:

- sft-consc-unconscious-process-001-external-consequence: predicted access-report-and-presence-nonidentity-preserved; source-derived access-report-and-presence-nonidentity-preserved; exact match True
- evidence disposition: mixed_boundary_constraint; directness: direct_observable_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero

- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:94f20d64eb351acee5dd83ca91e8640f1d4ef7b27ad9f06c1a577f1079810ddd; derivation seal sha256:b911d01194531216d9e94f01870d3d042a5c2794256ff08da7260028d3a975ec; engine receipt sha256:aa334010554c126808560e78e8af4788981244496b9ff10a670b2f06cdf08be0 at receipts/engine/model_admitted/SFT-CONSC-UNCONSCIOUS-PROCESS-001-aa334010554c1268.json; empirical validation sha256:3d30d12993ce42932954f2ae4e16d2e79810e670caadc0d99f6e3805b4e2f553; measurement receipt sha256:11f41e92a81b6e004a0355d02e28884586f85917d3083153df99762bce1550c5; isolation sha256:95c11b60c137b075d13529feed4009fa2777ffb0af99ad7de05beaabb33a3c4a; custody sha256:77814d4f3c96088d8067fa5719384546b7ff38c5830084c4fa3966a6b98d6d18.

13. Altered-state evidence boundary

Claim identity: SFT-CONSC-ALTERED-STATE-REPORT-BOUNDARY-001

Question and exact theorem. Reports of altered experience are preserved as first-person observations while their ontological interpretation requires a separately derived and tested bridge.

state-report-observation-word__report/ontology__source-bound-comparison__state-induction-report-controls__first-person-report-with-third-person-controls__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-MEASUREMENT-CONSCIOUSNESS-BOUNDARY-001
- SFT-CONSC-UNCONSCIOUS-PROCESS-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	state-report-observation-word	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	report/ontology	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	source-bound-comparison	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	state-induction-report-controls	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	first-person-report-with-third-person-controls	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: state-report-observation-word__report/ontology__source-bound-comparison__state-induction-report-controls__first-person-report-with-third-person-controls__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes reports of altered experience are preserved as first-person observations while their ontological interpretation requires a separately derived and tested bridge. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Access, report and phenomenal presence are three different coordinates. Content can be processed without integrated access; accessible content can lack an expression route; a report is evidence of reporting. The phenomenal coordinate is carried only at the interior-observation boundary and is never manufactured from confidence, fluency or correlation.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:3f2c1207edb232623e68ebfb75ec59ef846ce0e065d29ac5e77395f31a7f5e3c.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:38ee4306e9f1ae4d7142eba4da91de7fbdc22d18228179d413c4747e90b217b.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:062a976b92639df2de9064275a08cb96ecb2464c0751ae04ae6a1851a4cc1355.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:1e7aa9ed6c70bdbfc5f1c36e48db533116fe3e8e294c6548287f94a8ac526b29.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:71f300f44c1fdcb3ba665104d4b8aa7d9faef42b4c95e027f3c0de54c4d44845; independent certificate sha256:247b1534e91d5cd2a80d86b0b9e3f923b815202256a06b2a1726b5e8094d4f84; external-validation identity sha256:20ebd02cfef5f8b350446b594e3f9795f29fdec1465f0f0a3894f6fbce616185.

Post-seal empirical-boundary comparison. Directness: `participant_bound_empirical_constraint`. Disposition: `mixed_boundary_constraint`. Report/no-report and adversarial experiments test access and report correlates; phenomenal occurrence remains a separately named evidential coordinate.

The external target was opened only after the claim derivation seal. Predicted consequence: `access-report-and-presence-nonidentity-preserved`. Source-derived consequence: `access-report-and-presence-nonidentity-preserved`. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-NOREPORT-NETWORKS-2022 - Kronemer et al., Nature Communications; <https://pmc.ncbi.nlm.nih.gov/articles/PMC9707162/>; class: primary human report/no-report experiment; usable captures: 1; registered features present: Report + No-Report Paradigm, participants, perceived, not perceived; absent and preserved: none; adverse/boundary role: none.

- CONSC-NOREPORT-CRITIQUE-2022 - Duman et al., Frontiers in Human Neuroscience; <https://pmc.ncbi.nlm.nih.gov/articles/PMC9130851/>; class: registered methodological adverse evidence; usable captures: 1; registered features present: report-based paradigms, no-report paradigms, alternative interpretations; absent and preserved: none; adverse/boundary role: Preserves the finding that no-report indicators do not automatically establish phenomenal occurrence..
- CONSC-COGITATE-RESULT-2025 - Ferrante et al., Nature; <https://www.nature.com/articles/s41586-025-08888-1>; class: primary preregistered adversarial theory comparison; usable captures: 1; registered features present: adversarial collaboration, preregistered, participants, functional magnetic resonance imaging, magnetoencephalography; absent and preserved: none; adverse/boundary role: Preserves mixed and theory-adverse results instead of treating any established theory as the SFT law..

Comparison record:

- sft-consc-altered-state-report-boundary-001-external-consequence: predicted access-report-and-presence-nonidentity-preserved; source-derived access-report-and-presence-nonidentity-preserved; exact match True
- evidence disposition: mixed_boundary_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256: 44309232df5bb840158e62b0f2e50a7c128a2d634bd030d385b84c9991eb9842; derivation seal sha256: fc8580d1b71541de3695d5b64c9f8531e87b78276f4397ca54c4b9e64cd10b70; engine receipt sha256: bd8431fda7b9c23f2179b1f44fc3d624de0af2def23ea6443c70a2edcd9090de at receipts/engine/model_admitted/SFT-CONSC-ALTERED-STATE-REPORT-BOUNDARY-001-bd8431fda7b9c23f.json; empirical validation sha256: 019fa6b07171f3e6d45526e12135edc672ec4354291edc73a5dfb9a81b38ff18; measurement receipt

```
sha256:45c1d7612fc8dc1420c59f09bd796da55a876983a07095ec9ff9ef6e5f5a3eb1; isolation
sha256:2f5cd927b11a0dd14e91ff02437f2403d9f93f311021fa311930d581648e387b; custody
sha256:b11653cb1e8e276405faa4736d60e629e98cc45305f93fa2568fab33d1d07d9d.
```

17. Binding and Unity

Binding retains difference while forming a joint successor. Unity requires complete mutual participation in one closed process; synchrony and one-way influence are insufficient. Fragmentation is therefore an exact loss of joint closure, not a vague reduction in a scalar integration score.

14. Plural conscious contents

Claim identity: SFT-CONSC-CONTENT-PLURALITY-001

Question and exact theorem. Binding begins with multiple retained content identities; erasing their differences is not unity.

```
distinct-content-labels__plural/merged__held-joint-composition__content-identities
__phenomenal-discrimination__root-bound-forward-forcing__positive-finite-successor
-with-all-alternatives__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-ALTERED-STATE-REPORT-BOUNDARY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	distinct-content-labels	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	plural/merged	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	held-joint-composition	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	content-identities	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	phenomenal-discrimination	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `distinct-content-labels__plural/merged__held-joint-composition__content-identities__phenomenal-discrimination__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes binding begins with multiple retained content identities; erasing their differences is not unity. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Binding retains difference while forming a joint successor. Unity requires complete mutual participation in one closed process; synchrony and one-way influence are insufficient. Fragmentation is therefore an exact loss of joint closure, not a vague reduction in a scalar integration score.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:aa1627729a1ac81488952894a848fc89e9a24f93f76a1c99532e02b98bdcc4aa.

- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:07c70d9f37da38805379a81c31db66b90cdde3c377f4666174495adaf2d90957.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:5288384ee930c52e406fa8e68030274b68d23225fc73972365d4ec5db4da8cf5.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:26199870d93e2fa583a250343171ba5ddc10cd2222b31c8ca90525503a58c14.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:6bf4a074a6363c9d3b3195337c1c9f31e89151cda233dea1cc4648e4465989a9; independent certificate sha256:8c8b57b4fc9a0682658d08afd05dc071bd34b1a68f23e4ea655d15703a2f3d38; external-validation identity sha256:44d753215336a914a10d01b40a3d1c5fc90b6007b6702b3f2d0ca763add29f09.

Post-seal empirical-boundary comparison. Directness: participant_bound_empirical_constraint. Disposition: mixed_and_adverse_constraint. Split-brain and adversarial records test integration while preserving disagreement about whether a single phenomenal whole has been established.

The external target was opened only after the claim derivation seal. Predicted consequence:

integration-and-unity-evidence-boundary-preserved. Source-derived consequence:

integration-and-unity-evidence-boundary-preserved. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-SPLIT-INTEGRATION-2023 - de Haan et al., Brain Sciences; <https://pmc.ncbi.nlm.nih.gov/articles/PMC10667445/>; class: primary split-brain integration experiment; usable captures: 1; registered features present: split-brain patient, two experiments, automatic processing, deliberate; absent and preserved: none; adverse/boundary role: none.
- CONSC-SPLIT-BOUNDARY-2020 - de Haan et al., Neuropsychologia review; <https://pmc.ncbi.nlm.nih.gov/articles/PMC7305066/>; class: registered unity-evidence boundary; usable captures: 1; registered features present: whether dividing the brain divides consciousness, intermediate results, measures; absent and preserved: none; adverse/boundary role: Preserves disagreement and measurement ambiguity rather than selecting a unity theory..
- CONSC-COGITATE-RESULT-2025 - Ferrante et al., Nature; <https://www.nature.com/articles/s41586-025-08888-1>; class: primary preregistered adversarial theory comparison; usable captures: 1; registered features present: adversarial collaboration, preregistered, participants, functional magnetic resonance imaging, magnetoencephalography; absent and preserved: none; adverse/boundary role: Preserves mixed and theory-adverse results instead of treating any established theory as the SFT law..

Comparison record:

- sft-consc-content-plurality-001-external-consequence: predicted integration-and-unity-evidence-boundary-preserved; source-derived integration-and-unity-evidence-boundary-preserved; exact match True
- evidence disposition: mixed_and_adverse_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:ddc505d922b826f6728cc9d25ff93a3b0eb5777d7fcf08e26f234f459e32966f; derivation seal sha256:4f368f52ba0deb7d5483cfee4cf416774683ca9f1b3ce6122563107552546ab0; engine receipt sha256:e18cca317d9758e664c61de2384f6824e04ba53cd258f652d34219c7176ed90d at receipts/engine/model_admitted/SFT-CONSC-CONTENT-PLURALITY-001-e18cca317d9758e6.json; empirical validation sha256:300b670617962386f93afb5d5990bedf0fcfc95fe87eb235fbd796c1e4d96286; measurement receipt sha256:ed80a0e90b7c3c72d02abb251e19515e5c66d33c7d60edc7c0bc5da452ec8e29; isolation sha256:830ad8ee239c7a08074c741ae2dd1de128e061277b7000330da75f899770ec10; custody sha256:83864b9c93abbc717c7338ba29cfb73b0d85e7816ad88dee9d8615f6e57ba2ef.

15. Binding relation

Claim identity: SFT-CONSC-BINDING-001

Question and exact theorem. Binding is a generated joint relation in which distinct contents participate in one successor while remaining reconstructible from held labels.

joint-content-word__separate/joint__shared-successor-composition__component-and-joint-trace__phenomenal-and-operational__root-bound-forward-forcing__positive-finite__successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001

- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-ALTERED-STATE-REPORT-BOUNDARY-001
- SFT-CONSC-CONTENT-PLURALITY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	joint-content-word	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	separate/joint	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	shared-successor-composition	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	component-and-joint-trace	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	phenomenal-and-operational	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: joint-content-word__separate/joint__shared-successor-composition__component-and-joint-trace__phenomenal-and-operational__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. Binding is joint recurrence with reconstructible components. It creates a shared successor without erasing either contributing content identity.

Binding retains difference while forming a joint successor. Unity requires complete mutual participation in one closed process; synchrony and one-way influence are insufficient. Fragmentation is therefore an exact loss of joint closure, not a vague reduction in a scalar integration score.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:b80851dd39733b06d950827d88cba3cefcddad1176f520158255e9844b6e6519c.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:d416d3a2feb026e2fdb5f3893b7f6adee0d4923f48a06ce8c79eb0988a60f4da.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:a950782f9cf28de96334815fbe7ce7d6d744418545dca81309a7cb495c81a0f6.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:8f4ae17d08ad0ddf4a769b70728dc1352e88346c0dd5d291e22712b638a3ee7b.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:7628cbfea360a8c7f8e498ab87dd9bcc21b3deb4aff8f4497cd489d1a9b5be2; independent certificate sha256:621ae76c5469ed06e77a2d0be24791f9d33ee0b49ffdbf24d0876ab034948e34; external-validation identity
sha256:d616408d74324dfc36b41c47b9fa765a026558af432138e72f2ce420c9d9d4c8.

Post-seal empirical-boundary comparison. Directness: participant_bound_empirical_constraint. Disposition: mixed_and_adverse_constraint. Split-brain and adversarial records test integration while preserving disagreement about whether a single phenomenal whole has been established.

The external target was opened only after the claim derivation seal. Predicted consequence:

integration-and-unity-evidence-boundary-preserved. Source-derived consequence:

integration-and-unity-evidence-boundary-preserved. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-SPLIT-INTEGRATION-2023 - de Haan et al., Brain Sciences; <https://pmc.ncbi.nlm.nih.gov/articles/PMC10667445/>; class: primary split-brain integration experiment; usable captures: 1; registered features present: split-brain patient, two experiments, automatic processing, deliberate; absent and preserved: none; adverse/boundary role: none.
- CONSC-SPLIT-BOUNDARY-2020 - de Haan et al., Neuropsychologia review; <https://pmc.ncbi.nlm.nih.gov/articles/PMC7305066/>; class: registered unity-evidence boundary; usable captures: 1; registered features present: whether dividing the brain divides consciousness, intermediate results, measures; absent and preserved: none; adverse/boundary role: Preserves disagreement and measurement ambiguity rather than selecting a unity theory..
- CONSC-COGITATE-RESULT-2025 - Ferrante et al., Nature; <https://www.nature.com/articles/s41586-025-08888-1>; class: primary preregistered adversarial theory comparison; usable captures: 1; registered features present: adversarial collaboration, preregistered, participants, functional magnetic resonance imaging, magnetoencephalography; absent and preserved: none; adverse/boundary role: Preserves mixed and theory-adverse results instead of treating any established theory as the SFT law..

Comparison record:

- sft-consc-binding-001-external-consequence: predicted integration-and-unity-evidence-boundary-preserved; source-derived integration-and-unity-evidence-boundary-preserved; exact match True
- evidence disposition: mixed_and_adverse_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:14f4ae0e2fd7fd2756fde73aa0ca60c03891b4d9b0d017a37a07ef0ea0bb3d66; derivation seal sha256:01ced9f81d51c7ae3603f4e43d501d55ed7759d48f9e01502a3eb4949c9c2da5; engine receipt sha256:ffc2d41dd6abb0cc20efa97823488619db0adba759a3118ed0f6713c3d883a16 at

receipts/engine/model_admitted/SFT-CONSC-BINDING-001-ffc2d41dd6abb0cc.json; empirical validation sha256:f648d5d20b7aafb055e917d4c6659db0132affdd4001cfa8c60cd2d867a456; measurement receipt sha256:3be570c6dd027afebd75a9943de8514e9ae0ff6173b150e1f709e86cca62ce4a; isolation sha256:22f334e4f9267eb6c0694167c65889cd6e6aa7d4e4bc6421045b4282c4b383b1; custody sha256:574240775621e489b1f3ca892600b000acba75e0048e7264340cb078b7c1691f.

16. Unity of experience

Claim identity: SFT-CONSC-UNITY-001

Question and exact theorem. Unity is one closed interior process over complete participating content, not a heap, synchrony measure or unlabelled aggregate.

complete-bound-content__one-process/many-processes__joint-closure-to-one-carrier__ participation-and-boundary__first-person-unity__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENC-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-ALTERED-STATE-REPORT-BOUNDARY-001
- SFT-CONSC-BINDING-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	complete-bound-content	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	one-process/many-processes	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	joint-closure-to-one-carrier	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	participation-and-boundary	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	first-person-unity	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: complete-bound-content__one-process/many-processes__joint-closure-to-one-carrier__participation-and-boundary__first-person-unity__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. Unity is one closed interior process over all participating contents. A heap, a synchrony observation or a scalar correlation lacks the required mutual participation and fails.

Binding retains difference while forming a joint successor. Unity requires complete mutual participation in one closed process; synchrony and one-way influence are insufficient. Fragmentation is therefore an exact loss of joint closure, not a vague reduction in a scalar integration score.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:267e5f4b91b41271a5054cd31a66bdd1d2f0ee90f560d6634cd818eab6cbf694.

- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:2d1378cf06544b73858f998df06bb23591cdc8c665eaf7458ab82d982ab09b03.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:d4ed2374536edc5e7dee85ba2d311fb9781b60875a253c006b10fb15bdd60b85.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:38e066ac03b7344e5f71b87a24c9de9f546777976094e9a31bfa37ca346053d6.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:4df9b0a35ab7dc354c8fd6a9637b3c0e669ab986708cac8b2c3785fa329d6a12; independent certificate sha256:b53975bb80fa30028dee2cfaf9f0f2b0e9ddcb2224c1616151a57b7de43ddef1; external-validation identity sha256:bf5553d021098723321f1de67b6e78ea569b084cc2e5e4e6dc77711186818933.

Post-seal empirical-boundary comparison. Directness: participant_bound_empirical_constraint. Disposition: mixed_and_adverse_constraint. Split-brain and adversarial records test integration while preserving disagreement about whether a single phenomenal whole has been established.

The external target was opened only after the claim derivation seal. Predicted consequence:

integration-and-unity-evidence-boundary-preserved. Source-derived consequence:

integration-and-unity-evidence-boundary-preserved. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-SPLIT-INTEGRATION-2023 - de Haan et al., Brain Sciences; <https://pmc.ncbi.nlm.nih.gov/articles/PMC10667445/>; class: primary split-brain integration experiment; usable captures: 1; registered features present: split-brain patient, two experiments, automatic processing, deliberate; absent and preserved: none; adverse/boundary role: none.
- CONSC-SPLIT-BOUNDARY-2020 - de Haan et al., Neuropsychologia review; <https://pmc.ncbi.nlm.nih.gov/articles/PMC7305066/>; class: registered unity-evidence boundary; usable captures: 1; registered features present: whether dividing the brain divides consciousness, intermediate results, measures; absent and preserved: none; adverse/boundary role: Preserves disagreement and measurement ambiguity rather than selecting a unity theory..
- CONSC-COGITATE-RESULT-2025 - Ferrante et al., Nature; <https://www.nature.com/articles/s41586-025-08888-1>; class: primary preregistered adversarial theory comparison; usable captures: 1; registered features present: adversarial collaboration, preregistered, participants, functional magnetic resonance imaging, magnetoencephalography; absent and preserved: none; adverse/boundary role: Preserves mixed and theory-adverse results instead of treating any established theory as the SFT law..

Comparison record:

- sft-consc-unity-001-external-consequence: predicted integration-and-unity-evidence-boundary-preserved; source-derived integration-and-unity-evidence-boundary-preserved; exact match True
- evidence disposition: mixed_and_adverse_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:10f3be92852b46812f71b9d3e29d54f622d359eccc602fe025ed1b58b02e3dcb; derivation seal sha256:45b4fb0adcd903f0cc94a66ab809bfd8c2984543d55a64234ce3967253614c34; engine receipt sha256:389adaf57aaa2bbe491ab44a470c6ab63913c9ff1529b530ebc7207e4791826c at receipts/engine/model_admitted/SFT-CONSC-UNITY-001-389adaf57aaa2bbe.json; empirical validation sha256:1974949cbfee47c3a88b42b7236b4c6fe43bcbfc1129b42bb8894e7f731bf766; measurement receipt sha256:dd696d2c4d1c10863667ba7c57d3532ab76f7496ab934657e435d2e8af655dcf; isolation sha256:ca2b28ff89be21f5dcbf0a6c0185a43287da4d5da5f4047e7dbe1e1e55fc912f; custody sha256:9b39cefd6df565875bf1ca6638bd46881a9462d20354e3ddb43b2ecf94dc54d.

17. Integration boundary

Claim identity: SFT-CONSC-INTEGRATION-BOUNDARY-001

Question and exact theorem. Integration requires each participating process to enter the joint closure; a one-way influence does not establish one interior whole.

coupled-process-support__separate/integrated__complete-mutual-participation__coupling-and-transition-trace__perturbational-and-phenomenal__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001

- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-ALTERED-STATE-REPORT-BOUNDARY-001
- SFT-CONSC-UNITY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	coupled-process-support	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	separate/integrated	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	complete-mutual-participation	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	coupling-and-transition-trace	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	perturbational-and-phenomenal	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: coupled-process-support__separate/integrated__complete-mutual-participation__coupling-and-transition-trace__perturbational-and-phe

nomenal__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. This exact relation closes integration requires each participating process to enter the joint closure; a one-way influence does not establish one interior whole. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Binding retains difference while forming a joint successor. Unity requires complete mutual participation in one closed process; synchrony and one-way influence are insufficient. Fragmentation is therefore an exact loss of joint closure, not a vague reduction in a scalar integration score.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:77b12f6b7231f954eefe3a69a3033c0b5546cc34c5127d9996117272c4f499a6.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:304c79a083d4c2a05a35f578ac9552e99919bad87ab0228226448802d8827f46.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:0a8cdde24e2b1aaa6201f07aa663e8d9a0c9c5875466e4b58cc04e83a738504c.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:ce2905d8243a0f2a4be8e62510d8e6bfb938846c212bb947ab0a7f3b4b27b8c.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:c94295389fa638fe9f9633f5fbb0b2fa91f28917976b840574a8a5f45de55b95; independent certificate sha256:55b104149ce8855bfaa177d53dfb021ef32d9d2f9f9f516250a5746c25088992; external-validation identity sha256:efe0977cb49bdd27d773476ebcbcb64434784c7ced05a8fcf0fc5911f17763d.

Post-seal empirical-boundary comparison. Directness: participant_bound_empirical_constraint. Disposition: mixed_and_adverse_constraint. Split-brain and adversarial records test integration while preserving disagreement about whether a single phenomenal whole has been established.

The external target was opened only after the claim derivation seal. Predicted consequence:

integration-and-unity-evidence-boundary-preserved. Source-derived consequence:

integration-and-unity-evidence-boundary-preserved. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-SPLIT-INTEGRATION-2023 - de Haan et al., Brain Sciences; <https://pmc.ncbi.nlm.nih.gov/articles/PMC10667445/>; class: primary split-brain integration experiment; usable captures: 1; registered features present: split-brain patient, two experiments, automatic processing, deliberate; absent and preserved: none; adverse/boundary role: none.
- CONSC-SPLIT-BOUNDARY-2020 - de Haan et al., Neuropsychologia review; <https://pmc.ncbi.nlm.nih.gov/articles/PMC7305066/>; class: registered unity-evidence boundary; usable captures: 1; registered features present: whether dividing the brain divides consciousness, intermediate results, measures; absent and preserved: none; adverse/boundary role: Preserves disagreement and measurement ambiguity rather than selecting a unity theory..
- CONSC-COGITATE-RESULT-2025 - Ferrante et al., Nature; <https://www.nature.com/articles/s41586-025-08888-1>; class: primary preregistered adversarial theory comparison; usable captures: 1; registered features present: adversarial collaboration, preregistered, participants, functional magnetic resonance imaging, magnetoencephalography; absent and preserved: none; adverse/boundary role: Preserves mixed and theory-adverse results instead of treating any established theory as the SFT law..

Comparison record:

- sft-consc-integration-boundary-001-external-consequence: predicted integration-and-unity-evidence-boundary-preserved; source-derived integration-and-unity-evidence-boundary-preserved; exact match True
- evidence disposition: mixed_and_adverse_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:512595669f455d8c126a4a15d33bf55b35b30417d625b78ec7daed668b839cdf; derivation seal sha256:8108e4847702fb6ce81da1bdfed3021baa95495cf45517c2b60d1e02c7572ccd; engine receipt sha256:45a4d53110a1ab574c74fcd186be33dd10becf5e85807d819ec37802cccd5a98 at receipts/eng

ine/model_admitted/SFT-CONSC-INTEGRATION-BOUNDARY-001-45a4d53110a1ab57.json; empirical validation sha256:80e33219a5b016922279625b78403125a64ac816965bdac1148c4fc3283afebf; measurement receipt sha256:c4b0dalebca4f2c977f1fa425dff38c3afe2f620282ac67d157d5fc4c7f5d0a4; isolation sha256:d194e78ee260c9ea70cf2b1d3bd0e87ff498a05ff0f933b5267c1b8cecbcdac6; custody sha256:b0e3876aaf329a25a91eb871b2586b7dbeb407fa8c9f2145fc5d88cbbf040d4f.

18. Synchrony and binding distinction

Claim identity: SFT-CONSC-SYNCHRONY-BOUNDARY-001

Question and exact theorem. Synchrony may test a timing relation but does not alone prove binding unless joint compositional closure is independently shown.

timed-process-traces__same-time/same-whole__recurrence-alignment-test__phase-and-composition-record__biological-correlate-boundary__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENC-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-ALTERED-STATE-REPORT-BOUNDARY-001
- SFT-CONSC-INTEGRATION-BOUNDARY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	timed-process-traces	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	same-time/same-whole	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	recurrence-alignment-test	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	phase-and-composition-record	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	biological-correlate-boundary	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `timed-process-traces__same-time/same-whole__recurrence-alignment-test__phase-and-composition-record__biological-correlate-boundary__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes synchrony may test a timing relation but does not alone prove binding unless joint compositional closure is independently shown. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Binding retains difference while forming a joint successor. Unity requires complete mutual participation in one closed process; synchrony and one-way influence are insufficient. Fragmentation is therefore an exact loss of joint closure, not a vague reduction in a scalar integration score.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:45f91f709a7502c0b5764420568835f64aae9997a35fe216ed862003282d04d1.

- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:552fda98478e9b54a9570256c2740b6e96cef1e8e4e5fa7ee352f95fa4084493.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:fb3ac6deb373b806d88a54ed2299ee404060fbd1ad43817b925522568d7f180e.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:8b1be9cff7f86bd07f929afecfb055e672cad61b2edded5c4b72ee5970662cf1.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:6cf68b152d3fceed0cb4367ad27310d9003535ece8b4d9a47220868ccf54966e; independent certificate sha256:cfe0dd97be4c138a3451366533b5da71244b9053eeb5cb089ae308cf1050af9; external-validation identity sha256:3d9405d834e087d61c633935b2b9f2c955d643f13a5a173f0a10e98f18862874.

Post-seal empirical-boundary comparison. Directness: operational_or_record_constraint. Disposition: mixed_and_adverse_constraint. Split-brain and adversarial records test integration while preserving disagreement about whether a single phenomenal whole has been established.

The external target was opened only after the claim derivation seal. Predicted consequence:

integration-and-unity-evidence-boundary-preserved. Source-derived consequence:

integration-and-unity-evidence-boundary-preserved. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-SPLIT-INTEGRATION-2023 - de Haan et al., Brain Sciences; <https://pmc.ncbi.nlm.nih.gov/articles/PMC10667445/>; class: primary split-brain integration experiment; usable captures: 1; registered features present: split-brain patient, two experiments, automatic processing, deliberate; absent and preserved: none; adverse/boundary role: none.
- CONSC-SPLIT-BOUNDARY-2020 - de Haan et al., Neuropsychologia review; <https://pmc.ncbi.nlm.nih.gov/articles/PMC7305066/>; class: registered unity-evidence boundary; usable captures: 1; registered features present: whether dividing the brain divides consciousness, intermediate results, measures; absent and preserved: none; adverse/boundary role: Preserves disagreement and measurement ambiguity rather than selecting a unity theory..
- CONSC-COGITATE-RESULT-2025 - Ferrante et al., Nature; <https://www.nature.com/articles/s41586-025-08888-1>; class: primary preregistered adversarial theory comparison; usable captures: 1; registered features present: adversarial collaboration, preregistered, participants, functional magnetic resonance imaging, magnetoencephalography; absent and preserved: none; adverse/boundary role: Preserves mixed and theory-adverse results instead of treating any established theory as the SFT law..

Comparison record:

- sft-consc-synchrony-boundary-001-external-consequence: predicted integration-and-unity-evidence-boundary-preserved; source-derived integration-and-unity-evidence-boundary-preserved; exact match True
- evidence disposition: mixed_and_adverse_constraint; directness: operational_or_record_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:95fed58dbf0306dc36a670874d486cd8d5a3d5e67bee6378fa9f3f5ae415eff8; derivation seal sha256:a891c6ed73769906833c09cfc2d5aaee2d7147edccbacc0ad6af72feableec60; engine receipt sha256:ce19b1ec90c8508be91246272716bc450af0868ad37b983c3383ea3e961a988a at receipts/engine/model_admitted/SFT-CONSC-SYNCHRONY-BOUNDARY-001-ce19b1ec90c8508b.json; empirical validation sha256:c3cb1b84811b34585304822eb4ca8de17f85540ef0839fb63010d1423b397fab; measurement receipt sha256:993651cbc7c58e6ecdcaf05e70b2e609fbb2f99efd08f00be220fc1a5058565c; isolation sha256:3102bc6bf78346c6a75aca93ab3c488cb5f18382f938d9bafd545681221c7832; custody sha256:1079a9d3f7c19adbbdd5928a908cb95e3fee0fdab744fe2edaa93bd83d840c8a.

19. Fragmentation and divided access

Claim identity: SFT-CONSC-FRAGMENTATION-001

Question and exact theorem. Fragmentation is the loss of joint closure while component processes and their distinct access records remain represented.

formerly-joint-process__joint/separated__loss-of-joint-closure__component-continuation-traces__phenomenal-clinical-handoff__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001

- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-ALTERED-STATE-REPORT-BOUNDARY-001
- SFT-CONSC-SYNCHRONY-BOUNDARY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	formerly-joint-process	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	joint/separated	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	loss-of-joint-closure	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	component-continuation-traces	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	phenomenal-clinical-handoff	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: formerly-joint-process__joint/separated__loss-of-joint-closure__component-continuation-traces__phenomenal-clinical-handoff__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes fragmentation is the loss of joint closure while component processes and their distinct access records remain represented. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Binding retains difference while forming a joint successor. Unity requires complete mutual participation in one closed process; synchrony and one-way influence are insufficient. Fragmentation is therefore an exact loss of joint closure, not a vague reduction in a scalar integration score.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:cc2d3cba52cb63c20752a1027618af31fc1d5841ef425ca21c45ebf5e17cd302.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:6a0478bb6c5ddc81a0e84bf1d4ec20164797b120bca7c8ae5867e7e92dceba7a.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:7aea991a69cb090aab35a495906d11f2698e4bc85dbc7d899b71f55e6b2bcd9f.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:b530424a1798742343a939830ea649b924a8984f66fd62749f3b22ac00f2f653.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:3934f121f40af13c0bceb5392c9ee288a5e770f2fc6fcda1df8be2ea0d29cb8; independent certificate sha256:74bbf43340bb5fdde4a7e0d7c847aad4ebba6f25e8769623ec281ddaa9187cee; external-validation identity sha256:b4e813dc404da5f3075be0cdebc4d7228da888f5aac798e100d8205b75897aea.

Post-seal empirical-boundary comparison. Directness: `participant_bound_empirical_constraint`. Disposition: `mixed_and_adverse_constraint`. Split-brain and adversarial records test integration while preserving disagreement about whether a single phenomenal whole has been established.

The external target was opened only after the claim derivation seal. Predicted consequence: `integration-and-unity-evidence-boundary-preserved`. Source-derived consequence: `integration-and-unity-evidence-boundary-preserved`. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-SPLIT-INTEGRATION-2023 - de Haan et al., Brain Sciences;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10667445/>; class: primary split-brain integration experiment; usable captures: 1;

registered features present: split-brain patient, two experiments, automatic processing, deliberate; absent and preserved: none; adverse/boundary role: none.

- CONSC-SPLIT-BOUNDARY-2020 - de Haan et al., Neuropsychologia review; <https://pmc.ncbi.nlm.nih.gov/articles/PMC7305066/>; class: registered unity-evidence boundary; usable captures: 1; registered features present: whether dividing the brain divides consciousness, intermediate results, measures; absent and preserved: none; adverse/boundary role: Preserves disagreement and measurement ambiguity rather than selecting a unity theory..
- CONSC-COGITATE-RESULT-2025 - Ferrante et al., Nature; <https://www.nature.com/articles/s41586-025-08888-1>; class: primary preregistered adversarial theory comparison; usable captures: 1; registered features present: adversarial collaboration, preregistered, participants, functional magnetic resonance imaging, magnetoencephalography; absent and preserved: none; adverse/boundary role: Preserves mixed and theory-adverse results instead of treating any established theory as the SFT law..

Comparison record:

- sft-consc-fragmentation-001-external-consequence: predicted integration-and-unity-evidence-boundary-preserved; source-derived integration-and-unity-evidence-boundary-preserved; exact match True
- evidence disposition: mixed_and_adverse_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256: 9989a87a8984564f733030c56ed64b969fd709643f5f85b34ef2ce6d6c04b498; derivation seal sha256: d94965f4ccbf314957947594c03f56ea1082a317109930aa1bdbae04267ab12d; engine receipt sha256: 3a6d372a2c715e2d34799dcdf73304b503a18216935fd424efcb1112dcc75637 at receipts/engine/model_admitted/SFT-CONSC-FRAGMENTATION-001-3a6d372a2c715e2d.json; empirical validation sha256: ecbd05fa7689ef93838ba0483b0f82a57f285037772ce08853d2a57274010568;

measurement receipt

sha256:9bf74bc022d2fcb02359baa0e17f3460cdd242b355e813d6e0783f676c955e2c; isolation
 sha256:eb97efc17987462e60317bb20f6d34d1f66fffa46893aa613127126186055a9b; custody
 sha256:00013ff07ec0b09beca20d72e0ff2d6ac8495255ed5414f1a09aa97f95c437e5.

18. Subject, Perspective and Interiority

A perspective is a source-located access path. The subject is the continuing integrated relation holding that path while content changes. Interiority is the structural asymmetry between participating in self-observation and possessing its external image; phenomenal privacy follows when the image closes a predecessor label retained by the participant.

20. Perspective

Claim identity: SFT-CONSC-PERSPECTIVE-001

Question and exact theorem. A perspective is the exact observation path available from one retained source position and is not an unlocated view from nowhere.

located-observation-path__this-path/other-path__source-relative-access__path-and-s
 source-label__first-person-structural__root-bound-forward-forcing__positive-finite-
 successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENC-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-FRAGMENTATION-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	located-observation-path	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	this-path/other-path	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	source-relative-access	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	path-and-source-label	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	first-person-structural	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: located-observation-path__this-path/other-path__source-relative-access__path-and-source-label__first-person-structural__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. This exact relation closes a perspective is the exact observation path available from one retained source position and is not an unlocated view from nowhere. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

A perspective is a source-located access path. The subject is the continuing integrated relation holding that path while content changes. Interiority is the structural asymmetry between participating in self-observation and possessing its external image; phenomenal privacy follows when the image closes a predecessor label retained by the participant.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt

sha256:2b0f1d3b10331d000ab352a1b0931efeb1234afc006a784453e635279bf03509.

- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:676f0caedeb70147cb4fc06161cd427b94b595e392baa5e26045a99539e85982.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:11f6255f9ace7dce4650ea38c6aec80fd2b05885adb1dec23066f79af91c5440.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:15786813ae1383c3ddc1f8ca73e5f0ae2509820e58bf953951e0cd8916349c3e.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:1913eb131a314214f0df5481897429605bbe9c1ee4544a1721976b30257ced7b; independent certificate sha256:2dadf634bf32b8f1ebe51c5da0a0dcea932e8d5081caac61e2bdf06900e61692; external-validation identity sha256:af552c3f51a0823059a402da4ceeb41cd6898c5772f7900cb9df8509d712b3a4.

Post-seal empirical-boundary comparison. Directness: participant_bound_empirical_constraint. Disposition: boundary_constraint. External integration and no-report records constrain behavior and access but cannot replace the participant's first-person evidence class.

The external target was opened only after the claim derivation seal. Predicted consequence:

first-and-third-person-evidence-nonidentity-preserved. Source-derived consequence:

first-and-third-person-evidence-nonidentity-preserved. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-SPLIT-INTEGRATION-2023 - de Haan et al., Brain Sciences;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10667445/>; class: primary split-brain integration experiment; usable captures: 1; registered features present: split-brain patient, two experiments, automatic processing, deliberate; absent and preserved: none; adverse/boundary role: none.
- CONSC-NOREPORT-CRITIQUE-2022 - Duman et al., Frontiers in Human Neuroscience;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9130851/>; class: registered methodological adverse evidence; usable captures: 1; registered features present: report-based paradigms, no-report paradigms, alternative interpretations; absent and preserved: none; adverse/boundary role: Preserves the finding that no-report indicators do not automatically establish phenomenal occurrence..

Comparison record:

- sft-consc-perspective-001-external-consequence: predicted first-and-third-person-evidence-nonidentity-preserved; source-derived first-and-third-person-evidence-nonidentity-preserved; exact match True
- evidence disposition: boundary_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity

- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:dbd55de5acfd5946e6673368e0d83c92ab691822d14011f252bea25186fbe58f; derivation seal sha256:1df0d67a7e02225fcabc700bf33322c1ed884bd77fee77be523b60aa4bd124f07; engine receipt sha256:eb898f58b07150cfc4a3fbf96f1e32d6cf56c45d5b2cbc2a3e3cf0cb7f4be4ee at receipts/engine/model_admitted/SFT-CONSC-PERSPECTIVE-001-eb898f58b07150cf.json; empirical validation sha256:16dd6b1ef8478c2b082cd3598b7bb2903a9f2688ace27100fb0fe1254149aa09; measurement receipt sha256:c279bc9c699102294fdb15a9a6bb2697e436116bcd90e6177877bf8074bf3232; isolation sha256:e84d893a3c7510df955682b1dbd6fa66fc20c7975068a7e3b2622c8c82b4a302; custody sha256:e38d82324a1e567dde82af4038b29a0aa986553c21df451e589a995f81df5b1a.

21. Interiority

Claim identity: SFT-CONSC-INTERIORITY-001

Question and exact theorem. Interiority is the asymmetry between participating in a closed self-observation process and possessing an external description of it.

closed-self-observation-loop__inside/outside-description__state-participates-in-own-transition__internal-and-external-traces__phenomenal-bridge__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-FRAGMENTATION-001
- SFT-CONSC-PERSPECTIVE-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	closed-self-observation-loop	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	inside/outside-description	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	state-participates-in-own-transition	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	internal-and-external-traces	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	phenomenal-bridge	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: closed-self-observation-loop__inside/outside-description__state-participates-in-own-transition__internal-and-external-traces__phenomenal-bridge__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. Interiority is forced by asymmetric access: the participating state enters the transition that produces its image, while an external observer possesses the image and provenance but not the closed predecessor label as lived participation.

A perspective is a source-located access path. The subject is the continuing integrated relation holding that path while content changes. Interiority is the structural asymmetry between participating in self-observation and possessing its external image; phenomenal privacy follows when the image closes a predecessor label retained by the participant.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: f4e1bfc03ebe78398e213b7b3182393c7e1891d898ea10b651c8afac66940dc9.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256: 7ce6ef8ce262c9cf08bc66339115b7577f68134f64fa4146fd861aa53479fcc5.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 0ffe651d72c7a4199581061e0b426aba13a088a9f0d1e5b064969b55e0c18738.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 8bfb66a95b4078bae186f052544c548700c0bb9986846c20b754bce3e2598d1d.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256: eced28360b22a9c729d13e85dda914a8bb287837c230bc520c7f6851c4883525; independent certificate sha256: 643c1c174c7e1f6a1164c4686116300215a5b25fb427ad92719f0cf45032fc94; external-validation identity
sha256: cd73489d fae372f05a9748f6decb59ef1110c4ab87f48470ce56ebddb6f5c5eb.

Post-seal empirical-boundary comparison. Directness: `participant_bound_empirical_constraint`. Disposition: `boundary_constraint`. External integration and no-report records constrain behavior and access but cannot replace the participant's first-person evidence class.

The external target was opened only after the claim derivation seal. Predicted consequence:

`first-and-third-person-evidence-nonidentity-preserved`. Source-derived consequence:

`first-and-third-person-evidence-nonidentity-preserved`. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-SPLIT-INTEGRATION-2023 - de Haan et al., Brain Sciences;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10667445/>; class: primary split-brain integration experiment; usable captures: 1; registered features present: split-brain patient, two experiments, automatic processing, deliberate; absent and preserved: none; adverse/boundary role: none.

- CONSC-NOREPORT-CRITIQUE-2022 - Duman et al., Frontiers in Human Neuroscience;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9130851/>; class: registered methodological adverse evidence; usable captures: 1; registered features present: report-based paradigms, no-report paradigms, alternative interpretations; absent and preserved: none; adverse/boundary role: Preserves the finding that no-report indicators do not automatically establish phenomenal occurrence..

Comparison record:

- sft-consc-interiority-001-external-consequence: predicted first-and-third-person-evidence-nonidentity-preserved; source-derived first-and-third-person-evidence-nonidentity-preserved; exact match True
- evidence disposition: boundary_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:b382fc6746b5495bbe81688c28cc4c172ece3e74adce3f2ca12a18d2ecf45fe2; derivation seal sha256:b4a95d1b97c6171170af5dd34287015e327198d5f1313dbc3553ec681485110c; engine receipt sha256:59ae46e5a54732dea1b38a2466638e612f69076c72079035b27e14959270082b at receipts/engine/model_admitted/SFT-CONSC-INTERIORITY-001-59ae46e5a54732de.json; empirical validation sha256:e3f78889afb363d6a4e00b94bc76f728f2016b5b452b88e2c215a7bedfef665b; measurement receipt sha256:d4790c33324555df07a203255125b4d0ab0b80746502df38475416ba7c3028e2; isolation sha256:6dc0254372d12420b0461c024a45a320c2bce76bb8d61edc15c1d2db00b0c3d0; custody sha256:af5d448105ce35ae7f465038500edec039a776c756317dcf73ba64cb8eba8c1f.

22. Subject carrier

Claim identity: SFT-CONSC-SUBJECT-CARRIER-001

Question and exact theorem. The subject carrier is the continuing integrated relation that holds a perspective while its content states change.

continuing-integrated-loop__subject/content__holds-perspective-across-content__identity-and-content-traces__phenomenal-identity__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-FRAGMENTATION-001
- SFT-CONSC-INTERIORITY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	continuing-integrated-loop	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	subject/content	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
operation	operation-imported-fitted-or-missing	holds-perspective-across-content	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	identity-and-content-traces	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	phenomenal-identity	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `continuing-integrated-loop__subject/content__holds-perspective-across-content__identity-and-content-traces__phenomenal-identity__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes the subject carrier is the continuing integrated relation that holds a perspective while its content states change. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

A perspective is a source-located access path. The subject is the continuing integrated relation holding that path while content changes. Interiority is the structural asymmetry between participating in self-observation and possessing its external image; phenomenal privacy follows when the image closes a predecessor label retained by the participant.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:c11e3bb574ea5a48933a696d25e4f303b696fefc4f382623f57021b8c8460d2f.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:b9fd655b34d310cf1f1f5093463cb74e652f34874c20daf4e082790ddaafc49eb.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:7794113b0ae48ecf582450e9f88b6305045d19bf66edda12ec28cdf38605e3ee.

- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:7ac573c78795efb4a0e60621a197b3d0d12f6e6693de8df86fecb6719289e8a7.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:04bc08385cea72d62e38561fa5139b078e82c0d9c3c00e32dd8a97ca3ac40bdc; independent certificate sha256:ede6ab588017c7c01193008d119fe80d37991105a09da758641370779385be7f; external-validation identity
sha256:26c838084fb3bd415099bba493cacef342f1e5f5922167e6be6b3be2e163f33a.

Post-seal empirical-boundary comparison. Directness: participant_bound_empirical_constraint. Disposition: boundary_constraint. External integration and no-report records constrain behavior and access but cannot replace the participant's first-person evidence class.

The external target was opened only after the claim derivation seal. Predicted consequence:

first-and-third-person-evidence-nonidentity-preserved. Source-derived consequence:

first-and-third-person-evidence-nonidentity-preserved. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-SPLIT-INTEGRATION-2023 - de Haan et al., Brain Sciences;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10667445/>; class: primary split-brain integration experiment; usable captures: 1; registered features present: split-brain patient, two experiments, automatic processing, deliberate; absent and preserved: none; adverse/boundary role: none.
- CONSC-NOREPORT-CRITIQUE-2022 - Duman et al., Frontiers in Human Neuroscience;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9130851/>; class: registered methodological adverse evidence; usable captures: 1; registered features present: report-based paradigms, no-report paradigms, alternative interpretations; absent and preserved: none; adverse/boundary role: Preserves the finding that no-report indicators do not automatically establish phenomenal occurrence..

Comparison record:

- sft-consc-subject-carrier-001-external-consequence: predicted first-and-third-person-evidence-nonidentity-preserved; source-derived first-and-third-person-evidence-nonidentity-preserved; exact match True
- evidence disposition: boundary_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor

- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:273e723ec75a59ddfdb213f73b88d7199fca18862b4556f8ec3866890410135e; derivation seal sha256:e3162fa0764297b9b832cdc80852678a0b3d7badff8eaccde8bec0cb9b1a2d4f; engine receipt sha256:bfd669d8b5abef024f7cf001f7f1afeceac2b34a830eff777dlaab5d0cc61000 at receipts/engine/model_admitted/SFT-CONSC-SUBJECT-CARRIER-001-bfd669d8b5abef02.json; empirical validation sha256:a0858fd9ba80f6ce3ed0de04c6e3bd4d21f7a90a86224b848baaffc7b3969b2e; measurement receipt sha256:5e6b3f9f283b90dc600fa67637bb23340e994cd94545f8782c447c74941c6c8e; isolation sha256:4df20681ddf82bbffe889afdb33278148187104d2e63a945692edb89d41f1993; custody sha256:2e20b89c7be43a600c9008b5762076396d98abb32aa460343b8480a0dbd68c8b.

23. Self invariant

Claim identity: SFT-CONSC-SELF-INVARIANT-001

Question and exact theorem. A self invariant is the unique admitted relation preserved across the declared self-observation trace, not an imported substance.

self-observation-trace__changing-content/invariant-relation__fixed-relation-test__ordered-state-and-invariant__first-person-and-formal__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENC-001
- SFT-COMP-LEARN-REPRESENTATION-001

- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-FRAGMENTATION-001
- SFT-CONSC-SUBJECT-CARRIER-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	self-observation-trace	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	changing-content/invariant-relation	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	fixed-relation-test	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	ordered-state-and-invariant	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	first-person-and-formal	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: self-observation-trace__changing-content/invariant-relation__fixed-relation-test__ordered-state-and-invariant__first-person-and-formal__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.

- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes a self invariant is the unique admitted relation preserved across the declared self-observation trace, not an imported substance. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

A perspective is a source-located access path. The subject is the continuing integrated relation holding that path while content changes. Interiority is the structural asymmetry between participating in self-observation and possessing its external image; phenomenal privacy follows when the image closes a predecessor label retained by the participant.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 9f6096e24277dcd7896086dba9a6285d31675ecbaf9425227b6c6b8879c22302.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256: f266ee2b5261573e6b41076d489de60c49645ea57e8539b230ed750ef23aee9e.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: c34bfd5975fc0cab6bbbdcc7531883775ddc02be848789846e4eba8a15e53fef.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 89374dbcac28e0ddcc2188c9ad725f40bc08f458f75e9f364c3fdd3d9a7021fd.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256: b719269ec5dfb71009ed195e5f298dcad422b2e6fe098e7e1c906aafcde9ddae; independent certificate sha256: de1b0b40b0cdd002be96fb09bd204b9fd6f3c6216520be2d013b9d5c1e834566; external-validation identity sha256: 2e05c605092976779355f6fc0936ba96b5a8ad6f45f5ace2e6c5a0f5690a2af7.

Post-seal empirical-boundary comparison. Directness: `participant_bound_empirical_constraint`. Disposition: `boundary_constraint`. External integration and no-report records constrain behavior and access but cannot replace the participant's first-person evidence class.

The external target was opened only after the claim derivation seal. Predicted consequence:

`first-and-third-person-evidence-nonidentity-preserved`. Source-derived consequence:

`first-and-third-person-evidence-nonidentity-preserved`. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-SPLIT-INTEGRATION-2023 - de Haan et al., Brain Sciences;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10667445/>; class: primary split-brain integration experiment; usable captures: 1; registered features present: split-brain patient, two experiments, automatic processing, deliberate; absent and preserved: none; adverse/boundary role: none.
- CONSC-NOREPORT-CRITIQUE-2022 - Duman et al., Frontiers in Human Neuroscience;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9130851/>; class: registered methodological adverse evidence; usable captures: 1; registered features present: report-based paradigms, no-report paradigms, alternative interpretations; absent and preserved: none; adverse/boundary role: Preserves the finding that no-report indicators do not automatically establish phenomenal occurrence..

Comparison record:

- sft-consc-self-invariant-001-external-consequence: predicted first-and-third-person-evidence-nonidentity-preserved; source-derived first-and-third-person-evidence-nonidentity-preserved; exact match True
- evidence disposition: boundary_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:14c64bc1ebe0431b56deb38e5fc661f40b53f182fded68e3496718c58d0c9a46; derivation seal sha256:1ee053577a123263475084ccac29c5a0664a261667eff7f9ec96d48a2073bce7; engine receipt sha256:02aca8e3d823e59800bacba232cd787d69a1430b79fe4742bb2a753d73b1df01 at receipts/engine/model_admitted/SFT-CONSC-SELF-INVARIANT-001-02aca8e3d823e598.json; empirical validation sha256:da30dff620d8863cf408edeb9f9715f6bbf5075d9daca74609522649d7edb707; measurement receipt sha256:05f1f160f6aeadea0f5878ce2555d1dcabe06885d67b32d7dfa8828989f69ad4; isolation sha256:c4030123b6af1043d46d48c726c292d529b00199897aee33c597c60a453bd2c4; custody sha256:a83c0bd5aaa6fbb6c0bd329d6189c040c76cd3549e751b69dec9d2f6dd6e1692.

24. First- and third-person evidence

Claim identity: SFT-CONSC-FIRST-THIRD-PERSON-001

Question and exact theorem. First-person and third-person records may be paired but remain different evidence classes unless a bridge law is independently admitted.

paired-interior-and-external-records__experienced/observed__provenance-preserving-pairing__participant-observer-method__evidence-class-separation__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-FRAGMENTATION-001
- SFT-CONSC-SELF-INVARIANT-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	paired-interior-and-external-records	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	experienced/observed	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	provenance-preserving-pairing	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	participant-observer-method	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	evidence-class-separation	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: paired-interior-and-external-records__experienced/observed__provenance-preserving-pairing__participant-observer-method__evidence-class-separation__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. This exact relation closes first-person and third-person records may be paired but remain different evidence classes unless a bridge law is independently admitted. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

A perspective is a source-located access path. The subject is the continuing integrated relation holding that path while content changes. Interiority is the structural asymmetry between participating in self-observation and possessing its external image; phenomenal privacy follows when the image closes a predecessor label retained by the participant.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:011118dfaf212988b3edbea8aa55dc7309f30cc2fc1ee56a9aef86788fcf1272.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:0d501080d20d94a0a282a220a81cbad62b975ee452d052cf2ee9218687c5ab74.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:a3fe7940fb4cc21ed263ac73de5277b0e42aac7463e8475e8ecb82da8d24370c.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:f3a4250483d8d5cd9e4825079bfa8f4398da47cf61f8003d91a0a03969ae40ee.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:f50827bfd93e160007a0911dc1221683456067d7e434f84ff0b9b586bab2334a; independent

certificate sha256:7467c3c1baa28d120d70bf807eef7c0bdfc9544a93ad74d45b295fa0de485ade;
 external-validation identity
 sha256:71a0fd034bcc94494ea7599628bb5373640549c0103a6891a2949ee4bf8d51aa.

Post-seal empirical-boundary comparison. Directness: operational_or_record_constraint. Disposition: boundary_constraint. External integration and no-report records constrain behavior and access but cannot replace the participant's first-person evidence class.

The external target was opened only after the claim derivation seal. Predicted consequence: first-and-third-person-evidence-nonidentity-preserved. Source-derived consequence: first-and-third-person-evidence-nonidentity-preserved. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-SPLIT-INTEGRATION-2023 - de Haan et al., Brain Sciences;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10667445/>; class: primary split-brain integration experiment; usable captures: 1; registered features present: split-brain patient, two experiments, automatic processing, deliberate; absent and preserved: none; adverse/boundary role: none.
- CONSC-NOREPORT-CRITIQUE-2022 - Duman et al., Frontiers in Human Neuroscience;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9130851/>; class: registered methodological adverse evidence; usable captures: 1; registered features present: report-based paradigms, no-report paradigms, alternative interpretations; absent and preserved: none; adverse/boundary role: Preserves the finding that no-report indicators do not automatically establish phenomenal occurrence..

Comparison record:

- sft-consc-first-third-person-001-external-consequence: predicted first-and-third-person-evidence-nonidentity-preserved; source-derived first-and-third-person-evidence-nonidentity-preserved; exact match True
- evidence disposition: boundary_constraint; directness: operational_or_record_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience

- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:e8003e2a7e7445362c1718dab5393a543fcc4ab7ce45a6ee6334603e5e8624ac; derivation seal sha256:e32def3fb21b3d51529849be36ec1c6cdb91036ab5c0f1808689af8f333c8f80; engine receipt sha256:5e56f6376aaab75f73c9ee3764f408ecf18f4a049bd9cb398dd900ca833a01d5 at receipts/engine/model_admitted/SFT-CONSC-FIRST-THIRD-PERSON-001-5e56f6376aaab75f.json; empirical validation sha256:a04d14c388266bdf99fadd13e5172e2441c12ebc50c8bf4994ea34cc9522f26; measurement receipt sha256:9ac6e17faef367f82580b6323875b3da384966ded76ee418de6a0ba07ebf04a8; isolation sha256:7254be96a5c5f5a21665a0a5c1bc9e9dd2061fffd7aa571ae5c2c950e6895da5; custody sha256:f79327a29599dac556d56359e9caf073275615660a25a295f5c9ccf9afa7c3f6.

25. Phenomenal privacy

Claim identity: SFT-CONSC-PHENOMENAL-PRIVACY-001

Question and exact theorem. Phenomenal privacy follows where an external image closes a predecessor distinction that remains present to the participating interior trace.

interior-state-and-external-image__private/inaccessible-by-image__preimage-loss-boundary__retained-private-label__formal-and-first-person__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-FRAGMENTATION-001

- SFT-CONSC-FIRST-THIRD-PERSON-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	interior-state-and-external-image	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	private/inaccessible-by-image	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	preimage-loss-boundary	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	retained-private-label	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	formal-and-first-person	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: interior-state-and-external-image__private/inaccessible-by-image__preimage-loss-boundary__retained-private-label__formal-and-first-person__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. Privacy is not mysterious inaccessibility added by assertion. It follows from the two-to-one observation class when the participant retains a predecessor label that the external image closes.

A perspective is a source-located access path. The subject is the continuing integrated relation holding that path while content changes. Interiority is the structural asymmetry between participating in self-observation and possessing its external image; phenomenal privacy follows when the image closes a predecessor label retained by the participant.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:2b3637ddf30390a1db7d7814737f94b7b35e829237eff5f508fcd969cd3380da.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:44fbf01a66697a177b15f4cd7467727e39a28df2fb6cb1ebbbe9f4ea9d1b7e3.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:5536e675d83a1f16cfd4ccdee02ff158ce7c0d66f8d0caae16eee7bf6a459de.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:beead5267938e5f11553fa0e61ed36e502817576acbe01828a92e5a33230618a.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:71d8611128c5fc1ba05fa15e0bb9db46b00416ff35402afa1ff26cebd33d6282; independent certificate sha256:7c0532ecbf4993fdcec04727012c450ab9985a5b57f4963ab671a161a41501fa; external-validation identity
sha256:cd126928cd23f6683edeae259ac234d48921d3d32904d0c8c2597809cf62520b.

Post-seal empirical-boundary comparison. Directness: `participant_bound_empirical_constraint`. Disposition: `boundary_constraint`. External integration and no-report records constrain behavior and access but cannot replace the participant's first-person evidence class.

The external target was opened only after the claim derivation seal. Predicted consequence:

`first-and-third-person-evidence-nonidentity-preserved`. Source-derived consequence:

`first-and-third-person-evidence-nonidentity-preserved`. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-SPLIT-INTEGRATION-2023 - de Haan et al., Brain Sciences;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC10667445/>; class: primary split-brain integration experiment; usable captures: 1; registered features present: split-brain patient, two experiments, automatic processing, deliberate; absent and preserved: none; adverse/boundary role: none.
- CONSC-NOREPORT-CRITIQUE-2022 - Duman et al., Frontiers in Human Neuroscience;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9130851/>; class: registered methodological adverse evidence; usable captures: 1; registered features present: report-based paradigms, no-report paradigms, alternative interpretations; absent and preserved: none; adverse/boundary role: Preserves the finding that no-report indicators do not automatically establish phenomenal occurrence..

Comparison record:

- `sft-consc-phenomenal-privacy-001-external-consequence`: predicted `first-and-third-person-evidence-nonidentity-preserved`; source-derived `first-and-third-person-evidence-nonidentity-preserved`; exact match `True`
- evidence disposition: `boundary_constraint`; directness: `participant_bound_empirical_constraint`
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:8d95be38c0e8ff00c5629144d41b8365f14256be01ace56b3251668760852508; derivation seal sha256:6b1807fda8bf03e68bdfd38d9fd31beeb5026f9c0f518628b21d3488017fe9; engine receipt sha256:ee914882836a80f76b9b8d676adcf1ae6a56fedcae38ca9f9bff30e0ab079c7b at receipts/engine/model_admitted/SFT-CONSC-PHENOMENAL-PRIVACY-001-ee914882836a80f7.json; empirical validation sha256:96059968f825a83c6bcb93df95d6bcae9351e04dcc1cb92560f0ea505592f3ed; measurement receipt sha256:fc45bfb9262bc5c6c10a57d07bccee91107e02ccc40f280fcc4bba98210db3e3; isolation sha256:82406ab36d9ecca63570ade84e449e0d9927368764d9ab023f3efa5deb3a62c4; custody sha256:f90b6d813160ec6416b3d89092901672d7aca1f1e44e264e2b5b801b22d36e46.

19. Self-Observation and Introspection

Self-observation can be deterministic and still opaque to itself. A two-to-one image does not recover its predecessor without the fibre label. Each unretained act therefore closes one exact distinction; confabulation is a completion not supported by the retained self-history.

26. Self-observation closure

Claim identity: SFT-CONSC-SELF-OBSERVATION-001

Question and exact theorem. Self-observation is closed only when the generated observation image becomes an input to the same identified process.

```
process-state-and-self-image__system/object-of-system__image-reentry__complete-self-transition__formal-self-model__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001

- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-PHENOMENAL-PRIVACY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	process-state-and-self-image	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	system/object-of-system	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	image-reentry	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	complete-self-transition	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	formal-self-model	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `process-state-and-self-image__system/object-of-system__image-reentry__complete-self-transition__formal-self-model__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes self-observation is closed only when the generated observation image becomes an input to the same identified process. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Self-observation can be deterministic and still opaque to itself. A two-to-one image does not recover its predecessor without the fibre label. Each unretained act therefore closes one exact distinction; confabulation is a completion not supported by the retained self-history.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:22dd8fbca61a9580f735dcac026b965e6175f8ca5759d18397b4e270a2cd207b`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:bb6d301110c70528acb68ffaf45fd33abf2417de1aab8124fc8966aefc67f336`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
`sha256:1ff635d62c21f655e3f0fa8e5714393584cc29b8012f75e776bebcfab86d05fd`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
`sha256:e4a71ccf3f157b0a8a691396a3ef5b3c23be61f92cdc664a884190f17602b725`.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

`sha256:a60b240e64f505bef6190c20dd6f29b2a5172942a6fc546368d281f4a1461528`; independent certificate `sha256:5337a9553f715ca6dddf2c7eac759da5f28d93a6a23feb90793303bde37489e1`; external-validation identity
`sha256:6d25ead2e2cb3deeb83c2cb8a10036c736db653aef4b988985d2c8506bcbdddf4`.

Post-seal empirical-boundary comparison. Directness:

formal_reconstruction_with_external_boundary_only. Disposition:

direct_behavioural_constraint. Choice-blindness manipulation directly tests the boundary between reported reasons and the trace available to the reporter.

The external target was opened only after the claim derivation seal. Predicted consequence:

introspective-report-and-retained-history-nonidentity-preserved. Source-derived consequence:

introspective-report-and-retained-history-nonidentity-preserved. Exact correspondence: true.

Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-CHOICE-BLINDNESS-2017 - Sloman et al., PLOS ONE; <https://pmc.ncbi.nlm.nih.gov/articles/PMC5308842/>; class: primary introspection and confabulation experiment; usable captures: 1; registered features present: Choice Blindness Paradigm, confidence, confabulate, manipulated; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- sft-consc-self-observation-001-external-consequence: predicted introspective-report-and-retained-history-nonidentity-preserved; source-derived introspective-report-and-retained-history-nonidentity-preserved; exact match True
- evidence disposition: direct_behavioural_constraint; directness: formal_reconstruction_with_external_boundary_only
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:865191205a5a1d9128678766e784960af4d45bf1a94aa5901a97fef0cc53075d; derivation seal sha256:f31aedc68a2e7e7a1334d990c6d7ab01f40d677f97013c7cd2d9a598c683e7e7; engine receipt

sha256:02090663f6c9634fd3b5c5982db88999596f6122d45d95d5716a64c92c86bf8c at
 receipts/engine/model_admitted/SFT-CONSC-SELF-OBSERVATION-001-02090663f6c9634f.json;
 empirical validation sha256:efe895c7f94de2230872c1286be8df293220e9658f158f410a486ebed26a5173;
 measurement receipt
 sha256:857a291c1b40339223fc24e76f60fed769b6d2632bb5f72a57738275ace46765; isolation
 sha256:c24c5a810c756eb3d514928e6aeaf944368fcf62b555357fd3150465ab94cbde; custody
 sha256:97426fff49142f0db25d33e3ef97b24d0b6e7599b6b77acbb92aa361be0d7b6e.

27. Introspection blind spot

Claim identity: SFT-CONSC-INTROSPECTION-BLIND-SPOT-001

Question and exact theorem. Introspection from the image alone cannot recover which generated predecessor label was closed by the act.

two-preimage-observation-class__predecessor/image__unretained-fibre-closure__missing-fibre-explicit__formal-and-report__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-PHENOMENAL-PRIVACY-001
- SFT-CONSC-SELF-OBSERVATION-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	two-preimage-observation-class	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	predecessor/image	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	unretained-fibre-closure	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	missing-fibre-explicit	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	formal-and-report	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: two-preimage-observation-class__predecessor/image__unretained-fibre-closure__missing-fibre-explicit__formal-and-report__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. From the present image alone, the process cannot decide which of the two generated predecessors produced it. Exact self-knowledge therefore requires a retained fibre label rather than confidence.

Self-observation can be deterministic and still opaque to itself. A two-to-one image does not recover its predecessor without the fibre label. Each unretained act therefore closes one exact distinction; confabulation is a completion not supported by the retained self-history.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:c396b3d3b259a01475f9f1fa1a6336e83c14cbec887f8271d17746058f26163f.

- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:0c727838c24dc61e6c315dc7fab450ea74082ebcdc0eb549484479a73b3abf9.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:84feeb421c965a6cce5caec9348ea8c38bb9e537e477a236b244d965c8322042.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:16ea4f0473953157de73b753f1b33e439034887cc700a1e29a0b4db4a233edcc.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:1dc509e46fdc9b6e5b241793e95b09dd908d2787fda55cc425f18ccc704abc64; independent certificate sha256:1fdc46779f7af6b08e65c016ba37dcf2e35d301a8472c26424265b5f4ab6ac64; external-validation identity sha256:b93eac06f430a0440d39d96c41a1500a7d92adad2ec67bbf5908b003985b6a0b.

Post-seal empirical-boundary comparison. Directness:

formal_reconstruction_with_external_boundary_only. Disposition:

direct_behavioural_constraint. Choice-blindness manipulation directly tests the boundary between reported reasons and the trace available to the reporter.

The external target was opened only after the claim derivation seal. Predicted consequence:

introspective-report-and-retained-history-nonidentity-preserved. Source-derived consequence:

introspective-report-and-retained-history-nonidentity-preserved. Exact correspondence: true.

Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-CHOICE-BLINDNESS-2017 - Sloman et al., PLOS ONE; <https://pmc.ncbi.nlm.nih.gov/articles/PMC5308842/>; class: primary introspection and confabulation experiment; usable captures: 1; registered features present: Choice Blindness Paradigm, confidence, confabulate, manipulated; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- sft-consc-introspection-blind-spot-001-external-consequence: predicted introspective-report-and-retained-history-nonidentity-preserved; source-derived introspective-report-and-retained-history-nonidentity-preserved; exact match True
- evidence disposition: direct_behavioural_constraint; directness: formal_reconstruction_with_external_boundary_only
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter

- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:c37dbf7f199e4f780612634f1f589340f01fb2ed110734e76c03bddcf10bca87; derivation seal sha256:bf29e77a6c2b5ede91129d2ba9894e65232641fd02f95d4ece13f6c126bcfa5e; engine receipt sha256:8e4f5d01fd8763c983350d229332d74756945c465f7441ec7519e156275db91e at receipts/engine/model_admitted/SFT-CONSC-INTROSPECTION-BLIND-SPOT-001-8e4f5d01fd8763c9.json; empirical validation sha256:2142941e57bbe1a0623bfe7019edf962ca6dec41a975020e7f380797d00c1efd; measurement receipt sha256:d471ff5ac19565195c3e9be0ab2d5515b6c7cf5a1a631fc3b181231efe2a3436; isolation sha256:a78bcb954fb6735540f1283f839d69350d37cc5f3b7947406a4e4c950a14cf36; custody sha256:8997fce2752ffef6d241c0a211c27fc1e7f71c46c88cf17775c7f330315cb5be.

28. Self-history reconstruction

Claim identity: SFT-CONSC-SELF-HISTORY-RECONSTRUCTION-001

Question and exact theorem. An exact self-history is reconstructible only to the extent that each closed fibre label was separately retained.

present-state-and-held-labels__recoverable/unrecoverable-history__reverse-by-retained-fibre__one-label-per-act__formal-memory__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENC-001

- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-PHENOMENAL-PRIVACY-001
- SFT-CONSC-INTROSPECTION-BLIND-SPOT-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	present-state-and-held-labels	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	recoverable/unrecoverable-history	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	reverse-by-retained-fibre	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	one-label-per-act	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	formal-memory	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: present-state-and-held-labels__recoverable/unrecoverable-history__reverse-by-retained-fibre__one-label-per-act__formal-memory__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **one retained fibre label per reconstructed observation act**. Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes an exact self-history is reconstructible only to the extent that each closed fibre label was separately retained. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Self-observation can be deterministic and still opaque to itself. A two-to-one image does not recover its predecessor without the fibre label. Each unretained act therefore closes one exact distinction; confabulation is a completion not supported by the retained self-history.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:b676d65a044f11d8d31d16d31ca31897436073bd33877ed4751ece32eb4a50a4.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:9013a80e9d2005205673e757a1e61d9769f8c816fc8f084150b4a77f09669586.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:2d35c4ae94852a3c933a9c97320c68225593d23a134f9ec90af773f41141d6e3.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:5828f31962645b5e46c6bddbf81ccea453c4220272abbde94785b71c8c08aff.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:51c3c14e907f9973f9cc835ff80e985cf8ac45c4100620f236dec3540c1b2dd4; independent certificate sha256:a2e5d99cdd1d264a81c4f3a0f28a9bcb6c918c09ddc55e119a648815bd30dd2c; external-validation identity sha256:34c027e8e27a94760b15881c5b94a3c25a83e3d4113545e48d75c786f7e00cd6.

Post-seal empirical-boundary comparison. Directness:

`formal_reconstruction_with_external_boundary_only`. Disposition:

`direct_behavioural_constraint`. Choice-blindness manipulation directly tests the boundary between reported reasons and the trace available to the reporter.

The external target was opened only after the claim derivation seal. Predicted consequence:

`introspective-report-and-retained-history-nonidentity-preserved`. Source-derived consequence:

`introspective-report-and-retained-history-nonidentity-preserved`. Exact correspondence: `true`.

Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- `CONSC-CHOICE-BLINDNESS-2017` - Sloman et al., PLOS ONE; <https://pmc.ncbi.nlm.nih.gov/articles/PMC5308842/>; class: primary introspection and confabulation experiment; usable captures: 1; registered features present: Choice Blindness Paradigm, confidence, confabulate, manipulated; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- `sft-consc-self-history-reconstruction-001-external-consequence`: predicted `introspective-report-and-retained-history-nonidentity-preserved`; source-derived `introspective-report-and-retained-history-nonidentity-preserved`; exact match `True`
- `evidence disposition`: `direct_behavioural_constraint`; `directness`: `formal_reconstruction_with_external_boundary_only`

- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:330a433337c446fc668d0e2cb762eb7e891cb00f794391159b0451087f30a8e8; derivation seal sha256:af8a17029166ca32b2bf47ad896b6e394f63b14d1059234fef99eeefb3893366a; engine receipt sha256:ba47a6dbbb4cde85fd6eae2714053d301cd59a3b78a7c530315ef3a5b8a24aad at receipts/engine/model_admitted/SFT-CONSC-SELF-HISTORY-RECONSTRUCTION-001-ba47a6dbbb4cde85.json; empirical validation sha256:ae2fecc6eeac824a62f3c395766f38b12561b7a527cc8aa5a175f7ca82e83120; measurement receipt sha256:1a677bf296a59689ce2d0eabc17a0dc79f313616ba79954aaf686eb802bf2414; isolation sha256:08d3629d323bc8133a1f62102490f23e5bbe87377610455f9596cf63edb608c2; custody sha256:66e7afa2f1002ae8c689ac2290a06a070ec2f44d1b7c6fc822de4683571dc49f.

29. Accumulated introspection loss

Claim identity: SFT-CONSC-INTROSPECTION-LOSS-001

Question and exact theorem. Each unrecorded two-to-one self-observation closes one predecessor distinction; repeated loss is counted without negative information quantities.

```
ordered-self-observation-trace__retained/closed-predecessors__counted-label-closure__loss-ledger__formal-information__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001

- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-PHENOMENAL-PRIVACY-001
- SFT-CONSC-SELF-HISTORY-RECONSTRUCTION-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	ordered-self-observation-trace	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	retained/closed-predecessors	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	counted-label-closure	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	loss-ledger	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	formal-information	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `ordered-self-observation-trace__retained/closed-predecessors__counted-label-closure__loss-ledger__formal-information__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **one retained fibre label is required per reversed two-to-one observation act**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes each unrecorded two-to-one self-observation closes one predecessor distinction; repeated loss is counted without negative information quantities. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Self-observation can be deterministic and still opaque to itself. A two-to-one image does not recover its predecessor without the fibre label. Each unretained act therefore closes one exact distinction; confabulation is a completion not supported by the retained self-history.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:67026cd705e56a8f89f6189a79e5a48e7845258d9da2979dcee559252803e858.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:6ca20d8c66aa9d313b8205fe3b02304b83efa2ce3b2d73351138270201af175c.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:f8fb3784e56e20f14907da43d73cb777e73d00656fff0d85d30c10713bbd9f05.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:bef0ee1db045e5b093d52370dc7a1fc48c03dd5e3ec58f9e2601f646de957fd3.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:aeb9d92dc31b7a5dbb0211542139b231ad0cde6e27a1ff72f1ca0e47eba6c4f6; independent certificate sha256:88b325436e872faedc23a18a588fc29212c71a5f8b5ce885cb5373a3f9cf6c7d; external-validation identity

sha256:d6b5b75190efff0348c40fc7c43435fbcfbf202777d96a8eea3b7aefe736e7d82.

Post-seal empirical-boundary comparison. Directness:

formal_reconstruction_with_external_boundary_only. Disposition:

direct_behavioural_constraint. Choice-blindness manipulation directly tests the boundary between reported reasons and the trace available to the reporter.

The external target was opened only after the claim derivation seal. Predicted consequence:

introspective-report-and-retained-history-nonidentity-preserved. Source-derived consequence:

introspective-report-and-retained-history-nonidentity-preserved. Exact correspondence: true.

Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-CHOICE-BLINDNESS-2017 - Sloman et al., PLOS ONE; <https://pmc.ncbi.nlm.nih.gov/articles/PMC5308842/>; class: primary introspection and confabulation experiment; usable captures: 1; registered features present: Choice Blindness Paradigm, confidence, confabulate, manipulated; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- sft-consc-introspection-loss-001-external-consequence: predicted introspective-report-and-retained-history-nonidentity-preserved; source-derived introspective-report-and-retained-history-nonidentity-preserved; exact match True
- evidence disposition: direct_behavioural_constraint; directness: formal_reconstruction_with_external_boundary_only
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:e2acad589126a679333e8bca0f19ae6ffb901971474d7b042cd1a3e7593991dc; derivation seal sha256:a7e22cf4896cb839f4bb5cc56663d72d2355c0a83443c77b157835803ef65dbf; engine receipt

sha256:88fe5ba6e920ccee62d58ddf4f1a98bb57c0cd51f0be32a778a183778acfd7d at receipts/engine/model_admitted/SFT-CONSC-INTROSPECTION-LOSS-001-88fe5ba6e920ccee.json; empirical validation sha256:b5365f3789487c84809a1c07d726cff30b6285aeb3f9e8443044b5cc35762cd6; measurement receipt
 sha256:131aba8cfdd032643626666dd0dcc792bc334ad715fbe527f07d069904d7d4d; isolation sha256:b7a1b3b9639da8a47bab5a3f6959a9d04992552583b85b7ec66ea2dfa7bfa21e; custody sha256:4d36b1c0a12e1b9093cf3d0c7aa44c2a5e7dd74ccd62455045df7b25965274dc.

30. Confabulation boundary

Claim identity: SFT-CONSC-CONFABULATION-BOUNDARY-001

Question and exact theorem. A reported reason unsupported by the retained self-history is a generated completion, not recovered introspective evidence.

report-and-incomplete-history__reconstruction/invention__trace-membership-test__cl
 aimed-and-held-premises__behavioural-report__root-bound-forward-forcing__positive-
 finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENC-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-PHENOMENAL-PRIVACY-001
- SFT-CONSC-INTROSPECTION-LOSS-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	report-and-incomplete-history	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	reconstruction/invention	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	trace-membership-test	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	claimed-and-held-premises	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	behavioural-report	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `report-and-incomplete-history__reconstruction/invention__trace-membership-test__claimed-and-held-premises__behavioural-report__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes a reported reason unsupported by the retained self-history is a generated completion, not recovered introspective evidence. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Self-observation can be deterministic and still opaque to itself. A two-to-one image does not recover its predecessor without the fibre label. Each unretained act therefore closes one exact distinction; confabulation is a completion not supported by the retained self-history.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:1e35a69c4f7f9efc6d137a807c4ce3536b25e7260eb34513ee0c1aa905cc427b.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:ac8c8b57a2fd4deab10cd84566424c29fd242174842d45e4ce804ed5f5981efc.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:410086949bf0ca4d9c96ce1a6509e215eb873642b76812229f68c092bd50819c.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:8bc241b06f2db352b154d8fdcfaba0728b7bf6c7d25d118fb82b36350af4ac74.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:007af261c603ce9c9589ab90908284242e905bca0f705bd7009ec6be619b6a4b; independent certificate sha256:a7f81f8077ea2190444f8b8e2e2ddf362839962b6dfe5076f1988a99dee3a5db; external-validation identity sha256:bde6dd9a17e5012dc1a3698ad3f2cd7aaf8c6b3ecfd8809550ff12151773d36c.

Post-seal empirical-boundary comparison. Directness: `direct_observable_constraint`. Disposition: `direct_behavioural_constraint`. Choice-blindness manipulation directly tests the boundary between reported reasons and the trace available to the reporter.

The external target was opened only after the claim derivation seal. Predicted consequence:

introspective-report-and-retained-history-nonidentity-preserved. Source-derived consequence: introspective-report-and-retained-history-nonidentity-preserved. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-CHOICE-BLINDNESS-2017 - Sloman et al., PLOS ONE; <https://pmc.ncbi.nlm.nih.gov/articles/PMC5308842/>; class: primary introspection and confabulation experiment; usable captures: 1; registered features present: Choice Blindness Paradigm, confidence, confabulate, manipulated; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- sft-consc-confabulation-boundary-001-external-consequence: predicted introspective-report-and-retained-history-nonidentity-preserved; source-derived introspective-report-and-retained-history-nonidentity-preserved; exact match True
- evidence disposition: `direct_behavioural_constraint`; directness: `direct_observable_constraint`
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity

- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:334f4a67c3161f50fdcfe955a064621103ecd176c73a8ef98eb82d9767ca252; derivation seal sha256:a9438075fc5ca55a166e6fa9741967ee8cf1cba59ab9f525a8eddb8771df2c6f; engine receipt sha256:db98ee24f4b1d7e1113a7ab950e046b2d892719e453628622d63297a27edcc5a at receipts/engine/model_admitted/SFT-CONSC-CONFABULATION-BOUNDARY-001-db98ee24f4b1d7e1.json; empirical validation sha256:f2d5ba59bd43008a1d2eebc47c77c74a23c7a82f18f2e8119c1b8a424ba9986d; measurement receipt sha256:8db11664836dbdc445c68e2304601dd8f253c39ef673339b44bef4bdb2e0435f; isolation sha256:1f9c3a9295c40cab9e0b69282a91f436882d6bbb63d884222be93bf0b42b9378; custody sha256:eefd01c3cb4116fd539cca610f65a54f9e1390a87ae7d652d48a34e17a0045a0.

31. Determinism with self-opacity

Claim identity: SFT-CONSC-DETERMINISM-SELF-OPACITY-001

Question and exact theorem. A process can have one generated successor while remaining unable to identify its unretained predecessor from the present image.

```
functional-transition-and-lost-preimage__unique-successor/unknown-predecessor__for
ward-execute-backward-class__successor-and-fibre-ledger__formal-agency-boundary__r
oot-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-ext
ra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001

- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-PHENOMENAL-PRIVACY-001
- SFT-CONSC-CONFABULATION-BOUNDARY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	functional-transition-and-lost-preimage	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	unique-successor/unknown-predecessor	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	forward-execute-backward-classes	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	successor-and-fibre-ledger	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	formal-agency-boundary	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: functional-transition-and-lost-preimage__unique-successor/unknown-predecessor__forward-execute-backward-class__successor-and-fibre-ledger__formal-agency-boundary__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. Superdeterministic transition and self-opacity coexist: the forward successor is unique, but backward identification remains a two-member class without the missing label. No metaphysical randomness is required.

Self-observation can be deterministic and still opaque to itself. A two-to-one image does not recover its predecessor without the fibre label. Each unretained act therefore closes one exact distinction; confabulation is a completion not supported by the retained self-history.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:81babba8d750f3ed231fc5d3580358c221063c745f8597a8251a70207b795724.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:c84812139fc706cc836d5c8dba8f1920dfb8b1c0d585f4fae0c817056b08834c.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:dd3642cbcdf50d094f7b5df2ceb962e450b499c1e123f7373107864eae0d6aaf.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:3c6a8d8601d924814e5e4a7481cb2149d1205943f05aa4355c66abde71673439.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:dcb9alb3f201f37adfd58f1a2fd8b84958c2cdf2effb17dd60802bac2f71f7c2; independent certificate sha256:d2a4c2900fa78a0c097a20a7b8f1a634c3ef30efc0f55c34821708d5b9247d72; external-validation identity sha256:4aaa9070ba964da8c9ff2639aa6a7ea08dcbbba9ee86a212d4842be4bed7d4453.

Post-seal empirical-boundary comparison. Directness:

`formal_reconstruction_with_external_boundary_only`. Disposition:

`direct_behavioural_constraint`. Choice-blindness manipulation directly tests the boundary between reported reasons and the trace available to the reporter.

The external target was opened only after the claim derivation seal. Predicted consequence:

`introspective-report-and-retained-history-nonidentity-preserved`. Source-derived consequence: `introspective-report-and-retained-history-nonidentity-preserved`. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-CHOICE-BLINDNESS-2017 - Sloman et al., PLOS ONE; <https://pmc.ncbi.nlm.nih.gov/articles/PMC5308842/>; class: primary introspection and confabulation experiment; usable captures: 1; registered features present: Choice Blindness Paradigm, confidence, confabulate, manipulated; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- `sft-consc-determinism-self-opacity-001-external-consequence`: predicted `introspective-report-and-retained-history-nonidentity-preserved`; source-derived

introspective-report-and-retained-history-nonidentity-preserved; exact match True

- evidence disposition: direct_behavioural_constraint; directness: formal_reconstruction_with_external_boundary_only
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:80f4e5b3205c173bb8022bf6a6102c21b70cff582dfd6f68c37aa73d0c2d8c0b; derivation seal sha256:aaa369ba5a4c5b36c7814e3e95cf534bf53c426459b92080a7f4c1d4bc4ad454; engine receipt sha256:cfdc06be185fa623913811f25f3f3d1dc0195004d8b558b7bb926561e0982066 at receipts/engine/model_admitted/SFT-CONSC-DETERMINISM-SELF-OPACITY-001-cfdc06be185fa623.json; empirical validation sha256:04226343f90e5ae00355076a20f2949483771bf1731338575f825c58d7d588a8; measurement receipt sha256:efccfee339de8e3fbd58d1672f672f55a72ee19cd18ee7dfc2b74e3bc00572a7; isolation sha256:868c4cba1c225825cb2a7c4e416002334eb366ce24d50a3f3516b97b4279facf; custody sha256:26e81647fc3dea6b9e8ba040f8e9928c668a365a8add5c426e7728f415826e79.

20. Memory, Temporal Continuity and Identity

Memory needs a carrier, recurrence or active renewal. A later equal output is not automatically the same memory. Temporal continuity is the complete predecessor-successor chain, while identity continuity adds lineage and the retained subject invariant; similarity or copying alone is insufficient.

32. Memory carrier

Claim identity: SFT-CONSC-MEMORY-CARRIER-001

Question and exact theorem. Memory requires a retained carrier relation linking the earlier content to its later availability.

```
retained-state-relation__held/rederived__recurrent-or-recorded-continuation__carrier-provenance__cognitive-and-biological-handoff__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-DETERMINISM-SELF-OPACITY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	retained-state-relation	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	held/rederived	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	recurrent-or-recorded-continuation	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	carrier-provenance	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
evidence	report-behaviour-correlation-or-confidence-substituted	cognitive-and-biological-handoff	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: retained-state-relation__held/rederived__recurrent-or-recorded-continuation__carrier-provenance__cognitive-and-biological-handoff__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. This exact relation closes memory requires a retained carrier relation linking the earlier content to its later availability. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Memory needs a carrier, recurrence or active renewal. A later equal output is not automatically the same memory. Temporal continuity is the complete predecessor-successor chain, while identity continuity adds lineage and the retained subject invariant; similarity or copying alone is insufficient.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:095670568ec08d7ad6e8d0fd0b400a4f2782adead14598f2db363bdcae61276a.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:d70d6a9fe3a463f65e824dec8154b808bc1e15307baf5a1c9aa2b061fa21b880.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:465fc4613dc8da01ca49d96bf4672381546fc93b66dc4acbd341cf11d2d65d9.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:0455aeb1b55ea9fa91632adf928150c25f8c32a457c612eeb78bf4cb4aededf6a.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
 sha256:5461d6994e091719d49b1e29437f8e09197f2f211cc36baa6103d97641bd857d; independent
 certificate sha256:1291a6c6173a07a7cbcd8c213aa00e2c62960fca47342bec1dd94a44ddaeb992;
 external-validation identity
 sha256:79607cb9f848cb435004d48093d22452ebd994e8ec7f0cedc012280b1a188b9e.

Post-seal empirical-boundary comparison. Directness: cross_domain_correspondence_constraint. Disposition:
 direct_and_boundary_constraint. Dream and anesthesia-recovery records test changing availability and report
 across state transitions without making report alone numerical identity.

The external target was opened only after the claim derivation seal. Predicted consequence:
 memory-report-state-and-lineage-boundary-preserved. Source-derived consequence:
 memory-report-state-and-lineage-boundary-preserved. Exact correspondence: true. Phenomenal
 occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact:
 false.

Sources and complete feature accounting:

- CONSC-DREAM-DATABASE-2025 - Siclari et al., Scientific Data; <https://pmc.ncbi.nlm.nih.gov/articles/PMC12350935/>;
 class: primary open dream EEG and report database; usable captures: 1; registered features present: DREAM database,
 participants, awakenings, Experience without recall, No experience; absent and preserved: none; adverse/boundary role:
 none.
- CONSC-ANESTHESIA-RECOVERY-2021 - Mashour et al., eLife; <https://pmc.ncbi.nlm.nih.gov/articles/PMC8163502/>;
 class: primary controlled consciousness-recovery experiment; usable captures: 1; registered features present: multicenter
 study, healthy humans, general anesthesia, recovery of consciousness, cognitive; absent and preserved: none;
 adverse/boundary role: none.

Comparison record:

- sft-consc-memory-carrier-001-external-consequence: predicted memory-report-state-and-lineage-boundary-preserved;
 source-derived memory-report-state-and-lineage-boundary-preserved; exact match True
- evidence disposition: direct_and_boundary_constraint; directness: cross_domain_correspondence_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness
 fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical
 measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience

- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:4a8128852004d80775ccf249842189db7eff72562720911795d560f891c1f3bf; derivation seal sha256:d0c127cbb5645f3d96ab0905e6737cbb2f0dd0460c7cfbd9c213cef5dad85951; engine receipt sha256:999b2b38351f2dd258de5e156760751843ea946e9371811d2da361b2bba9736b at receipts/engine/model_admitted/SFT-CONSC-MEMORY-CARRIER-001-999b2b38351f2dd2.json; empirical validation sha256:43e209b9d651ece4afe010a265585cea02869d63eb98a33e87707e04b4ae59cd; measurement receipt sha256:1b2e663aa80d2d66f0705f2ea3317860055723cc51f0b48d586a7fe6daa2f551; isolation sha256:d7830254b143c0c4a81cc1877420760835d021947c24c471434c448218eec8f9; custody sha256:181d376681d1703011d499377551c34d3665f86a1396ae25361b728ee722f873.

33. Memory persistence

Claim identity: SFT-CONSC-MEMORY-PERSISTENCE-001

Question and exact theorem. Persistent memory requires a recurrent or actively renewed retained relation; a coincidentally equal later output is insufficient.

finite-recurrent-support__persistent/transient__exact-return-recurrence__return-period-and-state-word__formal-and-behavioural__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001

- SFT-MED-SYMP TOM-001
- SFT-CONSC-DETERMINISM-SELF-OPACITY-001
- SFT-CONSC-MEMORY-CARRIER-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	finite-recurrent-support	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	persistent/transient	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	exact-return-recurrence	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	return-period-and-state-word	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	formal-and-behavioural	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: finite-recurrent-support__persistent/transient__exact-return-recurrence__return-period-and-state-word__formal-and-behavioural__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **one-third and two-thirds form an exact period-two return support.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. One-third and two-thirds exchange under the Fold and return in period two. This supplies an exact minimal recurrence witness; realized memory additionally requires its label-bearing carrier and evidence bridge.

Memory needs a carrier, recurrence or active renewal. A later equal output is not automatically the same memory. Temporal continuity is the complete predecessor-successor chain, while identity continuity adds lineage and the retained subject invariant; similarity or copying alone is insufficient.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:a192f5dd05d549ee57753ffaf9563851e961fbcccc16261bd41c204efdafbdf6.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:d24a39c4c1c9e4f6ce79a9283ddbe9c6259ed73f1a6b69d4cffb240a3cb6ea9e.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:4b2d073d91484a829d0e9dd59e3adde14e1949874e8c6302e9c4fd8847087e4d.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:3ebadfb02cce21b2b3b3391797bc0376e0ca921cdafe0b43984b2e8aab0c939c.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:5e4b804a69f038c78c2e5fc6bc3327f9cb0cd26f2a523e3ee584516be9a6359f; independent certificate sha256:a21de5559ea851d65985cc735932463258eca1c383e072a881d8f62e2d5fcc8e; external-validation identity sha256:3ae63b37b32501dd55c259dae0f220dfde8debccd2c7070bf575f709e4191a17.

Post-seal empirical-boundary comparison. Directness: `direct_observable_constraint`. Disposition: `direct_and_boundary_constraint`. Dream and anesthesia-recovery records test changing availability and report across state transitions without making report alone numerical identity.

The external target was opened only after the claim derivation seal. Predicted consequence:

`memory-report-state-and-lineage-boundary-preserved`. Source-derived consequence:

`memory-report-state-and-lineage-boundary-preserved`. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-DREAM-DATABASE-2025 - Siclari et al., Scientific Data; <https://pmc.ncbi.nlm.nih.gov/articles/PMC12350935/>; class: primary open dream EEG and report database; usable captures: 1; registered features present: DREAM database, participants, awakenings, Experience without recall, No experience; absent and preserved: none; adverse/boundary role: none.
- CONSC-ANESTHESIA-RECOVERY-2021 - Mashour et al., eLife; <https://pmc.ncbi.nlm.nih.gov/articles/PMC8163502/>; class: primary controlled consciousness-recovery experiment; usable captures: 1; registered features present: multicenter study, healthy humans, general anesthesia, recovery of consciousness, cognitive; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- `sft-consc-memory-persistence-001-external-consequence`: predicted `memory-report-state-and-lineage-boundary-preserved`; source-derived `memory-report-state-and-lineage-boundary-preserved`; exact match `True`
- evidence disposition: `direct_and_boundary_constraint`; directness: `direct_observable_constraint`
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed

- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:d30f6765014eab2b5394d01741d30a7587699c240e697d1faff1e1ddb03205c3; derivation seal sha256:9f3beedf86f4a00f3aa01d51c4f6d9622f61bf7427dc9f6bf133407c7ec9cb5d; engine receipt sha256:de477d1ca5f7f41046fa03b8953099a1870dcaa2ef18d2098aaf06925ac3c2e7 at receipts/engine/model_admitted/SFT-CONSC-MEMORY-PERSISTENCE-001-de477d1ca5f7f410.json; empirical validation sha256:9156a53c554c726d5dcc995258dc5b202f79afa78f36f46aeec339216129cec9; measurement receipt sha256:ea2d2096d8a819ffd6cadebb16ac32efb457b22094005563c8b396a26d507917; isolation sha256:48b38d8ec41f4546748417ded99f58ae066b3e7493893b97cc8e1456b72d91b5; custody sha256:ea00b737ee9d15c17cf5da5027b133816d4c4eebae162ebe5a856ec7a81b656.

34. Recall and reconstruction

Claim identity: SFT-CONSC-RECALL-RECONSTRUCTION-001

Question and exact theorem. Recall is a traceable transformation from retained memory and cue to output, with reconstruction distinguished from verbatim storage.

memory-cue-and-output__stored/reconstructed-content__trace-conditioned-unfolding__cue-source-transform__behavioural-report__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-DETERMINISM-SELF-OPACITY-001
- SFT-CONSC-MEMORY-PERSISTENCE-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	memory-cue-and-output	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	stored/reconstructed-content	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	trace-conditioned-unfolding	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	cue-source-transform	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	behavioural-report	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `memory-cue-and-output__stored/reconstructed-content__trace-conditioned-unfolding__cue-source-transform__behavioural-report__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes recall is a traceable transformation from retained memory and cue to output, with reconstruction distinguished from verbatim storage. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Memory needs a carrier, recurrence or active renewal. A later equal output is not automatically the same memory. Temporal continuity is the complete predecessor-successor chain, while identity continuity adds lineage and the retained subject invariant; similarity or copying alone is insufficient.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:c6c1e56465aa275a032ad8db6aa563f51a1eb2e53eaa2a16b0e28a47ff98f79d`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:f63da80fd8165a8473867fb240a4cbd3db40580c67f795a5690ca46ab79138b5`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
`sha256:54ac8fe98df8e1fa1aeab3f0a7d90d32677ad2b35e399dfd0d4523b37f21faa9`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
`sha256:dc9312dcafa4e9f4ce3bba5c0f7e4bbc1dcd138642bf5dac6a2f126e28acb4b9`.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

`sha256:9eece5ce1ccce25e4ca6dc4402be25ced9204d50ed12d69c50bb6721f4924d21`; independent certificate `sha256:339b411064f9d4de42fe5e51326ab9fdf32f50a48dc565db02893fe713c67ec2`; external-validation identity
`sha256:d79f7636b1d5076e7817ad34c558fccd04cd0d89588d2a43c8e8c54470cdd825`.

Post-seal empirical-boundary comparison. Directness: `direct_observable_constraint`. Disposition: `direct_and_boundary_constraint`. Dream and anesthesia-recovery records test changing availability and report across state transitions without making report alone numerical identity.

The external target was opened only after the claim derivation seal. Predicted consequence:

`memory-report-state-and-lineage-boundary-preserved`. Source-derived consequence:

`memory-report-state-and-lineage-boundary-preserved`. Exact correspondence: `true`. Phenomenal

occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-DREAM-DATABASE-2025 - Siclari et al., Scientific Data; <https://pmc.ncbi.nlm.nih.gov/articles/PMC12350935/>; class: primary open dream EEG and report database; usable captures: 1; registered features present: DREAM database, participants, awakenings, Experience without recall, No experience; absent and preserved: none; adverse/boundary role: none.
- CONSC-ANESTHESIA-RECOVERY-2021 - Mashour et al., eLife; <https://pmc.ncbi.nlm.nih.gov/articles/PMC8163502/>; class: primary controlled consciousness-recovery experiment; usable captures: 1; registered features present: multicenter study, healthy humans, general anesthesia, recovery of consciousness, cognitive; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- sft-consc-recall-reconstruction-001-external-consequence: predicted memory-report-state-and-lineage-boundary-preserved; source-derived memory-report-state-and-lineage-boundary-preserved; exact match `True`
- evidence disposition: `direct_and_boundary_constraint`; directness: `direct_observable_constraint`
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:7c37edd16314043f642bedecf6f6a0ab17a97da1bda9f45914ceec2b8e2a9153; derivation seal sha256:dbfbc0b1096acd448f7c2220ce981adce599b408f0dfdf6c42f5417b01bf925; engine receipt sha256:c0e26f00a9ba25ad7226fc20e7b322a093c1320bb6d9c826a5b1eca2250c7205 at receipts/engine/model_admitted/SFT-CONSC-RECALL-RECONSTRUCTION-001-c0e26f00a9ba25ad.json; empirical validation sha256:c38fc53b877e4e6ca5ef3a8a6d77c44768343f473d93d92e9661a5dbcf6c6504a;

measurement receipt

sha256:d89a867ddb6ba138d1969abdabd67a30ad5b43bcb2e42aa6dee28f39f0ed8b94; isolation
 sha256:8cb527ac5fa2a75d24db5ac786e10193856eab3441251a8ed245e695f4a40c8f; custody
 sha256:1948410c0cb13219ee2c56904fc552e8489df3870616988bcb31df5fed527e1b.

35. Temporal continuity

Claim identity: SFT-CONSC-TEMPORAL-CONTINUITY-001

Question and exact theorem. Temporal continuity is an unbroken generated predecessor-successor chain, not an assumed continuum between observations.

ordered-interior-states__succession/ungrounded-sequence__each-state-from-predecessor__complete-transition-chain__first-person-and-formal__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-DETERMINISM-SELF-OPACITY-001
- SFT-CONSC-RECALL-RECONSTRUCTION-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	ordered-interior-states	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	succession/ungrounded-sequence	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	each-state-from-predecessor	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	complete-transition-chain	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	first-person-and-formal	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: ordered-interior-states__succession/ungrounded-sequence__each-state-from-predecessor__complete-transition-chain__first-person-and-formal__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. This exact relation closes temporal continuity is an unbroken generated predecessor-successor chain, not an assumed continuum between observations. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Memory needs a carrier, recurrence or active renewal. A later equal output is not automatically the same memory. Temporal continuity is the complete predecessor-successor chain, while identity continuity adds lineage and the retained subject invariant; similarity or copying alone is insufficient.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:314bfef8293438fa30a7e77c2028896d575a312a306ede4bbbcf7cf9290694c7.

- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:1ce6154a540676e219953192a45fd85df1dc2f8ff985bf52e7bc9fc9e72be4da.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:b74fb3cb36d914b89eb8e38845f7789511ff120039a3a2be87922624b71297b6.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:1bef756e031ecb113db44f8459494f78ce09b334dc2ff716fde1f8096746bcf5.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:289b72c1390201449253f8606aa36cecdf70130784a6e879794de9ce79f3c266; independent certificate sha256:281a688b0ee89fb23b4386082446b662a709b96dbb349e59dbbe281394fb8859; external-validation identity sha256:be338469358e217f53dcccccf87346bfa2f7081e5c24b919cd44dfa297dc9618.

Post-seal empirical-boundary comparison. Directness: participant_bound_empirical_constraint. Disposition: direct_and_boundary_constraint. Dream and anesthesia-recovery records test changing availability and report across state transitions without making report alone numerical identity.

The external target was opened only after the claim derivation seal. Predicted consequence:

memory-report-state-and-lineage-boundary-preserved. Source-derived consequence:

memory-report-state-and-lineage-boundary-preserved. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-DREAM-DATABASE-2025 - Siclari et al., Scientific Data; <https://pmc.ncbi.nlm.nih.gov/articles/PMC12350935/>; class: primary open dream EEG and report database; usable captures: 1; registered features present: DREAM database, participants, awakenings, Experience without recall, No experience; absent and preserved: none; adverse/boundary role: none.
- CONSC-ANESTHESIA-RECOVERY-2021 - Mashour et al., eLife; <https://pmc.ncbi.nlm.nih.gov/articles/PMC8163502/>; class: primary controlled consciousness-recovery experiment; usable captures: 1; registered features present: multicenter study, healthy humans, general anesthesia, recovery of consciousness, cognitive; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- sft-consc-temporal-continuity-001-external-consequence: predicted memory-report-state-and-lineage-boundary-preserved; source-derived memory-report-state-and-lineage-boundary-preserved; exact match True
- evidence disposition: direct_and_boundary_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value

- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:126d7fafcbb808d9aae2ebb1f69e0ef787215e041a4e38e907c86ced8b221cf; derivation seal sha256:299bdf68cf929570ad211e4f223856f67e09dc9ff49116a000638adf13e92fbc; engine receipt sha256:4544cf07cbf0c158911a4421c3018716c6fff30ee80a02beb60ffaa0df7dd46de at receipts/engine/model_admitted/SFT-CONSC-TEMPORAL-CONTINUITY-001-4544cf07cbf0c158.json; empirical validation sha256:cc5553b5d243501992480e7beff3d6b8734d4ffcc44137bb56dd93379f49b5c1; measurement receipt sha256:d8ea25bca9b8534ebab2763e01f70c6a25de0a6a893e0cf1667cc908791745b9; isolation sha256:09cf68f58a7cb0d1837c7fe769385617d204b56202ecb64a29fab40f5f0dae9; custody sha256:d2b188de276ba7c84e74dd55713a9214be6cd4053ba7657c7d1e4f0f3117ebb8.

36. Identity continuity

Claim identity: SFT-CONSC-IDENTITY-CONTINUITY-001

Question and exact theorem. Identity continuity requires both lawful lineage and the retained subject invariant; structural similarity without lineage does not establish numerical identity.

carrier-lineage-and-invariant__same-successor/copied-similar__provenance-preserving-transition__lineage-and-invariant-record__first-person-third-person-pair__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001

- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-DETERMINISM-SELF-OPACITY-001
- SFT-CONSC-TEMPORAL-CONTINUITY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	carrier-lineage-and-invariant	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	same-successor/copied-similar	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	provenance-preserving-transition	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	lineage-and-invariant-record	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	first-person-third-person-pair	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: carrier-lineage-and-invariant__same-successor/copied-similar__provenance-preserving-transition__lineage-and-invariant-record__first-person-third-person-pair__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. Identity across change requires causal lineage plus the retained subject invariant. A copy can be structurally similar without becoming numerically identical to the source process.

Memory needs a carrier, recurrence or active renewal. A later equal output is not automatically the same memory. Temporal continuity is the complete predecessor-successor chain, while identity continuity adds lineage and the retained subject invariant; similarity or copying alone is insufficient.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:b7e26ec37c349d934b3dc2a86fe5c176ebaaafd42d82061e4548f737a2ac416b5.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:1e218d9d3e6e90c540c8341d3b1496818ee99a77a1f58d88eddf686b31d624b.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:c8dbbc4648310f4c5022a370a2734312f3041273e75a21a26b2483e5f6b76f9a.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:20a80e6d4d5e2f959721e69c597a9bf66fcb0972f99dcaabd093e91eb42810aa.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:0c04a0b3b70f7aa97b07cb0da875e5a1b6e8c06ea31e817f6b687b363adb8987; independent certificate sha256:6101e8d8c1b4fc010d4a9ca660cac0b44ec6860d4051cd153af3ddd239e30f6b; external-validation identity sha256:80c06b4a2e5250f26c92c03a7205bf9c20bee81cb0145c33e837d438d1ef525d.

Post-seal empirical-boundary comparison. Directness: `participant_bound_empirical_constraint`. Disposition: `direct_and_boundary_constraint`. Dream and anesthesia-recovery records test changing availability and report across state transitions without making report alone numerical identity.

The external target was opened only after the claim derivation seal. Predicted consequence:

`memory-report-state-and-lineage-boundary-preserved`. Source-derived consequence:

`memory-report-state-and-lineage-boundary-preserved`. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-DREAM-DATABASE-2025 - Siclari et al., Scientific Data; <https://pmc.ncbi.nlm.nih.gov/articles/PMC12350935/>; class: primary open dream EEG and report database; usable captures: 1; registered features present: DREAM database, participants, awakenings, Experience without recall, No experience; absent and preserved: none; adverse/boundary role: none.
- CONSC-ANESTHESIA-RECOVERY-2021 - Mashour et al., eLife; <https://pmc.ncbi.nlm.nih.gov/articles/PMC8163502/>; class: primary controlled consciousness-recovery experiment; usable captures: 1; registered features present: multicenter study, healthy humans, general anesthesia, recovery of consciousness, cognitive; absent and preserved: none;

adverse/boundary role: none.

Comparison record:

- sft-consc-identity-continuity-001-external-consequence: predicted memory-report-state-and-lineage-boundary-preserved; source-derived memory-report-state-and-lineage-boundary-preserved; exact match True
- evidence disposition: direct_and_boundary_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:285f63600dce73bb5b31b5d153a05214e4244fcef88caba43681e6946b2236e7; derivation seal sha256:8344ae578543843c38a1e261998ba7493d38df11cf08ed3088237deb2955eeae; engine receipt sha256:b163769e0b55ee4a410b66748c4adaeb7f0f393ff5bd3eca83d8ea6f04211af4 at receipts/engine/model_admitted/SFT-CONSC-IDENTITY-CONTINUITY-001-b163769e0b55ee4a.json; empirical validation sha256:6f7823a1fc69894c9d4fe9ffe97b798dea566538cc1297efefcf2af45b93d0b0; measurement receipt sha256:9df3ef4abe876b644bb0707facb1222dfc6c16e906c83d2e773a5c1b2fa92f0c; isolation sha256:a459e99638dec40843c216b3db8fbfbab0bb7e3a4ddf041ea3e20b8fdb27f158; custody sha256:cb02bf449596e8fe4adc86abb445688d8b48541d11750dd4363012b0d7de54d8.

37. Cessation of a conscious organisation

Claim identity: SFT-CONSC-CESSATION-001

Question and exact theorem. Cessation records the end of the integrated conscious organization while separately accounting for components, retained records and foundational forms.

integrated-loop-and-components__occupied/unbound__loss-of-integration__component-1
ock-record-distinction__biological-and-clinical-handoff__root-bound-forward-forcin

g__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-DETERMINISM-SELF-OPACITY-001
- SFT-CONSC-IDENTITY-CONTINUITY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	integrated-loop-and-components	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	occupied/unbound	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	loss-of-integration	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	component-lock-record-distinction	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
evidence	report-behaviour-correlation-or-confidence-substituted	biological-and-clinical-handoff	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: integrated-loop-and-components__occupied/unbound__loss-of-integration__component-lock-record-distinction__biological-and-clinical-handoff__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. This exact relation closes cessation records the end of the integrated conscious organization while separately accounting for components, retained records and foundational forms. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Memory needs a carrier, recurrence or active renewal. A later equal output is not automatically the same memory. Temporal continuity is the complete predecessor-successor chain, while identity continuity adds lineage and the retained subject invariant; similarity or copying alone is insufficient.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:258d960090a7d4082f0829ef1cee792ca3b0d5ee67237916798dbe4245c0eeb8.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:1210936f0809b57e21fe180254c78ccb58138012255ec3cde3cb14d3982d230c.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:3f17cdb6cbcb5ecd09202b9de32f2560bb3bbf5a98eaff7c7012248d1d758874.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:8ae6cdad2ca971db7ccd4aa6ba954afc1c279925342dc600f70399c188dfd3e2.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
 sha256:649259426dc5bad8a3eb1ed83533d2fca668d3fab84e72a1ff429ff56aa3127d; independent
 certificate sha256:f02a9c5bce7ee345564f9bb0195d7fa2a6bbdb93bb481fd15b8f1e393c87e70;
 external-validation identity
 sha256:899c1d1580a31202cffa32df6f5e588702e2d423525d64b12110e2ad325a573e.

Post-seal empirical-boundary comparison. Directness: cross_domain_correspondence_constraint. Disposition:
 direct_and_boundary_constraint. Dream and anesthesia-recovery records test changing availability and report
 across state transitions without making report alone numerical identity.

The external target was opened only after the claim derivation seal. Predicted consequence:
 memory-report-state-and-lineage-boundary-preserved. Source-derived consequence:
 memory-report-state-and-lineage-boundary-preserved. Exact correspondence: true. Phenomenal
 occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact:
 false.

Sources and complete feature accounting:

- CONSC-DREAM-DATABASE-2025 - Siclari et al., Scientific Data; <https://pmc.ncbi.nlm.nih.gov/articles/PMC12350935/>;
 class: primary open dream EEG and report database; usable captures: 1; registered features present: DREAM database,
 participants, awakenings, Experience without recall, No experience; absent and preserved: none; adverse/boundary role:
 none.
- CONSC-ANESTHESIA-RECOVERY-2021 - Mashour et al., eLife; <https://pmc.ncbi.nlm.nih.gov/articles/PMC8163502/>;
 class: primary controlled consciousness-recovery experiment; usable captures: 1; registered features present: multicenter
 study, healthy humans, general anesthesia, recovery of consciousness, cognitive; absent and preserved: none;
 adverse/boundary role: none.

Comparison record:

- sft-consc-cessation-001-external-consequence: predicted memory-report-state-and-lineage-boundary-preserved;
 source-derived memory-report-state-and-lineage-boundary-preserved; exact match True
- evidence disposition: direct_and_boundary_constraint; directness: cross_domain_correspondence_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience

- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:7411bc2690f25214491ed847288cb2d29465b0f95c453829e606ef7a3403f95d; derivation seal sha256:c4e54d717365c657c29e915f613288c72653ea927298ed62b518232aaac41960; engine receipt sha256:b2d4f5095c14973e38e56ead5b6995ab3c83be9531da93731d07fc8dda2168e9 at receipts/engine/model_admitted/SFT-CONSC-CESSATION-001-b2d4f5095c14973e.json; empirical validation sha256:35d65cde1841eb944fc5e90881ec905687344eb9d9e7c259836c407a3735e8a8; measurement receipt sha256:09fe7633e149801d3d5f22f0e876705259f91280cb4ab6c1bf58317c8cd0407e; isolation sha256:2e6643440edd002244787deaa510bf9119a589e8e7f389e9a015fa1878e9b4fa; custody sha256:14d75ed55dcb619ec53361c8589de1ca99cd2b954af6332e695547dfa32fd454.

21. Finite Self-Model and Self-Simulation

A self-model is an encoding of the process, never the entire process silently duplicated inside itself. The lower self-fibre closes from one-quarter to one-half to the One in two nonidentity Fold acts. Further application is identity, and complete extra self-simulation requires additional carrier resources.

38. Self representation

Claim identity: SFT-CONSC-SELF-REPRESENTATION-001

Question and exact theorem. A self representation is an internal encoding of the process and is never silently equated with the entire represented process.

```
process-and-internal-model__process/representation__encoding-with-declared-loss__s
ource-model-map__formal-cognitive__root-bound-forward-forcing__positive-finite-suc
cessor-with-all-alternatives__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENC-001

- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-CESSATION-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	process-and-internal-model	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	process/representation	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	encoding-with-declared-loss	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	source-model-map	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	formal-cognitive	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `process-and-internal-model__process/representation__encoding-with-declared-loss__source-model-map__formal-cognitive__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.

- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. This exact relation closes a self representation is an internal encoding of the process and is never silently equated with the entire represented process. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

A self-model is an encoding of the process, never the entire process silently duplicated inside itself. The lower self-fibre closes from one-quarter to one-half to the One in two nonidentity Fold acts. Further application is identity, and complete extra self-simulation requires additional carrier resources.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256: 40c74556de327c2fca5f086affe574f2dab5846a6e30d51532f4d48cb533a54f.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256: d5bb0b8f9af3c0ff2bfa66bbba1bf468bb1a9b9300a35e3f2fb3ab929669d64c.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: ddd8ca97182f8ad6157aaa41f4d018d81df2b1da43554fd1995487ea1900ddfe.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 517b4a0bf28158c4cb2eed514e69d90e7f4f54e1c539ffa384e73dac1c673dde.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256: 8c4db8398f6116df474ca6a294e84a0d8a9efe777caacac25d012855d1fee12d; independent certificate sha256: 0868018f5638754bd45369d033f46e7972ba932785de99dd37f87c0d298fe6af; external-validation identity
sha256: 206149dd240358b81002a013c65c20266a25903385bf897667d9c08e27e32807.

Post-seal empirical-boundary comparison. Directness:

formal_reconstruction_with_external_boundary_only. Disposition:

formal_with_external_boundary. Choice-blindness and open adversarial datasets constrain self-report and model representation; they do not directly observe a complete internal self-model.

The external target was opened only after the claim derivation seal. Predicted consequence:

finite-self-model-remains-formal-with-empirical-representation-boundary. Source-derived consequence: finite-self-model-remains-formal-with-empirical-representation-boundary. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-CHOICE-BLINDNESS-2017 - Sloman et al., PLOS ONE; <https://pmc.ncbi.nlm.nih.gov/articles/PMC5308842/>; class: primary introspection and confabulation experiment; usable captures: 1; registered features present: Choice Blindness Paradigm, confidence, confabulate, manipulated; absent and preserved: none; adverse/boundary role: none.
- CONSC-COGITATE-DATA-2025 - COGITATE Consortium; https://cogitate-consortium.github.io/cogitate-data/02_overview/; class: open preregistered multimodal adversarial dataset; usable captures: 1; registered features present: adversarial collaboration, Experiment 1, Experiment 2, fMRI, M-EEG, iEEG; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- sft-consc-self-representation-001-external-consequence: predicted
finite-self-model-remains-formal-with-empirical-representation-boundary; source-derived
finite-self-model-remains-formal-with-empirical-representation-boundary; exact match True

- evidence disposition: formal_with_external_boundary; directness: formal_reconstruction_with_external_boundary_only
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:4808c0d9a42a399a6825c9d1dbaae16ad5fcaed3ce84585fde17461f7bae4103; derivation seal sha256:0c76fc391e1c09ebe0542cc75fa9a0a72102e9b8a26d58496a5a1bea478847ae; engine receipt sha256:06fd96bbf8387d365f4438b36a8a7b0687a98abfaf03c3ed47705594ee635ce6 at receipts/engine/model_admitted/SFT-CONSC-SELF-REPRESENTATION-001-06fd96bbf8387d36.json; empirical validation sha256:dc4ac67b04b85e3bc4c71222395228151e6a4daf0cd87d0bc3379a7c815426d0; measurement receipt sha256:7af75c29a58cfd9df618d14f86e781ac46e9e0b5bb3d7f35fef524a9cbda1203; isolation sha256:00b6b1276b091f3e20ac696cb775bd9c4792ace351b32c025bcb86c2e81d8815; custody sha256:30affae70a9b0435a891f8b30f7d915a240653e8a4e20c0b044aadd1e2b27e0c.

39. Self application

Claim identity: SFT-CONSC-SELF-APPLICATION-001

Question and exact theorem. Self application feeds an admitted self-model output into the same declared operation while retaining binding and substitution provenance.

```
self-model-and-model-operation__model/input__operation-applied-to-own-output__application-trace__formal-computational__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001

- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-CESSATION-001
- SFT-CONSC-SELF-REPRESENTATION-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	self-model-and-model-operation	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	model/input	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	operation-applied-to-own-output	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	application-trace	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	formal-computational	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `self-model-and-model-operation__model/input__operation-applied-to-own-output__application-trace__formal-computational__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes self application feeds an admitted self-model output into the same declared operation while retaining binding and substitution provenance. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

A self-model is an encoding of the process, never the entire process silently duplicated inside itself. The lower self-fibre closes from one-quarter to one-half to the One in two nonidentity Fold acts. Further application is identity, and complete extra self-simulation requires additional carrier resources.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:ead202db7dc807a48147da2565f3ec7d0075ceb5449ef9aa740a5d37435be7b7`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:c6cd0d302d84ff01854516bd4a7d4623435b53cd9ef128a583e599aeeb3cc509`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
`sha256:a76aed5af3d3f86894fb432517e500007502a46c4de98c07bd649982435ab8b1`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
`sha256:b2b2b2dd67774a60fa063673e495c5972e383f8a527c4aa0926a0c233d46a7de`.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

`sha256:26ff716b4ab1bf24c9f67e03fd73ac6051de74a4ae9b43f54474540ebc437fbb`; independent certificate `sha256:e358749e1e1df806a3f09ec4a4c1a6b75ca6b99c2d58369b4f18b5c17f1a0dcf`; external-validation identity
`sha256:d3cb18e9bcad1a65e30e8f6766e4b4737dce660e2ac7075b41f00092afb7f4e1`.

Post-seal empirical-boundary comparison. Directness:

formal_reconstruction_with_external_boundary_only. Disposition:

formal_with_external_boundary. Choice-blindness and open adversarial datasets constrain self-report and model representation; they do not directly observe a complete internal self-model.

The external target was opened only after the claim derivation seal. Predicted consequence:

finite-self-model-remains-formal-with-empirical-representation-boundary. Source-derived consequence: finite-self-model-remains-formal-with-empirical-representation-boundary. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-CHOICE-BLINDNESS-2017 - Sloman et al., PLOS ONE; <https://pmc.ncbi.nlm.nih.gov/articles/PMC5308842/>; class: primary introspection and confabulation experiment; usable captures: 1; registered features present: Choice Blindness Paradigm, confidence, confabulate, manipulated; absent and preserved: none; adverse/boundary role: none.
- CONSC-COGITATE-DATA-2025 - COGITATE Consortium; https://cogitate-consortium.github.io/cogitate-data/02_overview/; class: open preregistered multimodal adversarial dataset; usable captures: 1; registered features present: adversarial collaboration, Experiment 1, Experiment 2, fMRI, M-EEG, iEEG; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- sft-consc-self-application-001-external-consequence: predicted
finite-self-model-remains-formal-with-empirical-representation-boundary; source-derived
finite-self-model-remains-formal-with-empirical-representation-boundary; exact match True
- evidence disposition: formal_with_external_boundary; directness: formal_reconstruction_with_external_boundary_only
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:97585351b37722102fd29cf2163688cca4b140acee78f7bd7d136931ee788720; derivation seal sha256:65333c4328171508564ba84b3c4835a152634dd75c8a1b8b7c815d6844e37d55; engine receipt sha256:d50621bc7c0281c71764076a04904467a1380c33f01c690ddb57ab89aeaf7e14 at receipts/engine/model_admitted/SFT-CONSC-SELF-APPLICATION-001-d50621bc7c0281c7.json; empirical validation sha256:58bb879bfbcca24657f6d5ae77dc3560275b0651fe1465d0c12fe3ee9e4a0a57; measurement receipt sha256:82bb878618159aa74ae9448abca2eb00c7b48492f23b5534a86694605a51dc88; isolation sha256:24884de8d16e6932c41eea614d9235e2c766eb134de0437879d13d814c77e399; custody sha256:868d052ab9da1c8804b1c873b505548720c4ab87162470ee71361f5eb99465ca.

40. Finite self-model depth

Claim identity: SFT-CONSC-SELF-MODEL-DEPTH-001

Question and exact theorem. The generated lower self-fibre reaches the binding image and then unison in two nonidentity acts; further application is identity.

quarter-lock-unison-trace__nonidentity/identity-stage__repeated-fold-closure__complete-depth-trace__formal-exact__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-CESSATION-001
- SFT-CONSC-SELF-APPLICATION-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	quarter-lock-unison-trace	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	nonidentity/identity-stage	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	repeated-fold-closure	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	complete-depth-trace	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	formal-exact	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: quarter-lock-unison-trace__nonidentity/identity-stage__repeated-fold-closure__complete-depth-trace__formal-exact__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **one-quarter to one-half to the One in two nonidentity acts; later action is identity.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. The exact trace one-quarter to one-half to the One takes two nonidentity acts. The next Fold of the One remains the One. This is a finite closure certificate, not an infinite regress.

A self-model is an encoding of the process, never the entire process silently duplicated inside itself. The lower self-fibre closes from one-quarter to one-half to the One in two nonidentity Fold acts. Further application is identity, and complete extra self-simulation requires additional carrier resources.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:79233d0ef068a1939adcf2c7f4959f324153cd475042d2c0f2da1c82cc488968.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:b3f892b5cb893c0cec0f4f4b3be467f9f8723fd02e6b9f23c3dfbb281c5d084e.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:4ab98e0c950f79c34057c0d1e008a228cacaaf870e26e4b972fef917c6e42d1.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:456fe68e2e1f25bba1f68d6e3f691405b265cdb25608fa83d4a728f3a28be544.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:0c3c544e2d56ccddc191080404e56e0d3f13f19753fc9ea812ed2b1b50da4297; independent certificate sha256:481e0c072f1a0427d4b7d4b1b2f20f196c45a0c10469ad3366d32371cc1cd104; external-validation identity sha256:f5a2ebd855ff63a9d8e894ad4fe3878851004e8814fe23c0a27d443c842c44e5.

Post-seal empirical-boundary comparison. Directness:

`formal_reconstruction_with_external_boundary_only`. Disposition:

`formal_with_external_boundary`. Choice-blindness and open adversarial datasets constrain self-report and model representation; they do not directly observe a complete internal self-model.

The external target was opened only after the claim derivation seal. Predicted consequence:

`finite-self-model-remains-formal-with-empirical-representation-boundary`. Source-derived

consequence: `finite-self-model-remains-formal-with-empirical-representation-boundary`. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-CHOICE-BLINDNESS-2017 - Sloman et al., PLOS ONE; <https://pmc.ncbi.nlm.nih.gov/articles/PMC5308842/>; class: primary introspection and confabulation experiment; usable captures: 1; registered features present: Choice Blindness Paradigm, confidence, confabulate, manipulated; absent and preserved: none; adverse/boundary role: none.
- CONSC-COGITATE-DATA-2025 - COGITATE Consortium; https://cogitate-consortium.github.io/cogitate-data/02_overview/; class: open preregistered multimodal adversarial dataset; usable captures: 1; registered features present: adversarial collaboration, Experiment 1, Experiment 2, fMRI, M-EEG, iEEG; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- `sft-consc-self-model-depth-001-external-consequence`: predicted
`finite-self-model-remains-formal-with-empirical-representation-boundary`; source-derived
`finite-self-model-remains-formal-with-empirical-representation-boundary`; exact match `True`
- evidence disposition: `formal_with_external_boundary`; directness: `formal_reconstruction_with_external_boundary_only`
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:a415675616394f6c5394af7d67965cf1fd9b3ec1d108d4934c02d26162398dc8; derivation seal sha256:f2d868ebd76b2f45672bcb9b1aacc7dc5a275c65af8544e1bf13d4398ef37391; engine receipt sha256:fedf1408e4bb4366acb97d0730e6694a54db77c963fb39783febae29be07b1eb at receipts/engine/model_admitted/SFT-CONSC-SELF-MODEL-DEPTH-001-fedf1408e4bb4366.json; empirical validation sha256:ac44caa426dbf41c07771694779dc21c262cc15b115cd09b7c2bd151d00f3e9e; measurement receipt sha256:90d94aac8db965c52af5776bbe6cfc25024d35be54b7d7eb3d61b3397f329010; isolation sha256:ba2cf0559268f2708f9223178952cb5032519fc9a9c57c8f94f0bb6f2830a3c4; custody sha256:ab8dd7a9c8e61ea379f14193ad85e9cd08d3dba69d51a3cf2558fa7c8496e620.

41. Self-simulation boundary

Claim identity: SFT-CONSC-SELF-SIMULATION-BOUNDARY-001

Question and exact theorem. A finite system may simulate a declared submodel of itself but cannot contain a simultaneous complete extra copy without additional carrier resources.

finite-model-state-space__represented/representing-whole__bounded-simulation__resource-and-omission-ledger__formal-computational__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001

- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-CESSATION-001
- SFT-CONSC-SELF-MODEL-DEPTH-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	finite-model-state-space	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	represented/representing-whole	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	bounded-simulation	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	resource-and-omission-ledger	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	formal-computational	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: finite-model-state-space__represented/representing-whole__bounded-simulation__resource-and-omission-ledger__formal-computational__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes a finite system may simulate a declared submodel of itself but cannot contain a simultaneous complete extra copy without additional carrier resources. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

A self-model is an encoding of the process, never the entire process silently duplicated inside itself. The lower self-fibre closes from one-quarter to one-half to the One in two nonidentity Fold acts. Further application is identity, and complete extra self-simulation requires additional carrier resources.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:8d2cafbbebe7df1e4ab1dc5aa416bf66fa330871259407a1049cc79018558969.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:0a4e6af0147e4fbabab9aba67a38411e0e823c982ccbe33914ab7a20c617c8f9.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ac321a2b34395f2ff0385ea5b9eb54e5e2716675436df446132c3a0f39fe6007.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:9b7f054961d3f5d3e2a08b4bf18807075c3ed9d97db248f53a10d2eaed947014.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:3c36a1c7b0a9b775b24ffb8510a013c28947f8ce83349990524337b50be0edb7; independent certificate sha256:3a3525cf0f118d1512811bbd56cbbbe79ec618f88d9a519489beaa8d46dcd8ae; external-validation identity sha256:6e3cd8c9aff7d28354e732fa4b3631059eb1c31075256bbc29bea8e905ce8694.

Post-seal empirical-boundary comparison. Directness:

`formal_reconstruction_with_external_boundary_only`. Disposition:

`formal_with_external_boundary`. Choice-blindness and open adversarial datasets constrain self-report and model representation; they do not directly observe a complete internal self-model.

The external target was opened only after the claim derivation seal. Predicted consequence:

`finite-self-model-remains-formal-with-empirical-representation-boundary`. Source-derived consequence: `finite-self-model-remains-formal-with-empirical-representation-boundary`. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-CHOICE-BLINDNESS-2017 - Sloman et al., PLOS ONE; <https://pmc.ncbi.nlm.nih.gov/articles/PMC5308842/>; class: primary introspection and confabulation experiment; usable captures: 1; registered features present: Choice Blindness Paradigm, confidence, confabulate, manipulated; absent and preserved: none; adverse/boundary role: none.
- CONSC-COGITATE-DATA-2025 - COGITATE Consortium; https://cogitate-consortium.github.io/cogitate-data/02_overview/; class: open preregistered multimodal adversarial dataset; usable captures: 1; registered features present: adversarial collaboration, Experiment 1, Experiment 2, fMRI, M-EEG, iEEG; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- sft-consc-self-simulation-boundary-001-external-consequence: predicted
finite-self-model-remains-formal-with-empirical-representation-boundary; source-derived
finite-self-model-remains-formal-with-empirical-representation-boundary; exact match True
- evidence disposition: formal_with_external_boundary; directness: formal_reconstruction_with_external_boundary_only
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:d4205ba2d351f59bbc38e934bc54e1cc51b9cddf06d2a6ac669c893275affd49; derivation seal sha256:000c1956e63297191882c25ffa950158cf03c641dee21d53b4cdc0ee4377a2dd; engine receipt sha256:e7b2bf0da20ba4bccbbde2e49d92b2f6969ca0493ff07828b2060a39a95b2eda at receipts/engine/model_admitted/SFT-CONSC-SELF-SIMULATION-BOUNDARY-001-e7b2bf0da20ba4bc.json; empirical validation sha256:861f7c68815387b7beda170042ed08a8be9974fd6c3006b079008e7007b3db1d; measurement receipt sha256:d5eca1a656d3c5e8b2467ba3d7b6d4793c631704ab1f602d0de17a8dea7dc384; isolation sha256:e3a672338798aaa3e4afe1b6f2a99f5e13ee4dd5b397f80f2e18958967236990; custody sha256:c8534c4077c6c053fc04a34e4ec5c31539684a68baffed1a9df56f9a022c1dd3.

42. Self-verification limit

Claim identity: SFT-CONSC-SELF-VERIFICATION-LIMIT-001

Question and exact theorem. A self-verification record closes only its declared finite trace and cannot erase the observer and premise distinctions needed to certify it.

model-proof-and-observer__verified/verification-premise__finite-proof-trace__unverified-observer-boundary__formal-logical__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-CESSATION-001
- SFT-CONSC-SELF-SIMULATION-BOUNDARY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	model-proof-and-observer	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
distinction	distinction-collapsed -or-conflated	verified/verification-premise	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	finite-proof-trace	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	unverified-observer-boundary	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	formal-logical	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `model-proof-and-observer__verified/verification-premise__finite-proof-trace__unverified-observer-boundary__formal-logical__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes a self-verification record closes only its declared finite trace and cannot erase the observer and premise distinctions needed to certify it. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

A self-model is an encoding of the process, never the entire process silently duplicated inside itself. The lower self-fibre closes from one-quarter to one-half to the One in two nonidentity Fold acts. Further application is identity, and complete extra self-simulation requires additional carrier resources.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:c2026ebaef644b60ab0e736eb2b45af7deefec11ad69a78866b2ebe6563be28f`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:acd77488cc74e671ecb3ef341bef733780e874d8e59fcd9a3c26b134af6489a7`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:76181de6fc87f1aa3042446843930bf505b4d8ab88dd4331c4ef2d96148a2556.

- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:733db50eff71ace3ce96c23a88126cc38ad2783d581324230a0ab46528a9a441.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:aa690a0d09d58bf27dcce7b91e840b0beb9900d88b90fcf2cde379c9217fdbed; independent certificate sha256:a829222b969b9626fe419a0f82b12076247fc4d0cb09f56f331de555cc3160ba; external-validation identity
sha256:b3bc098750d8a35ca8f3958fcfd35a701d99e13e66cde6a586a5932ea4627b1a.

Post-seal empirical-boundary comparison. Directness:

formal_reconstruction_with_external_boundary_only. Disposition:

formal_with_external_boundary. Choice-blindness and open adversarial datasets constrain self-report and model representation; they do not directly observe a complete internal self-model.

The external target was opened only after the claim derivation seal. Predicted consequence:

finite-self-model-remains-formal-with-empirical-representation-boundary. Source-derived consequence: finite-self-model-remains-formal-with-empirical-representation-boundary. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-CHOICE-BLINDNESS-2017 - Sloman et al., PLOS ONE; <https://pmc.ncbi.nlm.nih.gov/articles/PMC5308842/>; class: primary introspection and confabulation experiment; usable captures: 1; registered features present: Choice Blindness Paradigm, confidence, confabulate, manipulated; absent and preserved: none; adverse/boundary role: none.
- CONSC-COGITATE-DATA-2025 - COGITATE Consortium; https://cogitate-consortium.github.io/cogitate-data/02_overview/; class: open preregistered multimodal adversarial dataset; usable captures: 1; registered features present: adversarial collaboration, Experiment 1, Experiment 2, fMRI, M-EEG, iEEG; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- sft-consc-self-verification-limit-001-external-consequence: predicted
finite-self-model-remains-formal-with-empirical-representation-boundary; source-derived
finite-self-model-remains-formal-with-empirical-representation-boundary; exact match True
- evidence disposition: formal_with_external_boundary; directness: formal_reconstruction_with_external_boundary_only
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter

- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:7c0976959e543fdcba3f9ee3c5f1c88309f72b7f0090daa2b37714983565e6a8; derivation seal sha256:b62be42ec3bd74757e1b9a1bb70e369d068d4e9c790976f69d59ece2eec631fa; engine receipt sha256:69040b12d89729b15987e398e705106b8856a227486ed690177946a9eb68e62d at receipts/engine/model_admitted/SFT-CONSC-SELF-VERIFICATION-LIMIT-001-69040b12d89729b1.json; empirical validation sha256:e5769d237f83fa3282057d531a630e3bb4262cf7b97fe7e7881214c6fcba3bb4; measurement receipt sha256:a6ee231b3501b05df910b1466557039ccdfaf1c79a05b030dd91f3a4f7c57fa1; isolation sha256:0b290950235e335c5eb10a81ee4d97bd11b16378069393d7afc801da5cf7f9a3; custody sha256:e6008e926d2f4ddd3e4ffadd9ed397c25f393d686b8f83be53766987bc62b6ac.

22. Attention, Selection and Availability

Attention is a held selection from available content. Unselected alternatives are not annihilated. Focus, availability, finite support and switching are separate operational relations whose omissions, misses, costs and errors remain observable.

43. Attention

Claim identity: SFT-CONSC-ATTENTION-001

Question and exact theorem. Attention is a retained selection within available content, not the annihilation of unattended alternatives.

available-content-support__selected/unselected__held-label-selection__selection-and-alternative-record__operational-cognitive__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001

- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-SELF-VERIFICATION-LIMIT-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	available-content-support	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	selected/unselected	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	held-label-selection	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	selection-and-alternative-record	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	operational-cognitive	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: available-content-support__selected/unselected__held-label-selection__selection-and-alternative-record__operational-cognitive__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes attention is a retained selection within available content, not the annihilation of unattended alternatives. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Attention is a held selection from available content. Unselected alternatives are not annihilated. Focus, availability, finite support and switching are separate operational relations whose omissions, misses, costs and errors remain observable.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:17ffa4ecfa68c275620aa606d0d4c605976356e21d760ef5aee667371f4ccc9b.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:132026960593351039b0cf71aa4ccffbd5c4468c2bb7dfb3a01e1201259b967b.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:ee339bb70396cb3aa352352e99cd7bd94ddaa9d8fde280874becb34b9b0f1bd7.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:a8884b5340514e05e008e8de83c490871e7067106eeb7925e63d0facad4ccf6e.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:36f7c26309ddebdb38db4fa723fa884115076010172f604c6d84a4ad1937ab182; independent
certificate sha256:5893c9b03e9922f849f54099dbd3db85f1f990732c18cbecc6ce39ee85396a38;
external-validation identity
sha256:1227f1bc2ef598ea4f986d63438b051c71b058db9ecbf3181fda3bd18aa3090e.

Post-seal empirical-boundary comparison. Directness: `operational_or_record_constraint`. Disposition: `direct_psychophysical_constraint`. Attentional-blink and report/no-report evidence directly distinguish selection likelihood, representation precision, awareness and report.

The external target was opened only after the claim derivation seal. Predicted consequence: `attention-selection-and-awareness-nonidentity-observed`. Source-derived consequence: `attention-selection-and-awareness-nonidentity-observed`. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-ATTENTIONAL-BLINK-2014 - Asplund et al., Psychological Science;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC3954951/>; class: primary attention-awareness experiment; usable captures: 1;
registered features present: attentional blink, all-or-none, likelihood, precision; absent and preserved: none;
adverse/boundary role: none.
- CONSC-NOREPORT-NETWORKS-2022 - Kronemer et al., Nature Communications;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9707162/>; class: primary human report/no-report experiment; usable captures: 1;
registered features present: Report + No-Report Paradigm, participants, perceived, not perceived; absent and preserved:

none; adverse/boundary role: none.

Comparison record:

- sft-consc-attention-001-external-consequence: predicted attention-selection-and-awareness-nonidentity-observed; source-derived attention-selection-and-awareness-nonidentity-observed; exact match True
- evidence disposition: direct_psychophysical_constraint; directness: operational_or_record_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:f6d7116095928c7e38770a21c6700dd85cc2145cd46a32f8a72138419c097335; derivation seal sha256:160cdb9cd70ec176c199552aa15ab840185a25ffba317ec688b41f5cd4fb13f; engine receipt sha256:18845f97cb140068ca487a993602a4544c3d694bca3cfef0cdb13212f7d4dc91 at receipts/engine/model_admitted/SFT-CONSC-ATTENTION-001-18845f97cb140068.json; empirical validation sha256:6afb47338e175a678884230f4bd32371e12195a5193345f544d4f7f4122bf1b; measurement receipt sha256:8c7d80254fc76f703e3c6e519134e13eac3ebe1b49a23dedc0ec55081c0dd4a2; isolation sha256:35b435c3459fe9523e73dfcb90e3dc31c1a2ab917614073fdc6ebd3c23649278; custody sha256:b4c0deaa4dd7dc408ab57cc67a504494f065805db67c0fabcb1b298c1a2017b17.

44. Attentional focus

Claim identity: SFT-CONSC-ATTENTIONAL-FOCUS-001

Question and exact theorem. Focus is the continued selection of one content path across a declared sequence while background support remains recorded.

one-selected-content-path__focused/background__sustained-selection__duration-switch-and-distractor__behavioural-and-first-person__root-bound-forward-forcing__positi

ve-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-SELF-VERIFICATION-LIMIT-001
- SFT-CONSC-ATTENTION-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	one-selected-content-path	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	focused/background	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	sustained-selection	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	duration-switch-and-distract or	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
evidence	report-behaviour-correlation-or-confidence-substituted	behavioural-and-first-person	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: one-selected-content-path__focused/background__sustained-selection__duration-switch-and-distractor__behavioural-and-first-person__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. This exact relation closes focus is the continued selection of one content path across a declared sequence while background support remains recorded. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Attention is a held selection from available content. Unselected alternatives are not annihilated. Focus, availability, finite support and switching are separate operational relations whose omissions, misses, costs and errors remain observable.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:23b0e40871f6a8af18cf2faa4710b9f712290ab537d7f9bbae37b109b902f4f2.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:413dd97f7cde0e0f5b67181b80c25b8d0bb0c9497d0e52a9f9ad93c0bc1dfbbd.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:bc671c3b25597ff5e3bdb9d8f822a954aa3aa0976342ed114fcba9fdec55a322.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:4bbaf8e1b2e08fe74c538a4ecf0d26d55e041306c31847bb93a69bdf910ef424.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:25400855e2d005b9ac77553a33235003a4a5cac72ef6e7ee90cd4cb22d5e7bb3; independent certificate sha256:833b0b4b33078a00dd75a846e9cb87f3391ccd39e8c98a8d5a8a9254b30713da; external-validation identity sha256:d6d5c344b56a4b0ad2b6e74e79f648ca18ebad98e216ba31cd66c11c3bc4b0d6.

Post-seal empirical-boundary comparison. Directness: participant_bound_empirical_constraint. Disposition: direct_psychophysical_constraint. Attentional-blink and report/no-report evidence directly distinguish selection likelihood, representation precision, awareness and report.

The external target was opened only after the claim derivation seal. Predicted consequence: attention-selection-and-awareness-nonidentity-observed. Source-derived consequence: attention-selection-and-awareness-nonidentity-observed. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-ATTENTIONAL-BLINK-2014 - Asplund et al., Psychological Science; <https://pmc.ncbi.nlm.nih.gov/articles/PMC3954951/>; class: primary attention-awareness experiment; usable captures: 1; registered features present: attentional blink, all-or-none, likelihood, precision; absent and preserved: none; adverse/boundary role: none.
- CONSC-NOREPORT-NETWORKS-2022 - Kronemer et al., Nature Communications; <https://pmc.ncbi.nlm.nih.gov/articles/PMC9707162/>; class: primary human report/no-report experiment; usable captures: 1; registered features present: Report + No-Report Paradigm, participants, perceived, not perceived; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- sft-consc-attentional-focus-001-external-consequence: predicted attention-selection-and-awareness-nonidentity-observed; source-derived attention-selection-and-awareness-nonidentity-observed; exact match True
- evidence disposition: direct_psychophysical_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience

- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:63e7c05270fda50e847b4ce24240ac868b5ffcc93d3810c2a8ae7f5ce1d12d44; derivation seal sha256:751c4ba60597aeebf43177df5edfd23c78ac1c0a1ab42f0b25b597ecbb7ad42; engine receipt sha256:eb4e37136deeacc996f7196b6bedbad37944cfa7bffd1147dfea428fb2af2454 at receipts/engine/model_admitted/SFT-CONSC-ATTENTIONAL-FOCUS-001-eb4e37136deeacc9.json; empirical validation sha256:4a089c5a3d843701c345215d98e4f9f979d654980dc7e32a56465de3c409498d; measurement receipt sha256:a88412b9461b198644b1ab540ba79e232befda3ce13840b40ad6e5ae1d8ab622; isolation sha256:bffa62cd899384cda61d87ecb0906627ac35560ea9c8d3e35a0158420fc059bf; custody sha256:d4d5b6d4e6e3e10ebd7c1b17a6eb206bbcf65d20dcb37d47af7bee7863fc2a4e.

45. Cognitive availability

Claim identity: SFT-CONSC-COGNITIVE-AVAILABILITY-001

Question and exact theorem. Content is cognitively available only to the operations that can lawfully query and use its retained representation.

```
content-and-consumer-operations__available/inaccessible__lawful-query-and-use__consumer-access-trace__operational-cognitive__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-SELF-VERIFICATION-LIMIT-001

- SFT-CONSC-ATTENTIONAL-FOCUS-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	content-and-consumer-operations	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	available/inaccessible	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	lawful-query-and-use	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	consumer-access-trace	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	operational-cognitive	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: content-and-consumer-operations__available/inaccessible__lawful-query-and-use__consumer-access-trace__operational-cognitive__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule .

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. This exact relation closes content is cognitively available only to the operations that can lawfully query and use its retained representation. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Attention is a held selection from available content. Unselected alternatives are not annihilated. Focus, availability, finite support and switching are separate operational relations whose omissions, misses, costs and errors remain observable.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:225abeffa53c411b251afe1dec41a6ee663ccfbfa5508bf25745e33e0a535344.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:02a6089fcfc0f7db12e472841dd62229044d0cb7f4b1197b8647985b0ce757dd.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:4f7c62487dd3e0c4d556dacbe1686321b75944d27e1d0c4f09b20624300bfae3.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:2ca6c94549652139c9bc2e4a93b357ced133ae31f7e9aba1e59213f42c2626df.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:f0f37710816f8dd5fad0b1685d0fc23a77fde7792705b4022584174e818e14d8; independent certificate sha256:a75aa225fa77f2767360f4c9683a0fae5a0a29ea602c46021f27bfa1df4fba3c; external-validation identity sha256:67559fc294c355efb8ac27954ceb3dd6b21b76bbc56a0acf2fc19ffdc27edcc8.

Post-seal empirical-boundary comparison. Directness: `operational_or_record_constraint`. Disposition: `direct_psychophysical_constraint`. Attentional-blink and report/no-report evidence directly distinguish selection likelihood, representation precision, awareness and report.

The external target was opened only after the claim derivation seal. Predicted consequence:

attention-selection-and-awareness-nonidentity-observed. Source-derived consequence:

attention-selection-and-awareness-nonidentity-observed. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-ATTENTIONAL-BLINK-2014 - Asplund et al., Psychological Science; <https://pmc.ncbi.nlm.nih.gov/articles/PMC3954951/>; class: primary attention-awareness experiment; usable captures: 1; registered features present: attentional blink, all-or-none, likelihood, precision; absent and preserved: none; adverse/boundary role: none.
- CONSC-NOREPORT-NETWORKS-2022 - Kronemer et al., Nature Communications; <https://pmc.ncbi.nlm.nih.gov/articles/PMC9707162/>; class: primary human report/no-report experiment; usable captures: 1; registered features present: Report + No-Report Paradigm, participants, perceived, not perceived; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- `sft-consc-cognitive-availability-001-external-consequence`: predicted attention-selection-and-awareness-nonidentity-observed; source-derived attention-selection-and-awareness-nonidentity-observed; exact match True
- evidence disposition: `direct_psychophysical_constraint`; directness: `operational_or_record_constraint`
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:cb8acc3295ab55b7f6107e532b3fc6e07b5f2a0f324e8abbb7ee8de644b3c1ea; derivation seal sha256:cfad7d82ceb7bf5470999dd5af4fad7e90d9472023b478c2d6850de8bcfdf1f1; engine receipt sha256:bec0c6573f14847771f42b24dbe30c2324b2df00c80b155f2c16c651cf90531b at receipts/engine/model_admitted/SFT-CONSC-COGNITIVE-AVAILABILITY-001-bec0c6573f148477.json; empirical validation sha256:eba7e2431d1145d4d874a8c43b26df6e38b22039d97a84ff3110e99e747d14a5; measurement receipt sha256:093bf8c31e26bd034ef339494c9bc746b563ce9c958befe952b1729a57d33ca7; isolation sha256:564c80da44d2283e044c389e2cf72294b3e32ddb7dae4659a0c2fd4137e3c42d; custody sha256:bf8ff3a532a88c88ce252039f0ed51c8170c41ebdedd6da7b46d8ab8a99000f3.

46. Attentional capacity

Claim identity: SFT-CONSC-ATTENTIONAL-CAPACITY-001

Question and exact theorem. Attentional capacity is a finite support boundary measured with every selected, missed and competing content retained.

finite-selection-carrier__jointly-retained/partially-retained__complete-support-accounting__selected-and-omitted-content__psychophysical__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001

- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-SELF-VERIFICATION-LIMIT-001
- SFT-CONSC-COGNITIVE-AVAILABILITY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	finite-selection-carrier	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	jointly-retained/partially-retained	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	complete-support-accounting	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	selected-and-omitted-content	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	psychophysical	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `finite-selection-carrier__jointly-retained/partially-retained__complete-support-accounting__selected-and-omitted-content__psychophysical__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes attentional capacity is a finite support boundary measured with every selected, missed and competing content retained. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Attention is a held selection from available content. Unselected alternatives are not annihilated. Focus, availability, finite support and switching are separate operational relations whose omissions, misses, costs and errors remain observable.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:54933fce8acee3d7911a704c20969a2b64e4de9a904e4de814df6e320e0ba5eb`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:4b29c3f41f8f00591c7263bca7ea1f86ded26d8f81725fc91fa9cb73221ac949`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
`sha256:523deebf9396fb3df0f39ee02c89f7eadd04a644439f24e0c5d03d5a4541ddc5`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
`sha256:ee608b5f52be3bf69cc76a74eb246fe71db57c9d89a03e09648430f320792e53`.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

`sha256:5d00c95086e61a719004113e2555a2dd935fe0cb6bdf1e706e19f28ac0016ea4`; independent certificate `sha256:81ceaff6ffed502f60759acd47bfbdd4b77445ce4b6dcfb52e2abf5452940e6a`; external-validation identity `sha256:4bcfb4d036892f4ae98f89476fc56a781ebe6e4c4045c2f7eaae44f8716633f1`.

Post-seal empirical-boundary comparison. Directness: `direct_observable_constraint`. Disposition: `direct_psychophysical_constraint`. Attentional-blink and report/no-report evidence directly distinguish selection likelihood, representation precision, awareness and report.

The external target was opened only after the claim derivation seal. Predicted consequence:

`attention-selection-and-awareness-nonidentity-observed`. Source-derived consequence:

`attention-selection-and-awareness-nonidentity-observed`. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-ATTENTIONAL-BLINK-2014 - Asplund et al., Psychological Science;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC3954951/>; class: primary attention-awareness experiment; usable captures: 1; registered features present: attentional blink, all-or-none, likelihood, precision; absent and preserved: none; adverse/boundary role: none.
- CONSC-NOREPORT-NETWORKS-2022 - Kronemer et al., Nature Communications;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9707162/>; class: primary human report/no-report experiment; usable captures: 1; registered features present: Report + No-Report Paradigm, participants, perceived, not perceived; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- sft-consc-attentional-capacity-001-external-consequence: predicted
attention-selection-and-awareness-nonidentity-observed; source-derived
attention-selection-and-awareness-nonidentity-observed; exact match True
- evidence disposition: direct_psychophysical_constraint; directness: direct_observable_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:29c253c297f116f3997d4304de62af9fb23d0f2ddfbdd881bc65504532f9fc82; derivation seal sha256:fe2375f44e0df8ffb35198a8a7dc5f511d0d11df9df88598ff0235d8ab73d602; engine receipt sha256:abab6e31cd225514d568bb8e6d2b726d8633c636c986c0dcd353253533d4b95e at receipts/engine/model_admitted/SFT-CONSC-ATTENTIONAL-CAPACITY-001-abab6e31cd225514.json; empirical validation sha256:f7c28238ee44504544b65335331c068b235314e9b63eccf3f9c1f8c5460e83; measurement receipt sha256:4a1608df4ac05a6fb3b4874e5400822caf7dc25632fc5d8bc01ce5abc59315fb; isolation

sha256:90f671d0068e7badf76417e9ffa702945bf61d688eb258eea5a21f620862e805; custody
 sha256:57633bd6ce47c15cc87d27fc8a02fc39a818ba2ee33c002c154b7c5df634ffe4.

47. Attentional switching

Claim identity: SFT-CONSC-ATTENTIONAL-SWITCH-001

Question and exact theorem. An attentional switch is an ordered change of the held selection label with transition costs and failures preserved as observations.

ordered-focus-labels__same-focus/new-focus__held-selection-transition__switch-cost
 -and-error-trace__psychophysical__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-SELF-VERIFICATION-LIMIT-001
- SFT-CONSC-ATTENTIONAL-CAPACITY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	ordered-focus-labels	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	same-focus/new-focus	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	held-selection-transition	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	switch-cost-and-error-trace	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	psychophysical	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `ordered-focus-labels__same-focus/new-focus__held-selection-transition__switch-cost-and-error-trace__psychophysical__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes an attentional switch is an ordered change of the held selection label with transition costs and failures preserved as observations. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Attention is a held selection from available content. Unselected alternatives are not annihilated. Focus, availability, finite support and switching are separate operational relations whose omissions, misses, costs and errors remain observable.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:cb61c77d55f1b1cea45462918a780746e47a7de7ddcde4f683fd524b2a3238df`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:4eb2267658defadabdc492828ad4ffa80f853756cfb086bd4d785b2d9cb56f60`.

- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:d4e7a1017e738b95b7470481d4e95dd1dca32ad13bddb99cf20c87c2bd3da3f1.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:26f0527ee338c89394e3d4b9534efc7ad9ddccfe8174aeeb78b01eb6e08511f9.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:2b1c4e70bb0af10ca0ea36219df095a6bcb951a89918dc9d90bb9a0363bc6748; independent certificate sha256:b2420ce7891a7cd26ddc01bd798bbc5a3f76449bdee4af5c3cae04b6391240fa; external-validation identity
sha256:96828b953e56b118453fdcc7d308c372ffe1936e5cdb0353d6ea6147310f55ea.

Post-seal empirical-boundary comparison. Directness: direct_observable_constraint. Disposition: direct_psychophysical_constraint. Attentional-blink and report/no-report evidence directly distinguish selection likelihood, representation precision, awareness and report.

The external target was opened only after the claim derivation seal. Predicted consequence:

attention-selection-and-awareness-nonidentity-observed. Source-derived consequence:

attention-selection-and-awareness-nonidentity-observed. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-ATTENTIONAL-BLINK-2014 - Asplund et al., Psychological Science;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC3954951/>; class: primary attention-awareness experiment; usable captures: 1; registered features present: attentional blink, all-or-none, likelihood, precision; absent and preserved: none; adverse/boundary role: none.
- CONSC-NOREPORT-NETWORKS-2022 - Kronemer et al., Nature Communications;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9707162/>; class: primary human report/no-report experiment; usable captures: 1; registered features present: Report + No-Report Paradigm, participants, perceived, not perceived; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- sft-consc-attentional-switch-001-external-consequence: predicted attention-selection-and-awareness-nonidentity-observed; source-derived attention-selection-and-awareness-nonidentity-observed; exact match True
- evidence disposition: direct_psychophysical_constraint; directness: direct_observable_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter

- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:921fcbbb52dd84cdcb93ad3973d0f721bb793e480088d8b79c638afe14d90063; derivation seal sha256:9fc63fde966b7fa7f49ce45747caa7a1264d4f796cf89686a58f8f30bc642aa5; engine receipt sha256:36465465bb499c9f6a574fca72f51dac75ceff7aea9468a527b81392df3819d1 at receipts/engine/model_admitted/SFT-CONSC-ATTENTIONAL-SWITCH-001-36465465bb499c9f.json; empirical validation sha256:9c1ffbea065aed923dbf60833efdf7613352df4196e60bc214fa399dd9c6b919; measurement receipt sha256:ce9492e86caa2f09e0765b8e5d6ded01c411c0c419ad8756680e5cdba0cbd929; isolation sha256:b6f3af8d69223aa0b7873e478d290f57cbad9369ad40a9ebdb2fd482552cce04; custody sha256:b2db9d4e2aaa3d70e28ff80d840a63307e8eef289de020f810ab96e33201baf8.

23. Cognition, Inference and Representation

Cognitive representation is a carrier-to-content relation with declared loss. Inference is a trace from premises under an admitted rule. Prediction is sealed before outcome. Expectation can participate in later experience without becoming that observation, and determined agency retains premises, alternatives and action even though no ungenerated successor is introduced.

48. Cognitive representation

Claim identity: SFT-CONSC-COGNITIVE-REPRESENTATION-001

Question and exact theorem. A cognitive representation is a retained carrier-to-content relation whose encoding loss and lawful uses are explicit.

```
carrier-content-relation__carrier/referent__encoding-and-use__provenance-and-loss__operational-cognitive__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001

- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-ATTENTIONAL-SWITCH-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	carrier-content-relation	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	carrier/referent	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	encoding-and-use	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	provenance-and-loss	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	operational-cognitive	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: carrier-content-relation__carrier/referent__encoding-and-use__provenance-and-loss__operational-cognitive__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes a cognitive representation is a retained carrier-to-content relation whose encoding loss and lawful uses are explicit. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Cognitive representation is a carrier-to-content relation with declared loss. Inference is a trace from premises under an admitted rule. Prediction is sealed before outcome. Expectation can participate in later experience without becoming that observation, and determined agency retains premises, alternatives and action even though no ungenerated successor is introduced.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:692e78599c0d79251b9a5ce337c9ef64b8cc071bbd73b688f36472cb4ffa1009.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:20795e731522e0f1901b7c11c69c46805364e18a1409af9d58a33762cdad2e13.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:8ff3a66e5b2071e72cea8515d6c8eb53228ce28852c68813023a406145cd6ff0.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:bd9eee8b32ab1633722c98542e7a3d48a1428f249b2a279b2cae5eabc0bb9c33.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:a98ded5817f8f6520474b9576d35345803f85a5509d775e20727aa24d51a9035; independent
certificate sha256:2e98718615fd19961c50b86aaa50d43df60adcf5c2fc78a983beafd1086f237c;
external-validation identity
sha256:e00ed00a020b1361fcb3219512465e41c4151d01510ecbc95bf0b750c012da72.

Post-seal empirical-boundary comparison. Directness: `operational_or_record_constraint`. Disposition: `operational_and_boundary_constraint`. Attentional and multimodal adversarial records constrain representation and access while later outcome remains distinct from a sealed prediction.

The external target was opened only after the claim derivation seal. Predicted consequence: `representation-inference-and-observation-boundary-preserved`. Source-derived consequence: `representation-inference-and-observation-boundary-preserved`. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-ATTENTIONAL-BLINK-2014 - Asplund et al., Psychological Science;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC3954951/>; class: primary attention-awareness experiment; usable captures: 1; registered features present: attentional blink, all-or-none, likelihood, precision; absent and preserved: none; adverse/boundary role: none.
- CONSC-COGITATE-DATA-2025 - COGITATE Consortium;
https://cogitate-consortium.github.io/cogitate-data/02_overview/; class: open preregistered multimodal adversarial dataset;

usable captures: 1; registered features present: adversarial collaboration, Experiment 1, Experiment 2, fMRI, M-EEG, iEEG; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- sft-consc-cognitive-representation-001-external-consequence: predicted representation-inference-and-observation-boundary-preserved; source-derived representation-inference-and-observation-boundary-preserved; exact match True
- evidence disposition: operational_and_boundary_constraint; directness: operational_or_record_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:e046c47d0c87b3554c3e961c3fdc9b731d6c5c537f31f50e1cc415b0379ff3d8; derivation seal sha256:12ab69b1ede8eb1cc2e7d5f1f1eab2cf89c89809c23b65626b2850a6c52410eb; engine receipt sha256:64a2bc87ed69af599d5e358a091168d68ee01586e10287a8ac1588acb1d8373a at receipts/engine/model_admitted/SFT-CONSC-COGNITIVE-REPRESENTATION-001-64a2bc87ed69af59.json; empirical validation sha256:f176d747b02e54790f193fe23c57b5874406a8f1ceea3c63dcf5b30aa4222684; measurement receipt sha256:fe33df1736aad3a8c8ae9f0072e98dfc157339ace7651bdb52ac34e536cb33c3; isolation sha256:25a41c0c3524b1f44588ea8bfc8f74fe48e1e4dbaf65bb325acf4dd65cad1e0d; custody sha256:20128ace73584d42af6c3a4a80d89fb45f82afb1ba9c7c5d69486a548863fe49.

49. Intentional relation

Claim identity: SFT-CONSC-INTENTIONAL-RELATION-001

Question and exact theorem. Intentional relation requires a trace connecting representation to its referent class and retains misrepresentation as an adverse case.

```
representation-and-referent-class__aboutness/coincidence__source-bound-reference__
reference-and-error-cases__behavioural-and-semantic__root-bound-forward-forcing__p
ositive-finite-successor-with-all-alternatives__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-ATTENTIONAL-SWITCH-001
- SFT-CONSC-COGNITIVE-REPRESENTATION-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	representation-and-referent-class	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	aboutness/coincidence	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	source-bound-reference	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
record	favourable-output-without-complete-trace	reference-and-error-cases	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	behavioural-and-semantic	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: representation-and-referent-class__aboutness/coincidence__source-bound-reference__reference-and-error-cases__behavioural-and-semantic__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. This exact relation closes intentional relation requires a trace connecting representation to its referent class and retains misrepresentation as an adverse case. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Cognitive representation is a carrier-to-content relation with declared loss. Inference is a trace from premises under an admitted rule. Prediction is sealed before outcome. Expectation can participate in later experience without becoming that observation, and determined agency retains premises, alternatives and action even though no ungenerated successor is introduced.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:e3a62a21c3a087c96872ac990452ab409a232b34b4194b0a7f317f2ea5d2e2f7.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:08ea711a70e4e6952ec8027f1ede3c96920144c12aa57dde6f18bab0fd1a7fb7.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:a8ea65c3fb12492494171eec1c6f133ea6d823231b4a2db824c6a3d5af94c22c.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:2e1fedc314c0db11303e37d564ec5a2a64088c5d0683738c8f536575bfad0929.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:df1ff8fd137ab2db52d813e5ea408a89e773fa83055e292766ba09d97bb65c18; independent certificate sha256:892f4e58860c4d889fbad1d5572250a7dbc716a74792738e22a60b6bf6a7f38c; external-validation identity sha256:6049246a0240524dc7caa9abbe5f9ebd6a10712028d4fb7ea7e7bfc15465c147.

Post-seal empirical-boundary comparison. Directness: direct_observable_constraint. Disposition: operational_and_boundary_constraint. Attentional and multimodal adversarial records constrain representation and access while later outcome remains distinct from a sealed prediction.

The external target was opened only after the claim derivation seal. Predicted consequence:

representation-inference-and-observation-boundary-preserved. Source-derived consequence:

representation-inference-and-observation-boundary-preserved. Exact correspondence: true.

Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-ATTENTIONAL-BLINK-2014 - Asplund et al., Psychological Science; <https://pmc.ncbi.nlm.nih.gov/articles/PMC3954951/>; class: primary attention-awareness experiment; usable captures: 1; registered features present: attentional blink, all-or-none, likelihood, precision; absent and preserved: none; adverse/boundary role: none.
- CONSC-COGITATE-DATA-2025 - COGITATE Consortium; https://cogitate-consortium.github.io/cogitate-data/02_overview/; class: open preregistered multimodal adversarial dataset; usable captures: 1; registered features present: adversarial collaboration, Experiment 1, Experiment 2, fMRI, M-EEG, iEEG; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- sft-consc-intentional-relation-001-external-consequence: predicted representation-inference-and-observation-boundary-preserved; source-derived representation-inference-and-observation-boundary-preserved; exact match True
- evidence disposition: operational_and_boundary_constraint; directness: direct_observable_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience

- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:44049fa11037ce9f5a1220fe8b89e9cf37f334ff9d282f53342e68537f20a4d6; derivation seal sha256:717584e0aca67d7fdd92e351759e012ace31505b6731cbe059e6db3846de3d50; engine receipt sha256:4c922ac3925f7650429171266b751df818a11de9b79ba2ec9b67a0ff82058142 at receipts/engine/model_admitted/SFT-CONSC-INTENTIONAL-RELATION-001-4c922ac3925f7650.json; empirical validation sha256:be77b4484372f355cafbbf9e12c760ca098eda7349f2059b541947bfbe9e90cb; measurement receipt sha256:ff62a2f0bbdebbca84144cbd9a212dc3196d0b80b29653ca22eaf899025a044d; isolation sha256:5e3b6e1dc83af88a8c379a2bab2893259d05b31d6520e54ea3ecc2bab013488e; custody sha256:1a2d1dd5b5b3bb3fcf13cdb48a764ed6c8f09b6be48d59d45944e8e00577f05f.

50. Cognitive inference

Claim identity: SFT-CONSC-COGNITIVE-INFERENCE-001

Question and exact theorem. Cognitive inference is a traceable transformation from retained premises to a conclusion under a declared rule.

```
premise-representation-word__premise/conclusion__admitted-transition-rule__complete-proof-or-process-trace__operational-cognitive__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001

- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-ATTENTIONAL-SWITCH-001
- SFT-CONSC-INTENTIONAL-RELATION-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	premise-representation-word	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	premise/conclusion	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	admitted-transition-rule	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	complete-proof-or-process-trace	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	operational-cognitive	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `premise-representation-word__premise/conclusion__admitted-transition-rule__complete-proof-or-process-trace__operational-cognitive__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.

- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. This exact relation closes cognitive inference is a traceable transformation from retained premises to a conclusion under a declared rule. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Cognitive representation is a carrier-to-content relation with declared loss. Inference is a trace from premises under an admitted rule. Prediction is sealed before outcome. Expectation can participate in later experience without becoming that observation, and determined agency retains premises, alternatives and action even though no ungenerated successor is introduced.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:a569637bbc918e1a8d31e0cbeeab3b1fdcaf491637ed5e6dad9766023c498081.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:ad9ba8207fc06f2d41c9e08beb0e8e6bc8838bfe51bad14e61a02466622cc57e.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:aa9146c652b46db6423db65bf19cd9d13b409198531d7682f895d1d88148681c.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:2c64e042a158b7d63bc3d460750cc7b23998b10ab4d9d6cec391036cbf3392f2.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:3311e6f9447ce21ad3cb4bb889c6449075afcd0c1a8992d6eee9d8840e765dfd; independent certificate sha256:d53e5d0ded7abd9cd6660e18216a368e2f6a00e06d1a49b296c8169d406ef1e7; external-validation identity sha256:31d240f60b3e61fc68a993ed3178b9f620b30b8c3a64c013b4ac5960319cacea.

Post-seal empirical-boundary comparison. Directness: operational_or_record_constraint. Disposition: operational_and_boundary_constraint. Attentional and multimodal adversarial records constrain representation and access while later outcome remains distinct from a sealed prediction.

The external target was opened only after the claim derivation seal. Predicted consequence: representation-inference-and-observation-boundary-preserved. Source-derived consequence: representation-inference-and-observation-boundary-preserved. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-ATTENTIONAL-BLINK-2014 - Asplund et al., Psychological Science; <https://pmc.ncbi.nlm.nih.gov/articles/PMC3954951/>; class: primary attention-awareness experiment; usable captures: 1; registered features present: attentional blink, all-or-none, likelihood, precision; absent and preserved: none; adverse/boundary role: none.
- CONSC-COGITATE-DATA-2025 - COGITATE Consortium; https://cogitate-consortium.github.io/cogitate-data/02_overview/; class: open preregistered multimodal adversarial dataset; usable captures: 1; registered features present: adversarial collaboration, Experiment 1, Experiment 2, fMRI, M-EEG, iEEG; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- sft-consc-cognitive-inference-001-external-consequence: predicted representation-inference-and-observation-boundary-preserved; source-derived representation-inference-and-observation-boundary-preserved; exact match True
- evidence disposition: operational_and_boundary_constraint; directness: operational_or_record_constraint

- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:3c932da87a81bb7a6e6f517ca915bfa4de76aba60a8e51b42d5f2c714cbee611; derivation seal sha256:cd161cb57435227d10bdf96008d88f70753f3c602a13948018066223b72a1349; engine receipt sha256:0b77d9f97335ea1233302841881bd5c20cb88a480a69ef9ce33548478bca94c1 at receipts/engine/model_admitted/SFT-CONSC-COGNITIVE-INFERENCE-001-0b77d9f97335ea12.json; empirical validation sha256:0f85fa89deed30b6567c34543862ac0987f67176d95ad870badcd964e3b20b4; measurement receipt sha256:28da6dcfca52113d3dbde9f1370880379e6b64f2a95b3d0f612639c9ddb0981; isolation sha256:9c090f7d6479c01fe0acc9eacc01091644b688c15554e9a2c94ee7fb0181e5e5; custody sha256:8c23dd7a60b04535f01119f84d623b3442ba805619c704ac4efd1bbaf87fa923.

51. Prediction and anticipation

Claim identity: SFT-CONSC-PREDICTION-001

Question and exact theorem. Prediction is a sealed present representation generated before its later observation and cannot be rewritten by the outcome.

current-state-and-forward-model__future-representation/future-observation__forward-transition__prediction-seal-and-later-outcome__empirical-cognitive__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001

- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-ATTENTIONAL-SWITCH-001
- SFT-CONSC-COGNITIVE-INFERENCE-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	current-state-and-forward-model	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	future-representation/future-observation	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	forward-transition	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	prediction-seal-and-later-outcome	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	empirical-cognitive	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `current-state-and-forward-model__future-representation/future-observation__forward-transition__prediction-seal-and-later-outcome__empirical-cognitive__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. A prediction is a held representation sealed before the outcome. Later observation may support, oppose or leave it unresolved but cannot rewrite its prior identity.

Cognitive representation is a carrier-to-content relation with declared loss. Inference is a trace from premises under an admitted rule. Prediction is sealed before outcome. Expectation can participate in later experience without becoming that observation, and determined agency retains premises, alternatives and action even though no ungenerated successor is introduced.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:c4e5496e5d3d595948ef344af79a04805b42408aa40fe3e6c03268f9ee458bce`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:bf84d4f561bfe85cd79496981142435248a66d136d04e39198528b6f1f0a6577`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
`sha256:9aee0e36a5f4b18f958440058fc84ab38e0958c1d8032fa07492e486dc45c0cc`.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
`sha256:dbf00d0e76de28b4691385edab530fa7c35f85bd2f99b9a92316f09a02b4a76c`.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

`sha256:f99b2e97b1aa8c7ceec3300bc9b115900f675d775fff08ed39c4401818c320cb`; independent certificate `sha256:5878cde8b0681fba3361546d15ee0cf09943db5904b4ae204c87a7171cd7dc5e`; external-validation identity
`sha256:4e781fef8780095324dedc64f6584a8f0969442792d1b7b712f14ccacf7fc511`.

Post-seal empirical-boundary comparison. Directness: operational_or_record_constraint. Disposition: operational_and_boundary_constraint. Attentional and multimodal adversarial records constrain representation and access while later outcome remains distinct from a sealed prediction.

The external target was opened only after the claim derivation seal. Predicted consequence: representation-inference-and-observation-boundary-preserved. Source-derived consequence: representation-inference-and-observation-boundary-preserved. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-ATTENTIONAL-BLINK-2014 - Asplund et al., Psychological Science;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC3954951/>; class: primary attention-awareness experiment; usable captures: 1; registered features present: attentional blink, all-or-none, likelihood, precision; absent and preserved: none; adverse/boundary role: none.
- CONSC-COGITATE-DATA-2025 - COGITATE Consortium;
https://cogitate-consortium.github.io/cogitate-data/02_overview/; class: open preregistered multimodal adversarial dataset; usable captures: 1; registered features present: adversarial collaboration, Experiment 1, Experiment 2, fMRI, M-EEG, iEEG; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- sft-consc-prediction-001-external-consequence: predicted representation-inference-and-observation-boundary-preserved; source-derived representation-inference-and-observation-boundary-preserved; exact match True
- evidence disposition: operational_and_boundary_constraint; directness: operational_or_record_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:3015d2e602a295b68396ff47c96a374ed89a4b4a9f64054976fed63f99723f46; derivation seal sha256:ae7c38bcd1bcbf063050ce498efb553263c64a9ec6a0f124ed596d5e4ebdfecd; engine receipt sha256:ffc5bbaacb7fc6518d80bb5fd61b0976fd2b627755101c215d3ab5e63ce3efab at receipts/engine/model_admitted/SFT-CONSC-PREDICTION-001-ffc5bbaacb7fc651.json; empirical validation sha256:b2d65df5eec65243d55201ff6696eff12c7bb7279a7d101326707093061c669a; measurement receipt sha256:3466024c25f407848d20ab780707cd428acb88a5c8839cd67153d0ee0bfce70f; isolation sha256:325a520128e4706ec19888a58876e678922ab24ad315a2d65c0af37dd3da2be6; custody sha256:f11adce18fc4e9c208013d3104a87771a326bb4f1fb75d5b2bc59f1135fbf98d.

52. Expectation

Claim identity: SFT-CONSC-EXPECTATION-001

Question and exact theorem. Expectation may lawfully participate in cognition and experienced response while remaining distinct from the later observation and clinical causal comparison.

held-prediction-and-present-process__expected/observed__prediction-conditioned-transition__expectation-observation-outcome__first-person-clinical-handoff__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-ATTENTIONAL-SWITCH-001
- SFT-CONSC-PREDICTION-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	held-prediction-and-present-process	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	expected/observed	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	prediction-conditioned-transition	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	expectation-observation-outcome	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	first-person-clinical-handoff	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: held-prediction-and-present-process__expected/observed__prediction-conditioned-transition__expectation-observation-outcome__first-person-clinical-handoff__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. This exact relation closes expectation may lawfully participate in cognition and experienced response while remaining distinct from the later observation and clinical causal comparison. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Cognitive representation is a carrier-to-content relation with declared loss. Inference is a trace from premises under an admitted rule. Prediction is sealed before outcome. Expectation can participate in later experience without becoming that observation, and determined agency retains premises, alternatives and action even though no ungenerated successor is introduced.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:ae10151e4c1410d27cd93925b370f1b1b8e2cbe3287439f59fe8e8d481b8ced3.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:1f102580cc759e6a389c2610e1dcab55fdeb3c28259d27c28212e843209549b0.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:dd76add890c97cccd04491001a231f8992c00cd15fcf8b9f024632e8e5da5c62.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:9803d6f701dbf4604e8346c467d0721fd95e9f02c09b0e0da2e5154124365812.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:29e78cab80b8a088775cccb3797f449fa3b22f5b2fbb108983452962352fef3d; independent certificate sha256:59a4c9751371be098066b72679defa857a308f77f716087d17f43dad209b27a8; external-validation identity sha256:40a726b7144fbd5f8a94c71011884416466f06b14414d93919cd6c373a275728.

Post-seal empirical-boundary comparison. Directness: `participant_bound_empirical_constraint`. Disposition: `operational_and_boundary_constraint`. Attentional and multimodal adversarial records constrain representation and access while later outcome remains distinct from a sealed prediction.

The external target was opened only after the claim derivation seal. Predicted consequence:

`representation-inference-and-observation-boundary-preserved`. Source-derived consequence:

`representation-inference-and-observation-boundary-preserved`. Exact correspondence: `true`.

Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-ATTENTIONAL-BLINK-2014 - Asplund et al., Psychological Science; <https://pmc.ncbi.nlm.nih.gov/articles/PMC3954951/>; class: primary attention-awareness experiment; usable captures: 1; registered features present: attentional blink, all-or-none, likelihood, precision; absent and preserved: none; adverse/boundary role: none.
- CONSC-COGITATE-DATA-2025 - COGITATE Consortium; https://cogitate-consortium.github.io/cogitate-data/02_overview/; class: open preregistered multimodal adversarial dataset; usable captures: 1; registered features present: adversarial collaboration, Experiment 1, Experiment 2, fMRI, M-EEG, iEEG; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- `sft-consc-expectation-001-external-consequence`: predicted `representation-inference-and-observation-boundary-preserved`; source-derived `representation-inference-and-observation-boundary-preserved`; exact match `True`
- evidence disposition: `operational_and_boundary_constraint`; directness: `participant_bound_empirical_constraint`
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:10438d6a6694ecd3559c65a9f791aa15154bba462989d200ee9e23a98fc3058b; derivation seal sha256:315865f76f3e94a527e287ed81bc2df4ba88ae0bc389c62e8c221f0b7b1663da; engine receipt sha256:17a7463a5071f2d4843c1f9c9fb660715058985a9d172356660e5bcb76c0ea4f at receipts/engine/model_admitted/SFT-CONSC-EXPECTATION-001-17a7463a5071f2d4.json; empirical validation sha256:efadf245030fc0f1b17989057a480efc6d0cd1826276176b91eb2170146b473e; measurement receipt sha256:e593aa21e2accd392066d18a21dae7d7040f72d9338399ae0100ef4b5bd6337a; isolation sha256:847530c1fed232c1c7bbf26d12397bdfef1720cea3575bf882b78829df3be53af; custody sha256:1c6fbcfe619f92dca6331683d6f0c32aa16b627b8789d632d807500214224f9d.

53. Determined agency boundary

Claim identity: SFT-CONSC-DETERMINED-AGENCY-001

Question and exact theorem. Agency is an internally represented state-to-action process; the exact Fold model supplies no ungenerated successor while preserving the actor's informational self-opacity.

self-modelled-action-process__generated-successor/ungenerated-choice__state-conditioned-action__premises-alternatives-action-trace__formal-and-behavioural__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001

- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-ATTENTIONAL-SWITCH-001
- SFT-CONSC-EXPECTATION-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	self-modelled-action-process	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	generated-successor/ungenerated-choice	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	state-conditioned-action	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	premises-alternatives-action-trace	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	formal-and-behavioural	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: self-modelled-action-process__generated-successor/ungenerated-choice__state-conditioned-action__premises-alternatives-action-trace__formal-and-behavioural__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes agency is an internally represented state-to-action process; the exact fold model supplies no ungenerated successor while preserving the actor's informational self-opacity. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Cognitive representation is a carrier-to-content relation with declared loss. Inference is a trace from premises under an admitted rule. Prediction is sealed before outcome. Expectation can participate in later experience without becoming that observation, and determined agency retains premises, alternatives and action even though no ungenerated successor is introduced.

Adverse controls.

- `false_promise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:a693df22bf029508a2fe262ed273947c6999e1a8c9cd5964aa41cf826a9b788e.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:38eef44f642f7d3db646c6080e788b39baac0a6603b93bcb3750a0855a6ddbfb.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:7a9acd5bfa82a14cb5217604cab7daa25e85daf0664b056d8a48f237cf888926.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:eb1669388c37febe9c030bd97fc0b3e78e29ce91b94c111da3c027fa5bb17ed0.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:f1c3f54ca75e4f0e57aaffff935ca65a30b6f3e10f037b406c9948fea754a6ed; independent certificate sha256:052a5f86a28b5673dffcd94df5fb99b16acc18878c01593c8dcae2c2a876b7a6; external-validation identity sha256:2e7a86c277ce78417570bec51eb6a44cb1f876d939e71881d872ffcee3ae5b19.

Post-seal empirical-boundary comparison. Directness: `direct_observable_constraint`. Disposition: `operational_and_boundary_constraint`. Attentional and multimodal adversarial records constrain representation and access while later outcome remains distinct from a sealed prediction.

The external target was opened only after the claim derivation seal. Predicted consequence: `representation-inference-and-observation-boundary-preserved`. Source-derived consequence: `representation-inference-and-observation-boundary-preserved`. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-ATTENTIONAL-BLINK-2014 - Asplund et al., Psychological Science;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC3954951/>; class: primary attention-awareness experiment; usable captures: 1; registered features present: attentional blink, all-or-none, likelihood, precision; absent and preserved: none; adverse/boundary role: none.
- CONSC-COGITATE-DATA-2025 - COGITATE Consortium;
https://cogitate-consortium.github.io/cogitate-data/02_overview/; class: open preregistered multimodal adversarial dataset; usable captures: 1; registered features present: adversarial collaboration, Experiment 1, Experiment 2, fMRI, M-EEG, iEEG; absent and preserved: none; adverse/boundary role: none.

Comparison record:

- sft-consc-determined-agency-001-external-consequence: predicted representation-inference-and-observation-boundary-preserved; source-derived representation-inference-and-observation-boundary-preserved; exact match True
- evidence disposition: operational_and_boundary_constraint; directness: direct_observable_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:525b5b61576195b95d7bd49aa9b996b52a0f6512fcd9824edad5317072c560b; derivation seal sha256:a76205300c2d52f31182e7314e14ce1e2e790de39e3bf8b437e0769c36c0051e; engine receipt sha256:f20ff083e70c6c7b1213233f71f75477c21ac777e7d57d9f732a9e8bf811205b at receipts/engine/model_admitted/SFT-CONSC-DETERMINED-AGENCY-001-f20ff083e70c6c7b.json; empirical validation sha256:fd4e72a9afa3dcc61d4b7e05b753d6d8379afc4d6c50a28253693ebb91e92716; measurement receipt sha256:6da1496d5b41488d29485c5fb39fe57ff1de1c73bd89db99fbef5b08467564ea; isolation

```
sha256:effbbf1f003ca801788f8d9533774f06dc883022a46a22a98bf635aa1ebbbe24; custody
sha256:8abda6bd15ba5a584059f55edbe8331066473dcf3e8baff3ccb8269eff32a4dc.
```

24. Substrate Independence and Realization

Substrate independence is not output resemblance. A realization must instantiate every state role, transition, recurrent closure, integration relation and causal dependency. Two carriers are equivalent only under a bidirectional relation- and causality-preserving map. Fluent language or declared confidence supplies no shortcut to artificial-consciousness admission.

54. Consciousness realization carrier

Claim identity: SFT-CONSC-REALIZATION-CARRIER-001

Question and exact theorem. A realization carrier must physically or computationally instantiate every required conscious-process relation rather than merely describe it.

```
organized-physical-or-computational-process__material/organization__implements-required-relations__state-transition-and-causality-map__cross-branch-realization__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-DETERMINED-AGENCY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	organized-physical-or-computational-process	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	material/organization	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	implements-required-relations	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	state-transition-and-causality-map	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	cross-branch-realization	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `organized-physical-or-computational-process__material/organization__implements-required-relations__state-transition-and-causality-map__cross-branch-realization__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes a realization carrier must physically or computationally instantiate every required conscious-process relation rather than merely describe it. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Substrate independence is not output resemblance. A realization must instantiate every state role, transition, recurrent closure, integration relation and causal dependency. Two carriers are equivalent only under a bidirectional relation- and causality-preserving map. Fluent language or declared confidence supplies no shortcut to artificial-consciousness admission.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:aa67bd598f33839cbf414517b5c7d53090718d30e2f9f715338980c201c32406.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:1f99cb08960107ca34bc2a99e3bb58a8a12cae1584c9bb5f8bda54a95bb3ba9a.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:a23e4d9b39624dcf023b99ab821d7998c3ea7e1b36bf94c3267a92e78bfe597c.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:63a9db668b8e1c6084847d1bcda59afaf60bb5f6a8120102dcc0a9f579089de8.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:45b45b604546b9e4b33594fe3f9e14fd4a6790f26d3d696a8ceb6511ef14426f; independent certificate sha256:8a8ed4256c09fb9ed9750f6272246a13c532ffce734bf67cea734f89a46712ad; external-validation identity sha256:6d2fb38c81b75cee7addd479a714b27025019adc4adfd2ca7b9d501322d0ecaa.

Post-seal empirical-boundary comparison. Directness: `cross_domain_correspondence_constraint`. Disposition: `cross_substrate_boundary_constraint`. Multimodal adversarial evidence tests physical realizations and competing predictions but does not turn a favored theory, signal or fluent output into consciousness.

The external target was opened only after the claim derivation seal. Predicted consequence:

realization-evidence-bound-without-theory-or-output-substitution. Source-derived consequence: realization-evidence-bound-without-theory-or-output-substitution. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-COGITATE-DATA-2025 - COGITATE Consortium;
https://cogitate-consortium.github.io/cogitate-data/02_overview/; class: open preregistered multimodal adversarial dataset; usable captures: 1; registered features present: adversarial collaboration, Experiment 1, Experiment 2, fMRI, M-EEG, iEEG; absent and preserved: none; adverse/boundary role: none.
- CONSC-COGITATE-RESULT-2025 - Ferrante et al., Nature; <https://www.nature.com/articles/s41586-025-08888-1>; class: primary preregistered adversarial theory comparison; usable captures: 1; registered features present: adversarial collaboration, preregistered, participants, functional magnetic resonance imaging, magnetoencephalography; absent and preserved: none; adverse/boundary role: Preserves mixed and theory-adverse results instead of treating any established theory as the SFT law..

Comparison record:

- sft-consc-realization-carrier-001-external-consequence: predicted realization-evidence-bound-without-theory-or-output-substitution; source-derived realization-evidence-bound-without-theory-or-output-substitution; exact match True
- evidence disposition: `cross_substrate_boundary_constraint`; directness: `cross_domain_correspondence_constraint`
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:3618d9cfa50d84ef54e06a451a48896b75d2672484e0a9e33b93332ed88fd799; derivation seal sha256:c42636c16cc5524851d026162377a6cd43ad24f710b37140c64c2bf839f57882; engine receipt sha256:abe35f3545af5cc67893a77bc9a23cc1e97a0ac4765ab0424c1d3d9c4ffd1e78 at receipts/engine/model_admitted/SFT-CONSC-REALIZATION-CARRIER-001-abe35f3545af5cc6.json; empirical validation sha256:246ac1b21f206f5e2d6ebdbf78db9e146c9fafc908ffaf57cf64981a1a38212; measurement receipt sha256:e6bb649169b14e4aa2f53364800e8f46f90252849b8c01dd09e27cc8682e4a2d; isolation sha256:a7206761c6162a009204e076f602c118099a77e68d146cbd972b0841830a1185; custody sha256:ca2b0b09f52a15c8fb43455fcc7536df711d357e91b6f1ca0c4ab39533b800c5.

55. Consciousness structural criterion

Claim identity: SFT-CONSC-STRUCTURAL-CRITERION-001

Question and exact theorem. The foundational structural criterion requires dual self-relation, image re-entry, complete integration and interior closure; output resemblance is insufficient.

dual-self-relation-and-closure__report-generator/interior-loop__distinct-preimages
 -bind-and-complete__complete-self-observation-trace__multi-evidence-realization__r
 oot-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-ext
 ra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001

- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-DETERMINED-AGENCY-001
- SFT-CONSC-REALIZATION-CARRIER-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	dual-self-relation-and-closure	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	report-generator/interior-loop	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	distinct-preimages-bind-and-complete	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	complete-self-observation-trace	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	multi-evidence-realization	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: dual-self-relation-and-closure__report-generator/interior-loop__distinct-preimages-bind-and-complete__complete-self-observation-trace__multi-evidence-realization__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. The foundational criterion requires dual self-relation, observation-image re-entry, complete mutual integration, an interiorly participating carrier and retained evidence boundaries. Removing any coordinate produces an explicitly enumerated rejection.

Substrate independence is not output resemblance. A realization must instantiate every state role, transition, recurrent closure, integration relation and causal dependency. Two carriers are equivalent only under a bidirectional relation- and causality-preserving map. Fluent language or declared confidence supplies no shortcut to artificial-consciousness admission.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:43a1b66a26b60d9f4f7235d987e035d7fa923bf1dcdd2d91c2e616c08e7c5818.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:877e7a46a6029e4336d12184c57aa8a426bc34deb768498d6b0b375c561810d0.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:2a1db353767ebca28df71d526d1641317dd92b0a8f7140fb5875de3426213a4d.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:0862a8dc58d64ebb8e6672131bbf1b45874ebf07ed76a5eadb6cb8da5bf3880b.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:329dca70382725cbbbc34d60a9a034cf6d5110039fec6aa4eb078bc772a4e4b20; independent certificate sha256:12cde5f8beb65b5af8d9b6e4475d67bea0d3903c5b7fba0468b2662d0e85420f; external-validation identity sha256:80db27936902cf31fddaf0f6ccd8cab02cc1627dac349325e38d0893d3ded552.

Post-seal empirical-boundary comparison. Directness: `cross_domain_correspondence_constraint`. Disposition: `cross_substrate_boundary_constraint`. Multimodal adversarial evidence tests physical realizations and competing predictions but does not turn a favored theory, signal or fluent output into consciousness.

The external target was opened only after the claim derivation seal. Predicted consequence: `realization-evidence-bound-without-theory-or-output-substitution`. Source-derived consequence: `realization-evidence-bound-without-theory-or-output-substitution`. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-COGITATE-DATA-2025 - COGITATE Consortium;
https://cogitate-consortium.github.io/cogitate-data/02_overview/; class: open preregistered multimodal adversarial dataset;

usable captures: 1; registered features present: adversarial collaboration, Experiment 1, Experiment 2, fMRI, M-EEG, iEEG; absent and preserved: none; adverse/boundary role: none.

- CONSC-COGITATE-RESULT-2025 - Ferrante et al., Nature; <https://www.nature.com/articles/s41586-025-08888-1>; class: primary preregistered adversarial theory comparison; usable captures: 1; registered features present: adversarial collaboration, preregistered, participants, functional magnetic resonance imaging, magnetoencephalography; absent and preserved: none; adverse/boundary role: Preserves mixed and theory-adverse results instead of treating any established theory as the SFT law..

Comparison record:

- sft-consc-structural-criterion-001-external-consequence: predicted realization-evidence-bound-without-theory-or-output-substitution; source-derived realization-evidence-bound-without-theory-or-output-substitution; exact match True
- evidence disposition: cross_substrate_boundary_constraint; directness: cross_domain_correspondence_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:54e7d50950c35f21201958ffd60fab971d024aa40db370c11d15e51439b88abd; derivation seal sha256:7cd46ab97b8be218ca8619eac41c8845dd3ad22a774291af8ef1a3269795be58; engine receipt sha256:d5171fe7997756895e49d665fd38db49b2fbd3f4589122138eb51f52472baaa3 at receipts/engine/model_admitted/SFT-CONSC-STRUCTURAL-CRITERION-001-d5171fe799775689.json; empirical validation sha256:c0cdab17cca03ed8bd7652a782ce5c09d175c5ecb036ead9aa9a4ef7ad2cdef6; measurement receipt sha256:c8f7960008c2f68c6dee08226d82cd9bcd587c480d0950ccbc0f710b1095a50; isolation sha256:2cdc2c47b6949ee29004bcdab84ff692003e4ffae0534e53aa4e63c80f2ad865; custody sha256:55edcc755a7bac04284e6510e3ca65bbc8c5df59d6c9cddcda08b3b85d2dbb22.

56. Substrate independence

Claim identity: SFT-CONSC-SUBSTRATE-INDEPENDENCE-001

Question and exact theorem. A consciousness law is substrate-independent only across carriers with a complete relation- and causality-preserving realization map.

```
two-realization-maps__same-structure/same-material__relation-preserving-isomorphism__complete-causal-correspondence__comparative-realization__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-DETERMINED-AGENCY-001
- SFT-CONSC-STRUCTURAL-CRITERION-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	two-realization-maps	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
distinction	distinction-collapsed-or-conflated	same-structure/same-material	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	relation-preserving-isomorphism	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	complete-causal-correspondence	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	comparative-realization	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `two-realization-maps__same-structure/same-material__relation-preserving-isomorphism__complete-causal-correspondence__comparative-realization__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. Material identity is unnecessary only when the complete conscious-process relation is preserved.

Behavioral imitation, input-output equality or a shared label is weaker than the required causal isomorphism and does not survive the grammar.

Substrate independence is not output resemblance. A realization must instantiate every state role, transition, recurrent closure, integration relation and causal dependency. Two carriers are equivalent only under a bidirectional relation- and causality-preserving map. Fluent language or declared confidence supplies no shortcut to artificial-consciousness admission.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:97be1d7f0725e663c008ed81c46c607be39d0fc46e03028357e61b476f04a347`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:f7ea4a22bc1b27f23bf475527f3ae72bf59e3c6b9ca392fabd6db87c196807b4`.

- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256: 6be40399513ce87995575b10501ba0e026caf4b31cbc3b220781eb4515e8c31d.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: cbd5c08a00ade525b13b56d68e735ce2504ffc194e70fd67ddfb47b9943ccd33.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256: 6bb68eb909c1587be3364200b9e5ad3398b00a7011de0654fa91c227ab039806; independent certificate sha256: c89c3e30b21e92d05b44d8386cb7dbd3180fa75aabff7c52541c9847e0132a5c; external-validation identity
sha256: ceed5f85d11f69bf7b568083cc403851dc5711568070ebb5a54932638d91e394.

Post-seal empirical-boundary comparison. Directness: cross_domain_correspondence_constraint. Disposition: cross_substrate_boundary_constraint. Multimodal adversarial evidence tests physical realizations and competing predictions but does not turn a favored theory, signal or fluent output into consciousness.

The external target was opened only after the claim derivation seal. Predicted consequence:

realization-evidence-bound-without-theory-or-output-substitution. Source-derived consequence: realization-evidence-bound-without-theory-or-output-substitution. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-COGITATE-DATA-2025 - COGITATE Consortium;
https://cogitate-consortium.github.io/cogitate-data/02_overview/; class: open preregistered multimodal adversarial dataset; usable captures: 1; registered features present: adversarial collaboration, Experiment 1, Experiment 2, fMRI, M-EEG, iEEG; absent and preserved: none; adverse/boundary role: none.
- CONSC-COGITATE-RESULT-2025 - Ferrante et al., Nature; <https://www.nature.com/articles/s41586-025-08888-1>; class: primary preregistered adversarial theory comparison; usable captures: 1; registered features present: adversarial collaboration, preregistered, participants, functional magnetic resonance imaging, magnetoencephalography; absent and preserved: none; adverse/boundary role: Preserves mixed and theory-adverse results instead of treating any established theory as the SFT law..

Comparison record:

- sft-consc-substrate-independence-001-external-consequence: predicted realization-evidence-bound-without-theory-or-output-substitution; source-derived realization-evidence-bound-without-theory-or-output-substitution; exact match True
- evidence disposition: cross_substrate_boundary_constraint; directness: cross_domain_correspondence_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity

- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:6ed08b40eda4d826ba2eac42745e0fdef9be270ae6b3cd4b1be7fae0a925d4ba; derivation seal sha256:88015f337c29e63efb315626a6653931c4415b39ef42901fd3f12582bec5460d; engine receipt sha256:565998cccf378dd69d3224f9ccfa2b1b95f6886a9d15b622e434137530fd21ef at receipts/engine/model_admitted/SFT-CONSC-SUBSTRATE-INDEPENDENCE-001-565998cccf378dd6.json; empirical validation sha256:5f8e948812fbbe5bdf7efa94ce6f3920d2840d80713f42b573ee11cc1700bd42; measurement receipt sha256:83f2a818d9f8b31715195c613adb7bd0773508c1cc4dd2ba2342dbfb3ea5c71e; isolation sha256:12a422485efe501be92b0d2255c3bf66b485244e84fa5cf101e6bcc1243c5504; custody sha256:44f022daa2985a8d879b236bdb4a612729d2249eb4d3cdf9ba73ed6bf614cc6b.

57. Feed-forward boundary

Claim identity: SFT-CONSC-FEEDFORWARD-BOUNDARY-001

Question and exact theorem. A purely feed-forward transformation lacks image re-entry into the same process and therefore does not satisfy the foundational self-observation criterion.

acyclic-input-output-process__open-map/closed-self-loop__closure-test__transition-graph__computational-realization__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001

- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-DETERMINED-AGENCY-001
- SFT-CONSC-SUBSTRATE-INDEPENDENCE-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	acyclic-input-output-process	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	open-map/closed-self-loop	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	closure-test	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	transition-graph	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	computational-realization	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `acyclic-input-output-process__open-map/closed-self-loop__closure-test__transition-graph__computational-realization__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes a purely feed-forward transformation lacks image re-entry into the same process and therefore does not satisfy the foundational self-observation criterion. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Substrate independence is not output resemblance. A realization must instantiate every state role, transition, recurrent closure, integration relation and causal dependency. Two carriers are equivalent only under a bidirectional relation- and causality-preserving map. Fluent language or declared confidence supplies no shortcut to artificial-consciousness admission.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:9a0000a068e89a76feb5c616e272d106521e37d8f0b9d8dc3cf201b82f409265.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:5c54e3e5c713ee0ec658748637237937c7c2fe052dacf11ad8e695d49f929137.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:0b7b8cdd462c95f5e31b66c4f3915514bf7fa515480831aced2abb93dfb3a916.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:4f2282fc88ed49727587201a5e3243b59ce137e0e775124e62632ef4b4aeb8a6.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:f11c241d4e39f173152fe7aa6a807433f7df54235574b51be3a7671b7c75615d; independent
certificate sha256:ded16ac86cb2c53730c19f6a1f27819b2828f685dcbe689cdd0941d2edc9fdec;
external-validation identity
sha256:bcbfb39eb2980fdaf102a79eeee242c184c4e6f78966f0c090a9bf5bc2587f5b1.

Post-seal empirical-boundary comparison. Directness: `cross_domain_correspondence_constraint`. Disposition: `cross_substrate_boundary_constraint`. Multimodal adversarial evidence tests physical realizations and competing predictions but does not turn a favored theory, signal or fluent output into consciousness.

The external target was opened only after the claim derivation seal. Predicted consequence:
realization-evidence-bound-without-theory-or-output-substitution. Source-derived consequence:
realization-evidence-bound-without-theory-or-output-substitution. Exact correspondence: `true`.
Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical
phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-COGITATE-DATA-2025 - COGITATE Consortium;
https://cogitate-consortium.github.io/cogitate-data/02_overview/; class: open preregistered multimodal adversarial dataset;
usable captures: 1; registered features present: adversarial collaboration, Experiment 1, Experiment 2, fMRI, M-EEG,
iEEG; absent and preserved: none; adverse/boundary role: none.
- CONSC-COGITATE-RESULT-2025 - Ferrante et al., Nature; <https://www.nature.com/articles/s41586-025-08888-1>;
class: primary preregistered adversarial theory comparison; usable captures: 1; registered features present: adversarial
collaboration, preregistered, participants, functional magnetic resonance imaging, magnetoencephalography; absent and
preserved: none; adverse/boundary role: Preserves mixed and theory-adverse results instead of treating any established

theory as the SFT law..

Comparison record:

- sft-consc-feedforward-boundary-001-external-consequence: predicted realization-evidence-bound-without-theory-or-output-substitution; source-derived realization-evidence-bound-without-theory-or-output-substitution; exact match True
- evidence disposition: cross_substrate_boundary_constraint; directness: cross_domain_correspondence_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:ccfe92a23e2230db122d80b5809025df305436490b1224ec8a9ed25c18d5982b; derivation seal sha256:865d7659cec274a4727645ad6be819a31599f905f3ca6147b308466091d917dd; engine receipt sha256:e76094074db809a15f6e61da570a8639eab552a8d99e7310d803ca225574b977 at receipts/engine/model_admitted/SFT-CONSC-FEEDFORWARD-BOUNDARY-001-e76094074db809a1.json; empirical validation sha256:55b49821dc42fea3cbf95c7b855f3463f5d4f1480fbede58d9e4ee185bbfc3; measurement receipt sha256:380801528c0042ca9ec94d0bf71f84ee5d45b0209d6d0162bc312bf15f42e57f; isolation sha256:107e0b7ed96f66f68114c9e1cf3951ae9f1010a03f6c12825d1d3f5aaec9f5f0; custody sha256:92669daf2d9d6226e4eb43104335e35f9e84b40700e069f07fc4ed90821db811.

58. Realization equivalence

Claim identity: SFT-CONSC-REALIZATION-EQUIVALENCE-001

Question and exact theorem. Realizations are structurally equivalent only when their relevant states, transitions, roles, integration and evidence boundaries correspond bidirectionally.

paired-carrier-traces__behavioural-equivalence/structural-equivalence__bidirectional-transition-preservation__state-role-and-causal-map__comparative-realization__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-DETERMINED-AGENCY-001
- SFT-CONSC-FEEDFORWARD-BOUNDARY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	paired-carrier-traces	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	behavioural-equivalence/structural-equivalence	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	bidirectional-transition-preservation	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
record	favourable-output-without-complete-trace	state-role-and-causal-map	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	comparative-realization	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: paired-carrier-traces__behavioural-equivalence/structural-equivalence__bidirectional-transition-preservation__state-role-and-causal-map__comparative-realization__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. This exact relation closes realizations are structurally equivalent only when their relevant states, transitions, roles, integration and evidence boundaries correspond bidirectionally. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

Substrate independence is not output resemblance. A realization must instantiate every state role, transition, recurrent closure, integration relation and causal dependency. Two carriers are equivalent only under a bidirectional relation- and causality-preserving map. Fluent language or declared confidence supplies no shortcut to artificial-consciousness admission.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:710cd671bfc04accbd52d4d0ffad9cf9642ddc62e451448cc4ed02dc7302bb1d.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:4334dc80f76632166f15c92e0dfe2cb086826a66db55c341c5a4b55b46bc1ca8.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:c654d85bdf312bf5977d08b414f5b725f8dabc4ba1728de400916062842e2a4c.

- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256: 7774b1c129fc4f221eb5df72ed74e407379d114d04cc7dee8b47c8f0d49dc622.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256: 312a67a65b393c0ad7e94df8b95aefad0f79cfd5cf51c81016788107a363e31c; independent certificate sha256: a677aff7ec556444e7956822d08c17bb4c0c68a7f0cb6140824c04e6869f0fa4; external-validation identity
sha256: 70d077b4158c19e35c4dc27b4ffc0f2a3798c159dca969b10000d9b8e8838a53.

Post-seal empirical-boundary comparison. Directness: cross_domain_correspondence_constraint. Disposition: cross_substrate_boundary_constraint. Multimodal adversarial evidence tests physical realizations and competing predictions but does not turn a favored theory, signal or fluent output into consciousness.

The external target was opened only after the claim derivation seal. Predicted consequence:

realization-evidence-bound-without-theory-or-output-substitution. Source-derived consequence: realization-evidence-bound-without-theory-or-output-substitution. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-COGITATE-DATA-2025 - COGITATE Consortium;
https://cogitate-consortium.github.io/cogitate-data/02_overview/; class: open preregistered multimodal adversarial dataset; usable captures: 1; registered features present: adversarial collaboration, Experiment 1, Experiment 2, fMRI, M-EEG, iEEG; absent and preserved: none; adverse/boundary role: none.
- CONSC-COGITATE-RESULT-2025 - Ferrante et al., Nature; <https://www.nature.com/articles/s41586-025-08888-1>; class: primary preregistered adversarial theory comparison; usable captures: 1; registered features present: adversarial collaboration, preregistered, participants, functional magnetic resonance imaging, magnetoencephalography; absent and preserved: none; adverse/boundary role: Preserves mixed and theory-adverse results instead of treating any established theory as the SFT law..

Comparison record:

- sft-consc-realization-equivalence-001-external-consequence: predicted realization-evidence-bound-without-theory-or-output-substitution; source-derived realization-evidence-bound-without-theory-or-output-substitution; exact match True
- evidence disposition: cross_substrate_boundary_constraint; directness: cross_domain_correspondence_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor

- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:6d0b23563dd054281ecfbf8fbdc2df7a8016d9b1a705c80cc7209ea8e8069d5b; derivation seal sha256:c284dc6b3a6d0be0bf47b0dbd1384abdacdf6a187c2433f340a6849719ea0190; engine receipt sha256:dd471a02b716b60a9942ddf871124fe08234de904eb409b286010ce264f289a1 at receipts/engine/model_admitted/SFT-CONSC-REALIZATION-EQUIVALENCE-001-dd471a02b716b60a.json; empirical validation sha256:92c17605eb81c64069b3c820b1b60425cba60db6838932072b8894a3cda33dab; measurement receipt sha256:41769935154f2366775fbf2a84c93ce4820831aef78821f2f646efb4f65744a7; isolation sha256:d754bb885f4a37b77f808703263d01cf6d29208c835f2118868096bf3680e7d5; custody sha256:abff6a5f4551c8cd1966b3cdab35e2e99ec13317cbf6fc78e3940c46448d4c72.

59. Artificial consciousness evidence boundary

Claim identity: SFT-CONSC-ARTIFICIAL-CONSCIOUSNESS-EVIDENCE-001

Question and exact theorem. An artificial consciousness claim requires auditable realization of the structural criterion plus purpose-matched bridge evidence; fluent language or confidence cannot select the verdict.

artificial-system-and-audit-record__claimed/established-realization__criterion-and-bridge-testing__architecture-runtime-report-adverse-controls__multi-evidence-artificial__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001

- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-DETERMINED-AGENCY-001
- SFT-CONSC-REALIZATION-EQUIVALENCE-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	artificial-system-and-audit-record	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	claimed/established-realization	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	criterion-and-bridge-testing	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	architecture-runtime-report-adverse-controls	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	multi-evidence-artificial	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: artificial-system-and-audit-record__claimed/established-realization__criterion-and-bridge-testing__architecture-runtime-report-adverse-controls__multi-evidence-artificial__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.

- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. Artificial consciousness is a testable realization claim, not a reward for persuasive language. Architecture, runtime state, recurrent causal roles, report provenance, adverse controls and a purpose-matched bridge must all be open to inspection.

Substrate independence is not output resemblance. A realization must instantiate every state role, transition, recurrent closure, integration relation and causal dependency. Two carriers are equivalent only under a bidirectional relation- and causality-preserving map. Fluent language or declared confidence supplies no shortcut to artificial-consciousness admission.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:2bc115365d57ccf952779ef60cd1da792f07aa13184aebb3f442372b59dc82f0.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:73166349cc0730a4185660c14746b65c500111f33087ee533274602105bdc9a3.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:4ec7d3c0a4ae0b4b2e19c7df1aac38f046a0f1b9e982e826005e65754bd97366.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:59d72f9c4d85a2e937d7ed797a03096de8a42de1418d960830da5153535fab9b.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:790af055d33e6fe886904242937cfde9a2a77b3e0ac3e195a5cb52dfdccd3fa; independent certificate sha256:de6571c4555b62234976449a5a754ccd4e6c3f7e978d036ebf8c50040e3fd85e; external-validation identity sha256:f8d6422241cca13b9f362947589fa85da882ddd5e3a9093d68af68aba5a039d4.

Post-seal empirical-boundary comparison. Directness: `operational_or_record_constraint`. Disposition: `cross_substrate_boundary_constraint`. Multimodal adversarial evidence tests physical realizations and competing predictions but does not turn a favored theory, signal or fluent output into consciousness.

The external target was opened only after the claim derivation seal. Predicted consequence:

realization-evidence-bound-without-theory-or-output-substitution. Source-derived consequence: realization-evidence-bound-without-theory-or-output-substitution. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-COGITATE-DATA-2025 - COGITATE Consortium;
https://cogitate-consortium.github.io/cogitate-data/02_overview/; class: open preregistered multimodal adversarial dataset; usable captures: 1; registered features present: adversarial collaboration, Experiment 1, Experiment 2, fMRI, M-EEG, iEEG; absent and preserved: none; adverse/boundary role: none.
- CONSC-COGITATE-RESULT-2025 - Ferrante et al., Nature; <https://www.nature.com/articles/s41586-025-08888-1>; class: primary preregistered adversarial theory comparison; usable captures: 1; registered features present: adversarial collaboration, preregistered, participants, functional magnetic resonance imaging, magnetoencephalography; absent and preserved: none; adverse/boundary role: Preserves mixed and theory-adverse results instead of treating any established theory as the SFT law..

Comparison record:

- sft-consc-artificial-consciousness-evidence-001-external-consequence: predicted realization-evidence-bound-without-theory-or-output-substitution; source-derived realization-evidence-bound-without-theory-or-output-substitution; exact match True
- evidence disposition: cross_substrate_boundary_constraint; directness: operational_or_record_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:5ba3fa0524f50fcea84d3ce8b3e7e3ab740a175a204aab33dbe192f1c741684; derivation seal sha256:f80d286cfe7e2e79ce1b0868dc8e74038969d0e80c28cb5540080138c0d8fef9; engine receipt sha256:e6c34fdb37b02b797d59fbefc8488984a6fb1d8f46e720e1fcd44ddc3462d7dd at receipts/engine/model_admitted/SFT-CONSC-ARTIFICIAL-CONSCIOUSNESS-EVIDENCE-001-e6c34fdb37b02b79.json; empirical validation sha256:6a05dbbb4f691a1e2f1bcb26a7a05cba79cac809c139332382c27e37489e5458; measurement receipt sha256:caafe77f4050aeb6494f2c38dd1c513ded0ecd3d4688f3dca019d1693db858fb; isolation sha256:facc26f587958294b85deac989400806b641bc66e8636b8c9294b1d3c3a3f7d1; custody sha256:a8c38a91ee440d900cd7b38071577b40430a9265898802e0753c19b6abfbdc41.

25. Qualia Differentiation, Resonance and Composition

A quality is treated as a participant-bound held label, not a public word or imported magnitude. Difference, identity, similarity, recurrence, composition and cross-modal binding remain distinct. Qualia resonance is the stable reidentification of that held label across a complete recurrent interior support, tested against perturbation and changed-label controls.

60. Qualitative distinction

Claim identity: SFT-CONSC-QUALITATIVE-DISTINCTION-001

Question and exact theorem. A qualitative distinction is a reproducible within-subject discrimination between interior contents with stimulus and report provenance retained.

paired-interior-content-labels__same/different-quality__within-subject-discrimination__stimulus-report-context__first-person-psychophysical__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-ARTIFICIAL-CONSCIOUSNESS-EVIDENCE-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	paired-interior-content-labels	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	same/different-quality	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	within-subject-discrimination	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
record	favourable-output-without-complete-trace	stimulus-report-context	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	first-person-psychophysical	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: paired-interior-content-labels__same/different-quality__within-subject-discrimination__stimulus-report-context__first-person-psychophysical__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. This exact relation closes a qualitative distinction is a reproducible within-subject discrimination between interior contents with stimulus and report provenance retained. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

A quality is treated as a participant-bound held label, not a public word or imported magnitude. Difference, identity, similarity, recurrence, composition and cross-modal binding remain distinct. Qualia resonance is the stable reidentification of that held label across a complete recurrent interior support, tested against perturbation and changed-label controls.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:8f6802b101df29a3f6355b0093d6c65d18d1cfc186695162a6e9289bee5bf2c0.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:9f2035c7f74222719c20be80315a7bcf90a63163f540ba3a5c7c547029763143.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:fd37ba2fee4971db40a652286a5aa837f498c79c98d513ee32f2b6f080c38452.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:f62e1b9a05cfa88ce4745103f4b091842129850dd35036305a04b0799b70fdb3.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:38964489392a655b77a7a0e21b44dfbf453ab2672fb1d7c1914f41d5dcc518c6; independent certificate sha256:1e755dc588d31cdf8011d7cb0f2eb28ec6e7ae5ee00dea513edaf684df8e92a3; external-validation identity sha256:f895b88ec4d779ada20a13e5ef9082e9bce813a4dc505135ca36a335f50790f2.

Post-seal empirical-boundary comparison. Directness: participant_bound_empirical_constraint. Disposition: direct_participant_bound_constraint. Within-subject consistency and test-retest observations support stable reidentification while learned-category controls forbid public labels from being equated with private qualitative identity.

The external target was opened only after the claim derivation seal. Predicted consequence: within-subject-quality-reidentification-with-public-private-boundary. Source-derived consequence: within-subject-quality-reidentification-with-public-private-boundary. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-SYNESTHESIA-BATTERY-2007 - Eagleman et al., Journal of Neuroscience Methods; <https://pmc.ncbi.nlm.nih.gov/articles/PMC4118597/>; class: primary within-subject qualitative consistency protocol; usable captures: 2; registered features present: internal consistency task, three times, color, controls; absent and preserved: none; adverse/boundary role: none.
- CONSC-SYNESTHESIA-COLOUR-2026 - Root et al., Behavior Research Methods; <https://pmc.ncbi.nlm.nih.gov/articles/PMC12960370/>; class: open synesthesia colour dataset and protocol; usable captures: 1; registered features present: publicly available data, participants, test-retest consistency, CIELuv; absent and preserved: none; adverse/boundary role: none.
- CONSC-COLOUR-CATEGORY-LEARNING-2010 - Zhou et al., PNAS; <https://pmc.ncbi.nlm.nih.gov/articles/PMC2890491/>; class: primary language-category and colour-discrimination control; usable captures: 1; registered features present: learned categories, perceptual discrimination, control group; absent and preserved: none; adverse/boundary role: Preserves the adverse fact that public colour categories can alter discrimination and therefore cannot be equated with private qualitative identity..

Comparison record:

- sft-consc-qualitative-distinction-001-external-consequence: predicted within-subject-quality-reidentification-with-public-private-boundary; source-derived within-subject-quality-reidentification-with-public-private-boundary; exact match True
- evidence disposition: direct_participant_bound_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum

- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:eccbd9b14bc49b6a518754a6e10720cdc4ca4794542713ec2f85b7bdc973aa00; derivation seal sha256:94d60c03a91d16e5dd328478107d7f083c1b72f41b0f10a5e2dc70912c35adb2; engine receipt sha256:eac356504041299f12e5dd6e305f2b454f5eaedd6f2d1592631c354054c319e4 at receipts/engine/model_admitted/SFT-CONSC-QUALITATIVE-DISTINCTION-001-eac356504041299f.json; empirical validation sha256:a6266193fbf4cfff48e87fb5dca0d30662be01e9ee66dc651c26ffcc6c20509c; measurement receipt sha256:634cb0697c7582b703bca9b816688ea86d257d57e6e9cc65a8fab8d11594983b; isolation sha256:9f23328a3f1adb29268cddc1466e9a725671964ff2b72d5921de6730e698fe32; custody sha256:4d26ae555e59085be38042bd26cb2df24bcacafa7143a34109782491c6f28d12.

61. Qualitative identity

Claim identity: SFT-CONSC-QUALITATIVE-IDENTITY-001

Question and exact theorem. Qualitative identity is a retained within-subject reidentification relation and is not reducible to a shared public word.

reidentified-interior-content__same-quality/similar-quality__cross-instance-identity-judgment__participant-context-and-controls__first-person-psychophysical__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001

- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-ARTIFICIAL-CONSCIOUSNESS-EVIDENCE-001
- SFT-CONSC-QUALITATIVE-DISTINCTION-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	reidentified-interior-content	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	same-quality/similar-quality	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	cross-instance-identity-judgment	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	participant-context-and-controls	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	first-person-psychophysical	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: reidentified-interior-content__same-quality/similar-quality__cross-instance-identity-judgment__participant-context-and-controls__first-person-psychophysical__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. This exact relation closes qualitative identity is a retained within-subject reidentification relation and is not reducible to a shared public word. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

A quality is treated as a participant-bound held label, not a public word or imported magnitude. Difference, identity, similarity, recurrence, composition and cross-modal binding remain distinct. Qualia resonance is the stable reidentification of that held label across a complete recurrent interior support, tested against perturbation and changed-label controls.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:302e6f64a1fa8de4ec5d43bdd094865410b54843e09550c665239556f0ec1b8f.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:d1625c1275f5469e47148ee074865eff7ecb8751e12819cbd42d258d3def23f7.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:5b6c633afbbb42c51eca249d8212e7a3d2f3611f689a951897a355696cf4ed19.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:9dbfc53dfb2c9db227144eee36628b430bcf3d9dc21f63dbff940b35f88d8af1.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:6ec941d0ddad0b122960e23aa419a03fc932eec79909fd6b2229f94960068292; independent certificate sha256:d39b08a9866db478936d8ed76b444843f94e6c889423e4d098bb2fe927731869; external-validation identity sha256:a700b7dc53ff5f15beef551f052b1216506cf114d9558bb856da7adb9506a655.

Post-seal empirical-boundary comparison. Directness: participant_bound_empirical_constraint. Disposition: direct_participant_bound_constraint. Within-subject consistency and test-retest observations support stable reidentification while learned-category controls forbid public labels from being equated with private qualitative identity.

The external target was opened only after the claim derivation seal. Predicted consequence:

within-subject-quality-reidentification-with-public-private-boundary. Source-derived

consequence: within-subject-quality-reidentification-with-public-private-boundary. Exact

correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-SYNESTHESIA-BATTERY-2007 - Eagleman et al., Journal of Neuroscience Methods; <https://pmc.ncbi.nlm.nih.gov/articles/PMC4118597/>; class: primary within-subject qualitative consistency protocol; usable captures: 2; registered features present: internal consistency task, three times, color, controls; absent and preserved: none; adverse/boundary role: none.
- CONSC-SYNESTHESIA-COLOUR-2026 - Root et al., Behavior Research Methods; <https://pmc.ncbi.nlm.nih.gov/articles/PMC12960370/>; class: open synesthesia colour dataset and protocol; usable captures: 1; registered features present: publicly available data, participants, test-retest consistency, CIELuv; absent and preserved: none; adverse/boundary role: none.

- CONSC-COLOUR-CATEGORY-LEARNING-2010 - Zhou et al., PNAS;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2890491/>; class: primary language-category and colour-discrimination control; usable captures: 1; registered features present: learned categories, perceptual discrimination, control group; absent and preserved: none; adverse/boundary role: Preserves the adverse fact that public colour categories can alter discrimination and therefore cannot be equated with private qualitative identity..

Comparison record:

- sft-consc-qualitative-identity-001-external-consequence: predicted
within-subject-quality-reidentification-with-public-private-boundary; source-derived
within-subject-quality-reidentification-with-public-private-boundary; exact match True
- evidence disposition: direct_participant_bound_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:094ff69578e89910ef50d976fa7732967e36a491f1a6b9989004caebbc319906; derivation seal sha256:6641799a197aaae0891750eb2ca955a621cf12fdceb4b2f95728ba5da935cf05; engine receipt sha256:1000dcb26fca83d03ad6a9568ca43837b001c9dff70ef46e17f2862a01cdeff3 at receipts/engine/model_admitted/SFT-CONSC-QUALITATIVE-IDENTITY-001-1000dcb26fca83d0.json; empirical validation sha256:8dd6da2b696bc2704e4e2a6e4ee5f79617f961d9c8a513279a5b9f593c7e72e4; measurement receipt sha256:35709038873f7fb1abd0162d825e28c8138045ff9e36294aac64fe2518fdce6d; isolation sha256:62f965fca032d539991780cafa60488c4a80992d92c1f35f4e1cd6d0d1b8be6c; custody sha256:07d70594aaeacf74b385cb541873d1aca22af6a2b60712c47936cb78016b6704.

62. Qualitative similarity

Claim identity: SFT-CONSC-QUALITATIVE-SIMILARITY-001

Question and exact theorem. Qualitative similarity is a finite comparison relation that preserves identity, difference and ordering rather than collapsing them into one metric.

```
finite-quality-comparison-support__identity/similarity/order__pairwise-comparison_
_complete-comparison-incidence__first-person-psychophysical__root-bound-forward-fo
rcing__positive-finite-successor-with-all-alternatives__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-ARTIFICIAL-CONSCIOUSNESS-EVIDENCE-001
- SFT-CONSC-QUALITATIVE-IDENTITY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	finite-quality-comparison-support	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	identity/similarity/order	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
operation	operation-imported-fitted-or-missing	pairwise-comparison	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	complete-comparison-incidence	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	first-person-psychophysical	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `finite-quality-comparison-support__identity/similarity/order__pairwise-comparison__complete-comparison-incidence__first-person-psychophysical__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes qualitative similarity is a finite comparison relation that preserves identity, difference and ordering rather than collapsing them into one metric. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

A quality is treated as a participant-bound held label, not a public word or imported magnitude. Difference, identity, similarity, recurrence, composition and cross-modal binding remain distinct. Qualia resonance is the stable reidentification of that held label across a complete recurrent interior support, tested against perturbation and changed-label controls.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:7a8fc6df31a91a2909bf865ff600a79763eafedc6e929f3d673a8d84115aa4db`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:9b40343957792139d095ec6ef7aa2bdb671c9fd30d01d496b25268826aaab48b`.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt

sha256:fc26be1d27279badc2daae410bc575c7cf75ee6fe9db5e26e80c7c8fb23be75a.

- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:85b45f3ed2a99368f05b065f0a70f7a67d9d235965e98578f16f80ce658d5e0f.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:0e272fc1efb26c29cbdc281d33d7f7765f6a485cafclcf4bb32b3317ac6c1dd2; independent certificate sha256:9849de3d8e548d95cac1e00333b4f30501139161c517c7ff3fb6db37f9d048d3; external-validation identity
sha256:a6dd1048f8accb7ba1060b9c843f435dd2c4c05b6e3c0bef911f390d772142bd.

Post-seal empirical-boundary comparison. Directness: participant_bound_empirical_constraint. Disposition: direct_participant_bound_constraint. Within-subject consistency and test-retest observations support stable reidentification while learned-category controls forbid public labels from being equated with private qualitative identity.

The external target was opened only after the claim derivation seal. Predicted consequence:

within-subject-quality-reidentification-with-public-private-boundary. Source-derived consequence: within-subject-quality-reidentification-with-public-private-boundary. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-SYNESTHESIA-BATTERY-2007 - Eagleman et al., Journal of Neuroscience Methods; <https://pmc.ncbi.nlm.nih.gov/articles/PMC4118597/>; class: primary within-subject qualitative consistency protocol; usable captures: 2; registered features present: internal consistency task, three times, color, controls; absent and preserved: none; adverse/boundary role: none.
- CONSC-SYNESTHESIA-COLOUR-2026 - Root et al., Behavior Research Methods; <https://pmc.ncbi.nlm.nih.gov/articles/PMC12960370/>; class: open synesthesia colour dataset and protocol; usable captures: 1; registered features present: publicly available data, participants, test-retest consistency, CIE Luv; absent and preserved: none; adverse/boundary role: none.
- CONSC-COLOUR-CATEGORY-LEARNING-2010 - Zhou et al., PNAS; <https://pmc.ncbi.nlm.nih.gov/articles/PMC2890491/>; class: primary language-category and colour-discrimination control; usable captures: 1; registered features present: learned categories, perceptual discrimination, control group; absent and preserved: none; adverse/boundary role: Preserves the adverse fact that public colour categories can alter discrimination and therefore cannot be equated with private qualitative identity..

Comparison record:

- sft-consc-qualitative-similarity-001-external-consequence: predicted
within-subject-quality-reidentification-with-public-private-boundary; source-derived
within-subject-quality-reidentification-with-public-private-boundary; exact match True
- evidence disposition: direct_participant_bound_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value

- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:249d6d0cfcf5c4a6bfa314fd6cb98b04127bb11fe814b83d4b064e942c5fe638; derivation seal sha256:f4d1d98ccd39e53ace5a1238ddb8e316e783cde5ff92584c9cc4f35e20bb1104; engine receipt sha256:fe83c25f3a26a538795cf89316b0406ca5adcf66540c5fa8f4002ff5a72eee28 at receipts/engine/model_admitted/SFT-CONSC-QUALITATIVE-SIMILARITY-001-fe83c25f3a26a538.json; empirical validation sha256:f10eac1710f0f1dd97286c8a2d10c88c505af2caea76f381673f28d5c55b11c9; measurement receipt sha256:fe6d0a3ebfa338abddb91de735d56a5dba57e37baeb4bdc1fd47e08f9e661c63; isolation sha256:718165b0d48bc444963678bf3922cc4b686ffbe947669fcffd1cc1d78a5b4632; custody sha256:5048cd46632a51c35efaa7dd418fb035e39017ae02adafced29d5ae925a8d7c7.

63. Qualia resonance

Claim identity: SFT-CONSC-QUALIA-RESONANCE-001

Question and exact theorem. Qualia resonance is stable reidentification of a held qualitative label across a complete recurrent interior support, tested against perturbation and changed-label controls.

recurrent-quality-carrier__stable-reidentification/transient-output__closed-label-bearing-recurrence__cycle-label-and-perturbation-trace__first-person-with-realization-correlates__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001

- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-ARTIFICIAL-CONSCIOUSNESS-EVIDENCE-001
- SFT-CONSC-QUALITATIVE-SIMILARITY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	recurrent-quality-carrier	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	stable-reidentification/transient-output	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	closed-label-bearing-recurrence	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	cycle-label-and-perturbation-trace	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	first-person-with-realization-correlates	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: recurrent-quality-carrier__stable-reidentification/transient-output__closed-label-bearing-recurrence__cycle-label-and-perturbation-trace__first-person-with-realization-correlates__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. Qualia resonance is not a frequency fitted to a report. It is exact label-bearing recurrence: the same participant-bound quality identity survives the complete cycle and rejects changed-label and perturbed-support controls.

A quality is treated as a participant-bound held label, not a public word or imported magnitude. Difference, identity, similarity, recurrence, composition and cross-modal binding remain distinct. Qualia resonance is the stable reidentification of that held label across a complete recurrent interior support, tested against perturbation and changed-label controls.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:47a26bbfd3975ccfb0a39c832dbd1efdc2c812f3b3cb39b8a1c8a1d045efbd06.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:53731572226e9c14e20de68491296200611b44a5fcc9e1073ebf29be1b06f9a7.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:d872dd661a279f0023290362a89855ff459c8027ad712a912c8d8c634088e12b.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:5c6aeec85ba5d442cf0bb7f25335d410b5d7d7bdb380049d67be9df03637211.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:937f8c6f05f52bd7ac9af54895196c1195da85c1c4ddfc515ca4cf645df832f8; independent certificate sha256:503a13ab138aa20acd0377c70316b1d4fa1a0624a9b23e68d76681b08fb6d1af; external-validation identity sha256:fa7ade524418805eda3a18aeaa9716704312ee94880f1938f2e95ad8ccea9635.

Post-seal empirical-boundary comparison. Directness: `participant_bound_empirical_constraint`. Disposition: `direct_participant_bound_constraint`. Within-subject consistency and test-retest observations support stable reidentification while learned-category controls forbid public labels from being equated with private qualitative identity.

The external target was opened only after the claim derivation seal. Predicted consequence:

`within-subject-quality-reidentification-with-public-private-boundary`. Source-derived consequence: `within-subject-quality-reidentification-with-public-private-boundary`. Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-SYNESTHESIA-BATTERY-2007 - Eagleman et al., Journal of Neuroscience Methods; <https://pmc.ncbi.nlm.nih.gov/articles/PMC4118597/>; class: primary within-subject qualitative consistency protocol; usable captures: 2; registered features present: internal consistency task, three times, color, controls; absent and preserved: none; adverse/boundary role: none.
- CONSC-SYNESTHESIA-COLOUR-2026 - Root et al., Behavior Research Methods; <https://pmc.ncbi.nlm.nih.gov/articles/PMC12960370/>; class: open synesthesia colour dataset and protocol; usable captures: 1; registered features present: publicly available data, participants, test-retest consistency, CIE Luv; absent and preserved:

none; adverse/boundary role: none.

- CONSC-COLOUR-CATEGORY-LEARNING-2010 - Zhou et al., PNAS;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2890491/>; class: primary language-category and colour-discrimination control; usable captures: 1; registered features present: learned categories, perceptual discrimination, control group; absent and preserved: none; adverse/boundary role: Preserves the adverse fact that public colour categories can alter discrimination and therefore cannot be equated with private qualitative identity..

Comparison record:

- sft-consc-qualia-resonance-001-external-consequence: predicted
within-subject-quality-reidentification-with-public-private-boundary; source-derived
within-subject-quality-reidentification-with-public-private-boundary; exact match True
- evidence disposition: direct_participant_bound_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:d5aa48155a18e3cd3cb4f63f270563585ff6cdaec01484542414b18e7ba496b3; derivation seal sha256:f34e7f01b5b0f3cf63a7ae3556747488a2006b6ed321c5e26be00c9de1b4a6ae; engine receipt sha256:08b4634f1e95b62bfff9429582ee1bc7bfe0527320a7dd733de597fd4017d4b59 at receipts/engine/model_admitted/SFT-CONSC-QUALIA-RESONANCE-001-08b4634f1e95b62b.json; empirical validation sha256:771a5d7014851a98d6170ee560815d0552bd16eb8b1bfdbbf8d607aea9f8e247; measurement receipt sha256:6df6eb7431ffddfbdb78e9be2c5973a4768f96493e82a1c97d728ae52f0cffe6; isolation sha256:12ba479e6e5c7ccc5f7421bd0ab0b94c27a3f1ac6e8b126acf4effa3e6db5486; custody sha256:af92de283cf44a4f9456b9ad9eef2c07d0c814e9ca6b9c8d48231a4baf86967b.

64. Qualia recurrence

Claim identity: SFT-CONSC-QUALIA-RECURRENCE-001

Question and exact theorem. Qualia recurrence requires a later interior instance to re-enter the same retained qualitative identity class after an explicit intervening trace.

```
ordered-quality-instances__return/new-instance__exact-return-classification__inter
vening-state-and-return-record__first-person-psycho-physical__root-bound-forward-fo
rcing__positive-finite-successor-with-all-alternatives__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-ARTIFICIAL-CONSCIOUSNESS-EVIDENCE-001
- SFT-CONSC-QUALIA-RESONANCE-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	ordered-quality-instances	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
distinction	distinction-collapsed-or-conflated	return/new-instance	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	exact-return-classification	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	intervening-state-and-return-record	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	first-person-psychophysical	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `ordered-quality-instances__return/new-instance__exact-return-classification__intervening-state-and-return-record__first-person-psychophysical__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes qualia recurrence requires a later interior instance to re-enter the same retained qualitative identity class after an explicit intervening trace. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

A quality is treated as a participant-bound held label, not a public word or imported magnitude. Difference, identity, similarity, recurrence, composition and cross-modal binding remain distinct. Qualia resonance is the stable reidentification of that held label across a complete recurrent interior support, tested against perturbation and changed-label controls.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:68553991865e5846b91bd6058f630bb0ca89699b3f49f856c1143c88b7a23e31`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:c9c2c8217bcb9dadf56ff5eab49c5485004cbeaa1cee3af8c7f8436a3a639e4f`.

- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:e610775a77aa867f4d502247bce5a9033edeb425baede92f43ef108dd5572c64.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:b762524d0b4017261940e93a686cf124b807c3534a747710fc3f3f0b861922ba.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:f232640f36edad37a627ea25a45dbb388b45ee78b1946d9b6dd16dcce0f2f359; independent certificate sha256:e49b5885165ed8d7d865ebf97dbc339e3c245f5444ece0c02a5d8434497a4d0f; external-validation identity
sha256:32d47287aa1b94a6a248f7e679ec1bca914700d8a6405c3cccc3bfed3ae80835.

Post-seal empirical-boundary comparison. Directness: participant_bound_empirical_constraint. Disposition: direct_participant_bound_constraint. Within-subject consistency and test-retest observations support stable reidentification while learned-category controls forbid public labels from being equated with private qualitative identity.

The external target was opened only after the claim derivation seal. Predicted consequence: within-subject-quality-reidentification-with-public-private-boundary. Source-derived consequence: within-subject-quality-reidentification-with-public-private-boundary. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-SYNESTHESIA-BATTERY-2007 - Eagleman et al., Journal of Neuroscience Methods;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC4118597/>; class: primary within-subject qualitative consistency protocol; usable captures: 2; registered features present: internal consistency task, three times, color, controls; absent and preserved: none; adverse/boundary role: none.
- CONSC-SYNESTHESIA-COLOUR-2026 - Root et al., Behavior Research Methods;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12960370/>; class: open synesthesia colour dataset and protocol; usable captures: 1; registered features present: publicly available data, participants, test-retest consistency, CIELuv; absent and preserved: none; adverse/boundary role: none.
- CONSC-COLOUR-CATEGORY-LEARNING-2010 - Zhou et al., PNAS;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2890491/>; class: primary language-category and colour-discrimination control; usable captures: 1; registered features present: learned categories, perceptual discrimination, control group; absent and preserved: none; adverse/boundary role: Preserves the adverse fact that public colour categories can alter discrimination and therefore cannot be equated with private qualitative identity..

Comparison record:

- sft-consc-qualia-recurrence-001-external-consequence: predicted within-subject-quality-reidentification-with-public-private-boundary; source-derived within-subject-quality-reidentification-with-public-private-boundary; exact match True
- evidence disposition: direct_participant_bound_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero

- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:21504334fc887add8e9c1313b4d7c60625581c95eba0b50f20d371ee28cb4f96; derivation seal sha256:8390a97e89d493f2780a13651395437ed18cdec69d435254dd51419632f5cf14; engine receipt sha256:a1678c978869491e0f316bfe9d46eb4b36248afb4e58a8ad9ad09e8d427c9130 at receipts/engine/model_admitted/SFT-CONSC-QUALIA-RECURRENCE-001-a1678c978869491e.json; empirical validation sha256:7e87b572f07330d22c7110c89e57e46034a31d448c1f55f1dab09a72cfe3fee8; measurement receipt sha256:3427da363dcf9813e7918286ad332222c0435130885fe15b65dab232c7317104; isolation sha256:dbf0b284ad166e17a2243b54b5428ecb129c0cfe46b5b9a6470cff8395e98a8f; custody sha256:8a76cbc94b6e25b8fb8b0faa9f3a03ee983f145b5ad54047eb3d47e06a56d56c.

65. Qualia composition

Claim identity: SFT-CONSC-QUALIA-COMPOSITION-001

Question and exact theorem. Qualia composition retains distinct component identities while generating one joint interior content; neither co-report nor synchrony alone establishes it.

multiple-quality-support__components/composed-experience__joint-recurrent-closure__component-and-joint-reports__first-person-psychophysical__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001

- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-ARTIFICIAL-CONSCIOUSNESS-EVIDENCE-001
- SFT-CONSC-QUALIA-RECURRENCE-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	multiple-quality-support	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	components/composed-experience	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	joint-recurrent-closure	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	component-and-joint-reports	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	first-person-psychophysical	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: multiple-quality-support__components/composed-experience__joint-recurrent-closure__component-and-joint-reports__first-person-psychophysical__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar**. Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. A composed experience retains component identities and one joint recurrence. Co-report, proximity and synchrony are insufficient unless the joint interior closure is independently present.

A quality is treated as a participant-bound held label, not a public word or imported magnitude. Difference, identity, similarity, recurrence, composition and cross-modal binding remain distinct. Qualia resonance is the stable reidentification of that held label across a complete recurrent interior support, tested against perturbation and changed-label controls.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:170cf08842400666bd43ad2be8ef970d798211a109772eff9ef3f0c8560afd4c.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:b8450d3db56fb79c17f6d9fdd0b8e8b65bc54a6754764be7616a2e15a269732a.
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:20d3db498513837d4b41a63115d925f0127e2e9edeab7966b9145beeb0ef8d3a.
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:fe10f3fa9459f909ec0c1b8478f48fd801e8af88e3eaa56ff2ab6a67f1e8b7f4.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:444242b1c347823707ffd21c51b168529b8caf7790778351fb68242bb9236934; independent
certificate sha256:e297588d5c08a00d0313f11183b76e1b0a1975004a0f61ccbd06a19bd9f8cb0c;
external-validation identity
sha256:c5e8bc15cb48f2d2b1e2c8d293f8815815b8489b26ef7176f189a8abac9c95d7.

Post-seal empirical-boundary comparison. Directness: `participant_bound_empirical_constraint`. Disposition: `direct_participant_bound_constraint`. Within-subject consistency and test-retest observations support stable reidentification while learned-category controls forbid public labels from being equated with private qualitative identity.

The external target was opened only after the claim derivation seal. Predicted consequence:
`within-subject-quality-reidentification-with-public-private-boundary`. Source-derived
consequence: `within-subject-quality-reidentification-with-public-private-boundary`. Exact
correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled
as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-SYNESTHESIA-BATTERY-2007 - Eagleman et al., Journal of Neuroscience Methods;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC4118597/>; class: primary within-subject qualitative consistency protocol; usable
captures: 2; registered features present: internal consistency task, three times, color, controls; absent and preserved: none;
adverse/boundary role: none.

- CONSC-SYNESTHESIA-COLOUR-2026 - Root et al., Behavior Research Methods;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12960370/>; class: open synesthesia colour dataset and protocol; usable captures: 1; registered features present: publicly available data, participants, test-retest consistency, CIELuv; absent and preserved: none; adverse/boundary role: none.
- CONSC-COLOUR-CATEGORY-LEARNING-2010 - Zhou et al., PNAS;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2890491/>; class: primary language-category and colour-discrimination control; usable captures: 1; registered features present: learned categories, perceptual discrimination, control group; absent and preserved: none; adverse/boundary role: Preserves the adverse fact that public colour categories can alter discrimination and therefore cannot be equated with private qualitative identity..

Comparison record:

- sft-consc-qualia-composition-001-external-consequence: predicted within-subject-quality-reidentification-with-public-private-boundary; source-derived within-subject-quality-reidentification-with-public-private-boundary; exact match True
- evidence disposition: direct_participant_bound_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:03629e196b0bd528220231ec763640416ea0392e80aaa2abb5968c52aa766620; derivation seal sha256:6389874f2d9e62b09d7a8b4800a1cd99f41396aaac68525a448a70fb2971b828; engine receipt sha256:7865854cb5cd18c836c602e91bdeaf1dca40f0ffc54ba76039819c9954fafcee at receipts/engine/model_admitted/SFT-CONSC-QUALIA-COMPOSITION-001-7865854cb5cd18c8.json; empirical validation sha256:80e55c8c96d42321303cdfcdcc8d4704b3d406545e9670aaf8886fe4f850d417; measurement receipt sha256:9a74a42610deb93b6c06e437c7ddc255829d67a9984e385f86b077f8b8d1f299; isolation

sha256:32c4d7f89ecee41a21e6f044a8f338b7048facc41ede593c61872970cd848cfc; custody
 sha256:e8c78cebc9c6381b10ce0093a06dc2e7704db544aea50a017ac849ead0a52c1f.

66. Cross-modal qualitative binding

Claim identity: SFT-CONSC-CROSS-MODAL-QUALIA-001

Question and exact theorem. Cross-modal qualitative binding is a joint interior composition whose distinct source modalities and qualitative identities remain traceable.

qualities-from-distinct-modal-sources__same-modal/cross-modal__shared-observation-
 class-composition__source-modality-component-joint-record__first-person-psychophys-
 ical__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives_
 _no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-ARTIFICIAL-CONSCIOUSNESS-EVIDENCE-001
- SFT-CONSC-QUALIA-COMPOSITION-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	qualities-from-distinct-modal-sources	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	same-modal/cross-modal	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	shared-observation-class-composition	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	source-modality-component-joint-record	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	first-person-psychophysical	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: `qualities-from-distinct-modal-sources__same-modal/cross-modal__shared-observation-class-composition__source-modality-component-joint-record__first-person-psychophysical__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule`.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. This exact relation closes cross-modal qualitative binding is a joint interior composition whose distinct source modalities and qualitative identities remain traceable. The result is structural at its declared boundary; participant-, method-, realization- and context-dependent observations remain source-bound records rather than universal fitted magnitudes.

A quality is treated as a participant-bound held label, not a public word or imported magnitude. Difference, identity, similarity, recurrence, composition and cross-modal binding remain distinct. Qualia resonance is the stable reidentification of that held label across a complete recurrent interior support, tested against perturbation and changed-label controls.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:a2b3da35f261db17dc6fd5cf759735f357f4568fb2bb4562cf3cbfceb6b2d38d`.
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt

sha256:646df2f415cb66042b8aab4814ec88ed52a1f179a46e5b5751ddce2854d98335.

- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:03280c7e8bb758fb6572d42e5f8757b6549d4c0dc6433cbeca7d4d89b57e721b.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:c7807f7d87912f28bd5df22be739f10198e4cad69fdf50dd406f054335157a42.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:1851e0ea83b0b39f678e839627a33f850805efbe27c477617cec8f706151135d; independent certificate sha256:cb850035bed462513ff4521e07af7f7ddc94a7859182a0ed99cc3eaeafd1a29b5; external-validation identity

sha256:ad12ec7a6866f3b48ee818f0bd19fa51152cb5d6d301fe7275a50d88d342d54e.

Post-seal empirical-boundary comparison. Directness: participant_bound_empirical_constraint. Disposition: direct_participant_bound_constraint. Within-subject consistency and test-retest observations support stable reidentification while learned-category controls forbid public labels from being equated with private qualitative identity.

The external target was opened only after the claim derivation seal. Predicted consequence:

within-subject-quality-reidentification-with-public-private-boundary. Source-derived

consequence: within-subject-quality-reidentification-with-public-private-boundary. Exact

correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-SYNESTHESIA-BATTERY-2007 - Eagleman et al., Journal of Neuroscience Methods;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC4118597/>; class: primary within-subject qualitative consistency protocol; usable captures: 2; registered features present: internal consistency task, three times, color, controls; absent and preserved: none; adverse/boundary role: none.
- CONSC-SYNESTHESIA-COLOUR-2026 - Root et al., Behavior Research Methods;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC12960370/>; class: open synesthesia colour dataset and protocol; usable captures: 1; registered features present: publicly available data, participants, test-retest consistency, CIELuv; absent and preserved: none; adverse/boundary role: none.
- CONSC-COLOUR-CATEGORY-LEARNING-2010 - Zhou et al., PNAS;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2890491/>; class: primary language-category and colour-discrimination control; usable captures: 1; registered features present: learned categories, perceptual discrimination, control group; absent and preserved: none; adverse/boundary role: Preserves the adverse fact that public colour categories can alter discrimination and therefore cannot be equated with private qualitative identity..

Comparison record:

- sft-consc-cross-modal-qualia-001-external-consequence: predicted
within-subject-quality-reidentification-with-public-private-boundary; source-derived
within-subject-quality-reidentification-with-public-private-boundary; exact match True
- evidence disposition: direct_participant_bound_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:b0360577d8bba39fd502e519143858f07b846ec79195c76961c8644df044343a; derivation seal sha256:8982181d321a4dfb245f51540f8eef0f67b7f760bcaa29834b2d3afad10169e7; engine receipt sha256:6db5ce7ce9393018e365451896a07dc3fa2bc15ee574e9e2809d2c89c593090a at receipts/engine/model_admitted/SFT-CONSC-CROSS-MODAL-QUALIA-001-6db5ce7ce9393018.json; empirical validation sha256:3390117fb47a2531a9fbb4c5b0e0d34d4524034b65169c515f930bafc1c1d29c; measurement receipt sha256:6e5c4e6001f6827f888d91be52d9582d2dbb60bfa6b9c209706fd34277d22cb4; isolation sha256:d71a3d9ddefa835185bf3d76b0c5d0ff77e5431ddd5701fa4cbdd152b75ee8cd; custody sha256:15b33ca74cd19f4ade3a70459c02adcf20417b22bc146ddf25335b26928f54b4.

26. The Red-of-Red

The red-of-red is the specific self-observed reidentification of the red qualitative carrier. Physical red stimulus, public token, participant quality and report about that quality remain separate. The exact formal carrier is the least registered three-position recurrence: one-seventh, two-sevenths and four-sevenths return in period three and compose exactly to the One.

67. Red stimulus and experience boundary

Claim identity: SFT-CONSC-RED-STIMULUS-BOUNDARY-001

Question and exact theorem. A physically classified red stimulus and the experienced quality remain distinct records connected only by a tested presentation-and-report bridge.

physical-stimulus-and-interior-label__stimulus/quality__source-bound-presentation__spectral-display-observer-context__physics-to-first-person__root-bound-forward-forwarding__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001

- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-CROSS-MODAL-QUALIA-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	physical-stimulus-and-interior-label	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	stimulus/quality	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	source-bound-presentation	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	spectral-display-observer-context	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	physics-to-first-person	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: physical-stimulus-and-interior-label__stimulus/quality__source-bound-presentation__spectral-display-observer-context__physics-to-first-person__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. CIE-classified stimulus and the experience of red are connected by a controlled presentation-and-report bridge but remain different records. Wavelength, display coordinate or public name is not the private quality.

The red-of-red is the specific self-observed reidentification of the red qualitative carrier. Physical red stimulus, public token, participant quality and report about that quality remain separate. The exact formal carrier is the least registered three-position recurrence: one-seventh, two-sevenths and four-sevenths return in period three and compose exactly to the One.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:f9c08dd2dbdd8b7fd7f7dd50add60f6b7353329b5dccc4b01d5ce2ceb79c573.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:d5c630a16ff769109cc398e5808231c57c1a395f7deb209179483b90377e5432.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:94c4fb341bc5cfd46dee6e1c3567a62b5893469780cb2df183c49319f53056be.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:fa3f1db018796a257df94e84d67e9e9412737b638549c893a3dbbed9dd5d80a4.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:092030d6bd470452800ad28ed4f3f0fa593e32afcc8c359f7b86cde78fd5d2e7; independent certificate sha256:1e54b25c37c112ad41be1e3fb08d2f2b4771451c0b8f70b96fca06daa9438bdd; external-validation identity sha256:73f5011b2ff196460adb5c5e1bee120d97bd2975bcd419eea532f9ba1fa637f4.

Post-seal empirical-boundary comparison. Directness: participant_bound_empirical_constraint. Disposition: direct_participant_bound_and_boundary_constraint. CIE stimulus records, participant consistency, synesthetic-colour evidence and learned-category controls jointly test reidentification while direct third-person possession of another subject's red remains unclaimed.

The external target was opened only after the claim derivation seal. Predicted consequence:

physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence.

Source-derived consequence:

physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence. Exact

correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-CIE-1931-CMF - International Commission on Illumination;
<https://cie.co.at/datatable/cie-1931-colour-matching-functions-2-degree-observer>; class: official physical colour-measurement standard; usable captures: 2; registered features present: colour-matching functions, standard colorimetric observer, one nm wavelength steps; absent and preserved: none; adverse/boundary role: none.
- CONSC-SYNESTHESIA-BATTERY-2007 - Eagleman et al., Journal of Neuroscience Methods;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC4118597/>; class: primary within-subject qualitative consistency protocol; usable captures: 2; registered features present: internal consistency task, three times, color, controls; absent and preserved: none; adverse/boundary role: none.
- CONSC-SYNESTHETIC-COLOUR-MATCH-2008 - Hong and Blake, Vision Research;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2423348/>; class: primary participant-specific colour matching experiment; usable captures: 1; registered features present: synesthetic color; absent and preserved: asymmetric matching, spectroradiometer, observers; adverse/boundary role: none.
- CONSC-COLOUR-CATEGORY-LEARNING-2010 - Zhou et al., PNAS;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2890491/>; class: primary language-category and colour-discrimination control; usable captures: 1; registered features present: learned categories, perceptual discrimination, control group; absent and preserved: none; adverse/boundary role: Preserves the adverse fact that public colour categories can alter discrimination and therefore cannot be equated with private qualitative identity..

Comparison record:

- sft-consc-red-stimulus-boundary-001-external-consequence: predicted
physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence; source-derived
physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence; exact match True
- evidence disposition: direct_participant_bound_and_boundary_constraint; directness:
participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience

- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:840460ff452c8accf3529b685800aff30708678d1a4c80c9bffffd8d2200a6605; derivation seal sha256:6e432cbf8aa10b8f60ad0493bbe6453a56fe354a1608c942488ff77b01996bc0; engine receipt sha256:6b3c57163aaf0b0ce1009040c55b4c20375d03416b677eb0855e695f3822410d at receipts/engine/model_admitted/SFT-CONSC-RED-STIMULUS-BOUNDARY-001-6b3c57163aaf0b0c.json; empirical validation sha256:7db6dcc6ce828f35296d388667aebb576e7c1a9ff1c532b8fa4649fd84e1e79a; measurement receipt sha256:1f21fld47a32c347599f7fe547842aa6537392aef0c6aac5887b394971276118; isolation sha256:3f261794f288966a6cf88480c6902c0caf9e1741bf9b5e816dfbc0d7085f6396; custody sha256:71a740694209f09d40eba03bc5dd6371265c0b6ef6fa819013fc160ff5ab8982.

68. Red qualitative identity

Claim identity: SFT-CONSC-RED-QUALITATIVE-IDENTITY-001

Question and exact theorem. Red qualitative identity is the within-subject reidentification of a specific quality across instances, not the public token 'red'.

within-subject-red-instances__red-label/red-quality__identity-discrimination__instance-context-and-alternatives__first-person-psychophysical__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Routed dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001

- SFT-CONSC-CROSS-MODAL-QUALIA-001
- SFT-CONSC-RED-STIMULUS-BOUNDARY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	within-subject-red-instances	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	red-label/red-quality	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	identity-discrimination	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	instance-context-and-alternatives	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	first-person-psychophysical	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: within-subject-red-instances__red-label/red-quality__identity-discrimination__instance-context-and-alternatives__first-person-psychophysical__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. Red identity is within-subject reidentification across instances. The public token red coordinates reports; it does not supply or replace the participant's held qualitative label.

The red-of-red is the specific self-observed reidentification of the red qualitative carrier. Physical red stimulus, public token, participant quality and report about that quality remain separate. The exact formal carrier is the least registered three-position recurrence: one-seventh, two-sevenths and four-sevenths return in period three and compose exactly to the One.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:f7bacf97cd91fd0a7058a26d354d31f45ff5e6eb0802d58b99dc8305f9f83efa.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:54dbca0cf1de8c783635ab71256051f076c9742458900c9bfe76c9b0a158a7f8.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:f1e660842ffb93e1d04b426b0e7b9c932699a9696845b40bf100627356217f54.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:826be2f65b64c94a72d8be8b3e0f91c9b1a9675820c824d784a52afa303bbbb7.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
sha256:2b59934a7811621c1233936788cc034e05839e400aa1038dd2b486bf36ffc9f7; independent
certificate sha256:a294e5bd4c5adf00cd2f54e49def605105dc9f72743a2483a4e1445b2ebd92bc;
external-validation identity
sha256:2bc01ffcaabda80481133ba2cf00089e24fa06e4c29b062c669d925d1f035ae8.

Post-seal empirical-boundary comparison. Directness: participant_bound_empirical_constraint. Disposition: direct_participant_bound_and_boundary_constraint. CIE stimulus records, participant consistency, synesthetic-colour evidence and learned-category controls jointly test reidentification while direct third-person possession of another subject's red remains unclaimed.

The external target was opened only after the claim derivation seal. Predicted consequence:

physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence.

Source-derived consequence:

physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-CIE-1931-CMF - International Commission on Illumination;
<https://cie.co.at/datatable/cie-1931-colour-matching-functions-2-degree-observer>; class: official physical colour-measurement standard; usable captures: 2; registered features present: colour-matching functions, standard colorimetric observer, one nm wavelength steps; absent and preserved: none; adverse/boundary role: none.
- CONSC-SYNESTHESIA-BATTERY-2007 - Eagleman et al., Journal of Neuroscience Methods;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC4118597/>; class: primary within-subject qualitative consistency protocol; usable captures: 2; registered features present: internal consistency task, three times, color, controls; absent and preserved: none; adverse/boundary role: none.
- CONSC-SYNESTHETIC-COLOUR-MATCH-2008 - Hong and Blake, Vision Research;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2423348/>; class: primary participant-specific colour matching experiment; usable captures: 1; registered features present: synesthetic color; absent and preserved: asymmetric matching, spectroradiometer, observers; adverse/boundary role: none.
- CONSC-COLOUR-CATEGORY-LEARNING-2010 - Zhou et al., PNAS;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2890491/>; class: primary language-category and colour-discrimination control; usable captures: 1; registered features present: learned categories, perceptual discrimination, control group; absent and preserved: none; adverse/boundary role: Preserves the adverse fact that public colour categories can alter discrimination

and therefore cannot be equated with private qualitative identity..

Comparison record:

- sft-consc-red-qualitative-identity-001-external-consequence: predicted physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence; source-derived physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence; exact match True
- evidence disposition: direct_participant_bound_and_boundary_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:b6a61ccb6c21312cdbl1e176cb1de743544ea183529b975010dc31721c2e14c3f; derivation seal sha256:8d4b53a17c834b5ff66f38ae99e7bceec48747aad447f2a43aa2c33d61f53ee0; engine receipt sha256:bb195ff50462932f27db6de2dd26798d8157138a4630227db0b7c74f20c3124d at receipts/engine/model_admitted/SFT-CONSC-RED-QUALITATIVE-IDENTITY-001-bb195ff50462932f.json; empirical validation sha256:bde7c8e11bd048e28693acd71b9ae5392f82ae065bc8ce2f0674c5040c2c0ddf; measurement receipt sha256:14b08858b4840ebacb491ec8d2d9c71ad0b060c386638288575375c179cf8c94; isolation sha256:549f36d881820031381daf099df3699bf778c57feaa7ff6bf40644aa93523118; custody sha256:888f2ac578ea28319d09eb990cd34bf38f0936af0cea6623fd8d11889218e005.

69. The red-of-red self-reidentification

Claim identity: SFT-CONSC-RED-OF-RED-001

Question and exact theorem. The red-of-red is the self-observed reidentification of the red qualitative carrier across complete recurrence, retaining the difference between the quality and a report about it.

```
red-quality-and-self-observation__quality/report-about-quality__label-bearing-recu
rrent-self-observation__source-instance-recurrence-self-report__first-person-speci
fic-qualia__root-bound-forward-forcing__positive-finite-successor-with-all-alterna
tives__no-extra-rule
```

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-CROSS-MODAL-QUALIA-001
- SFT-CONSC-RED-QUALITATIVE-IDENTITY-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	red-quality-and-self-observation	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	quality/report-about-quality	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	label-bearing-recurrent-self-observation	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
record	favourable-output-without-complete-trace	source-instance-recurrence-self-report	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	first-person-specific-qualia	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: red-quality-and-self-observation__quality/report-about-quality__label-bearing-recurrent-self-observation__source-instance-recurrence-self-report__first-person-specific-qualia__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **label-bearing period-three self-reidentification with quality and report held apart.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. The red-of-red is red re-entering a self-observation loop as the quality identified again, with the quality, the self-observation and the report about it separately held. This reconstructs the stronger prior result rather than reducing it to generic binding.

The red-of-red is the specific self-observed reidentification of the red qualitative carrier. Physical red stimulus, public token, participant quality and report about that quality remain separate. The exact formal carrier is the least registered three-position recurrence: one-seventh, two-sevenths and four-sevenths return in period three and compose exactly to the One.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:d00fa4e552082a5fff55834ebc7714e1b157955d317c99cdc6d0cc2540a8135e.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:1c6970234edfe130ec53e2630024837b1acff9eb97556dcb6377907131ab2b03.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:cf0fe1b663c34f1dbad10606f8415082cd7e76f9de854312443dc84449187687.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:86ed06e2a17aa52aac081bf6ac623abb965c71c483e0dd1147b5b25126537ee6.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:8158f8863d2218b1fec951f0401a280f446ad018a240320a0a059767582ddf2; independent certificate sha256:1281fc7bb8a58f840e345f197af163221cc467f43fb530ef786f642c35935599; external-validation identity sha256:43f56a2b69113a405f10e4c4b0bdalc2b8d31b7dc44edb1328aae82212c5783a.

Post-seal empirical-boundary comparison. Directness: participant_bound_empirical_constraint. Disposition: direct_participant_bound_and_boundary_constraint. CIE stimulus records, participant consistency, synesthetic-colour evidence and learned-category controls jointly test reidentification while direct third-person possession of another subject's red remains unclaimed.

The external target was opened only after the claim derivation seal. Predicted consequence:

physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence.

Source-derived consequence:

physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-CIE-1931-CMF - International Commission on Illumination;
<https://cie.co.at/datatable/cie-1931-colour-matching-functions-2-degree-observer>; class: official physical colour-measurement standard; usable captures: 2; registered features present: colour-matching functions, standard colorimetric observer, one nm wavelength steps; absent and preserved: none; adverse/boundary role: none.
- CONSC-SYNESTHESIA-BATTERY-2007 - Eagleman et al., Journal of Neuroscience Methods;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC4118597/>; class: primary within-subject qualitative consistency protocol; usable captures: 2; registered features present: internal consistency task, three times, color, controls; absent and preserved: none; adverse/boundary role: none.
- CONSC-SYNESTHETIC-COLOUR-MATCH-2008 - Hong and Blake, Vision Research;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2423348/>; class: primary participant-specific colour matching experiment; usable captures: 1; registered features present: synesthetic color; absent and preserved: asymmetric matching, spectroradiometer, observers; adverse/boundary role: none.
- CONSC-COLOUR-CATEGORY-LEARNING-2010 - Zhou et al., PNAS;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2890491/>; class: primary language-category and colour-discrimination control; usable captures: 1; registered features present: learned categories, perceptual discrimination, control group; absent and preserved: none; adverse/boundary role: Preserves the adverse fact that public colour categories can alter discrimination and therefore cannot be equated with private qualitative identity..

Comparison record:

- sft-consc-red-of-red-001-external-consequence: predicted physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence; source-derived physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence; exact match True
- evidence disposition: direct_participant_bound_and_boundary_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:885b3ad549cc9a166fe8c605d48052bca6bcb6cb38f141ed36193f77efe13e0e; derivation seal sha256:9fcd374fd3bfacec0c6d745f6c7f085d917d0d8b89dbd8d144cd1ac7c815d7f0; engine receipt sha256:69b23e1cd2dcd9545059c9a8528a3c8085ea04f816ad1cf6c8366fd437f4542d at receipts/engine/model_admitted/SFT-CONSC-RED-OF-RED-001-69b23e1cd2dcd954.json; empirical validation sha256:70358af657aeaf9c9a4735016971eb02906a0f0aebfef852f27d4aa687aa4f55; measurement receipt sha256:8705cfae084007e40bcb0f866150722178887d544b289e4656ac31bc2cdb7bc3; isolation sha256:9c05fcb4db46aa52e10f028695ac7dcd230df7a3caf52dd1c6030ea6265e1dcc; custody sha256:4ab2e38b9b9f2a1a7cf902c27474c282a89030c657b84dabeb40ae587e396a90.

70. Red recurrence

Claim identity: SFT-CONSC-RED-RECURRENCE-001

Question and exact theorem. The least registered three-position qualitative support is the exact one-seventh, two-sevenths, four-sevenths recurrence whose parts compose to the One; empirical evidence tests whether red identity is retained across the cycle.

three-position-quality-support__stable-return/changed-quality__one-seven-two-seven
 -four-seven-cycle__complete-cycle-and-label__formal-plus-first-person__root-bound-
 forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001

- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-CROSS-MODAL-QUALIA-001
- SFT-CONSC-RED-OF-RED-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	three-position-quality-support	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	stable-return/changed-quality	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	one-seven-two-seven-four-seven-cycle	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	complete-cycle-and-label	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	formal-plus-first-person	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: three-position-quality-support__stable-return/changed-quality__one-seven-two-seven-four-seven-cycle__complete-cycle-and-label__formal-plus-first-person__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **one-seventh, two-sevenths, four-sevenths: exact period three and exact composition to the One.**
Closure scope: `depth_independent`; minimality and named-shape uniqueness both pass.

Operational witnesses.

- `claim-carrier`: the claim retains a nonempty named carrier; passed `true`.
- `distinction-operation`: the retained distinction is not collapsed into its operation; passed `true`.
- `evidence-boundary`: the evidence class does not silently become the carrier; passed `true`.
- `exact-fold-witness`: the family's exact Fold construction reproduces; passed `true`.

Scientific meaning. The generated support one-seventh, two-sevenths and four-sevenths is period three under the Fold and sums exactly to the One. It is the formal carrier of the red-of-red test; empirical work determines whether a participant's red label remains stable across its realization protocol.

The red-of-red is the specific self-observed reidentification of the red qualitative carrier. Physical red stimulus, public token, participant quality and report about that quality remain separate. The exact formal carrier is the least registered three-position recurrence: one-seventh, two-sevenths and four-sevenths return in period three and compose exactly to the One.

Adverse controls.

- `false_premise`: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
`sha256:63f74a14865b3f03455a1d42515b985b1f23c5986c519e7d2c2f7f38d2979b72.`
- `tampered_source`: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
`sha256:9ac5d2c4068ea75d5c3da95fa876f3e7408a3f46d69799602e855ad9d6961212.`
- `tampered_artifact`: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
`sha256:9805bf50e36a2993e87f17bc8ad755d1ffb75b0ad7e569c727d221debcae99a0.`
- `boundary`: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
`sha256:26eed86d51b938a70a3daaa447bfbf7d068c83f7a2b9ff3c04ddd97cfa068d66.`

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation
`sha256:993265eb650a631b95e6098f290dc9dc27b14db1b297b3e49eaf9f83f66cc667`; independent
 certificate `sha256:90551c10a8b554f34f251f218ffb6f9c1d172607c54b6c81780085727bc47b04`;
 external-validation identity
`sha256:37caf6215d4abc5808983b56ad1850b2f10fcdf5f6b37b93b79752ae854b01e2.`

Post-seal empirical-boundary comparison. Directness: `participant_bound_empirical_constraint`. Disposition: `direct_participant_bound_and_boundary_constraint`. CIE stimulus records, participant consistency, synesthetic-colour evidence and learned-category controls jointly test reidentification while direct third-person possession of another subject's red remains unclaimed.

The external target was opened only after the claim derivation seal. Predicted consequence:

`physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence.`

Source-derived consequence:

`physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence.` Exact correspondence: `true`. Phenomenal occurrence claimed as directly third-person possessed: `false`. Formal structure relabelled as empirical phenomenal fact: `false`.

Sources and complete feature accounting:

- CONSC-CIE-1931-CMF - International Commission on Illumination;
<https://cie.co.at/datatable/cie-1931-colour-matching-functions-2-degree-observer>; class: official physical colour-measurement standard; usable captures: 2; registered features present: colour-matching functions, standard colorimetric observer, one nm wavelength steps; absent and preserved: none; adverse/boundary role: none.
- CONSC-SYNESTHESIA-BATTERY-2007 - Eagleman et al., Journal of Neuroscience Methods;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC4118597/>; class: primary within-subject qualitative consistency protocol; usable captures: 2; registered features present: internal consistency task, three times, color, controls; absent and preserved: none; adverse/boundary role: none.
- CONSC-SYNESTHETIC-COLOUR-MATCH-2008 - Hong and Blake, Vision Research;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2423348/>; class: primary participant-specific colour matching experiment; usable captures: 1; registered features present: synesthetic color; absent and preserved: asymmetric matching, spectroradiometer, observers; adverse/boundary role: none.
- CONSC-COLOUR-CATEGORY-LEARNING-2010 - Zhou et al., PNAS;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2890491/>; class: primary language-category and colour-discrimination control; usable captures: 1; registered features present: learned categories, perceptual discrimination, control group; absent and preserved: none; adverse/boundary role: Preserves the adverse fact that public colour categories can alter discrimination and therefore cannot be equated with private qualitative identity..

Comparison record:

- sft-consc-red-recurrence-001-external-consequence: predicted physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence; source-derived physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence; exact match True
- evidence disposition: direct_participant_bound_and_boundary_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:66d1b6372a1a8b16ef38a611c1ca07d830d4a2950311c00cef0cb5725c7d86e2; derivation seal sha256:089ed8aa2b3434c4f974e93c6cd1616c78114506631a6db6629e669ce178bebd; engine receipt sha256:2ea40f0832299ee54c30e0aa3f200adc3ee53965d3a4979416674260d160ad29 at receipts/engine/model_admitted/SFT-CONSC-RED-RECURRENCE-001-2ea40f0832299ee5.json; empirical validation sha256:fb26a08302a7205113cb3f812cf5e569690646efc7d8e4d27530b0a215da8b9; measurement receipt sha256:6ca125d9b8d512c0838f1ad0a941bbbe49f32e786d1d641f42132f155f362155; isolation sha256:e577c55c631eb3c92241f1fe6839961f3c35ed60414036c0c5c3e74619aa5296; custody sha256:465a68454db65f8e42b6441b1a685326cff3c7dc78d788deec24581b484084a8.

71. Red-of-red contrast controls

Claim identity: SFT-CONSC-RED-CONTROLS-001

Question and exact theorem. The red-of-red test preserves matched, mismatched, absent and uncertain rows and must reject a changed label, changed stimulus or report-only surrogate.

red-and-nonred-comparison-set__identity/mismatch__changed-label-and-changed-stimulus-tests__all-match-mismatch-absence-uncertain-rows__first-person-psychophysical__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-CROSS-MODAL-QUALIA-001
- SFT-CONSC-RED-RECURRENCE-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	red-and-nonred-comparison-set	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	identity/mismatch	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	changed-label-and-changed-stimulus-tests	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.
record	favourable-output-without-complete-trace	all-match-mismatch-absence-uncertain-rows	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	first-person-psychophysical	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: red-and-nonred-comparison-set__identity/mismatch__changed-label-and-changed-stimulus-tests__all-match-mismatch-absence-uncertain-rows__first-person-psychophysical__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. Matched, mismatched, absent and uncertain rows remain in the test. Changing the stimulus, held label or report-only surrogate must fail; no favorable participant row can erase the others.

The red-of-red is the specific self-observed reidentification of the red qualitative carrier. Physical red stimulus, public token, participant quality and report about that quality remain separate. The exact formal carrier is the least registered three-position recurrence: one-seventh, two-sevenths and four-sevenths return in period three and compose exactly to the One.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:027f5480c78257127ae89d2fe425c3d2b7a53dcfb7b385084dde1dbfe2b74135.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:2b1d0722a79ef8d8e71533e4f29458b82f36cfb0ec6b643cc4d6aa3b8af4cfc6.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:96bf410598478668055e9d7bd51495340a8e7b6153289eabf33300fba84a53f1.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt
sha256:9699578a444b97f6134474115933943c90517a562c81580d6463288f2c0ff423.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:c9e0b321077f5b8b7efc4da586bb04a650fcef3c13c915198256d3b90cdcebad3; independent certificate sha256:e24b7e40f0cea8ccb88e495b3dc17e9bbe0fea3ba5cb710731107832e1f83ad0; external-validation identity sha256:33b7d67117a738509d30b2cc305270d141ed118e60d8ab61b020a4675bb4449f.

Post-seal empirical-boundary comparison. Directness: participant_bound_empirical_constraint. Disposition: direct_participant_bound_and_boundary_constraint. CIE stimulus records, participant consistency, synesthetic-colour evidence and learned-category controls jointly test reidentification while direct third-person possession of another subject's red remains unclaimed.

The external target was opened only after the claim derivation seal. Predicted consequence:

physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence.

Source-derived consequence:

physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-CIE-1931-CMF - International Commission on Illumination;
<https://cie.co.at/datatable/cie-1931-colour-matching-functions-2-degree-observer>; class: official physical colour-measurement standard; usable captures: 2; registered features present: colour-matching functions, standard colorimetric observer, one nm wavelength steps; absent and preserved: none; adverse/boundary role: none.
- CONSC-SYNESTHESIA-BATTERY-2007 - Eagleman et al., Journal of Neuroscience Methods;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC4118597/>; class: primary within-subject qualitative consistency protocol; usable captures: 2; registered features present: internal consistency task, three times, color, controls; absent and preserved: none; adverse/boundary role: none.
- CONSC-SYNESTHETIC-COLOUR-MATCH-2008 - Hong and Blake, Vision Research;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2423348/>; class: primary participant-specific colour matching experiment; usable captures: 1; registered features present: synesthetic color; absent and preserved: asymmetric matching, spectroradiometer, observers; adverse/boundary role: none.
- CONSC-COLOUR-CATEGORY-LEARNING-2010 - Zhou et al., PNAS;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2890491/>; class: primary language-category and colour-discrimination control; usable captures: 1; registered features present: learned categories, perceptual discrimination, control group; absent and preserved: none; adverse/boundary role: Preserves the adverse fact that public colour categories can alter discrimination

and therefore cannot be equated with private qualitative identity..

Comparison record:

- sft-consc-red-controls-001-external-consequence: predicted
physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence; source-derived
physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence; exact match True
- evidence disposition: direct_participant_bound_and_boundary_constraint; directness:
participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source
manifest sha256:45d1bd7e3ed70d3f6602e0e3f4b177d99a6ea721e2d14fcf1759f550391cbe02; derivation
seal sha256:2f13c3e6ac0237cc0ca9434391bc02f80c5851b0eec6247e6ed6506ce9b646b0; engine receipt
sha256:c76466e545afa551a0afb93163b6bf337a5d1a523dbb28df74d7a3c1b3803c55 at
receipts/engine/model_admitted/SFT-CONSC-RED-CONTROLS-001-c76466e545afa551.json;
empirical validation sha256:660f5406aff5d9b017cdb5f385e1c5dca6158d20537eed52cf3a41fe87e56d55;
measurement receipt
sha256:c1f16822b526be23ae5b72bd0a158d5f8a54ece3ad5ab89f1544bd7db660d0ec; isolation
sha256:7ae781eaca678ab1e055caa6d91e96bcf958742d234a6e0b1d48776b9be7071f; custody
sha256:c77fd7128cc5678c5c9998365e1e1e8a5d870092e143e0bc0b972302f4d3dc58.

72. Red-of-red empirical boundary

Claim identity: SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001

Question and exact theorem. Empirical closure of red-of-red is participant- and protocol-bound: public evidence may establish stable discrimination and reidentification without claiming direct third-person possession of another subject's quality.

participant-specific-observation-record__private-identity/public-generalization__replicated-within-subject-protocol__participant-stimulus-response-method__first-person-with-third-person-custody__root-bound-forward-forcing__positive-finite-success-or-with-all-alternatives__no-extra-rule

Rooted dependency chain. The registration names SFT-ROOT-THERE-IS-NO-NOTHING, zero axioms and zero free parameters. It requires these already admitted receipts:

- SFT-FOUNDATION-ONE-001
- SFT-FOUNDATION-FOLD-001
- SFT-FOUNDATION-FOLD-DYNAMICS-001
- SFT-MATH-DYNAMICAL-SYSTEMS-001
- SFT-MATH-LOGIC-PROOF-001
- SFT-MATH-CATEGORY-TYPE-COMPOSITION-001
- SFT-INFO-SYMBOL-DISTINCTION-001
- SFT-INFO-QUANTITY-001
- SFT-INFO-MUTUAL-CONDITIONAL-001
- SFT-INFO-CONSERVATION-LOSS-001
- SFT-COMP-FORM-STATE-TRANSITION-001
- SFT-COMP-FORM-COMPOSITION-001
- SFT-COMP-FORM-RECURSIVE-FUNCTION-001
- SFT-COMP-LEARN-INFERENCE-001
- SFT-COMP-LEARN-REPRESENTATION-001
- SFT-PHYS-MEAS-OBSERVATION-CARRIER-001
- SFT-BIO-INSIDE-OUTSIDE-001
- SFT-BIO-LIFE-IDENTITY-TURNOVER-001
- SFT-MED-SYMPTOM-001
- SFT-CONSC-CROSS-MODAL-QUALIA-001
- SFT-CONSC-RED-CONTROLS-001

The dependency graph reaches the premise-free root through actual receipts; a citation or branch label never substitutes for that graph.

Generated grammar. Generate the literal Cartesian product of all eight registered binary preservation dimensions before any external consciousness, cognitive, neural, behavioural or colour outcome is opened.

Boundary: Exactly eight binary structural and evidential dimensions, giving 256 forms. The form closure is depth-independent under the stated positive-finite successor, while empirical correspondence remains source- and protocol-bound.

The exact product contains 256 candidates, 256 stored identities and 256 one-for-one decisions. Exactly one survives; 255 fail at least one required coordinate.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
carrier	carrier-absent-or-description-only	participant-specific-observation-record	A missing carrier or mere description cannot instantiate the claimed relation. The complete declared carrier is retained.
distinction	distinction-collapsed-or-conflated	private-identity/public-generalization	Collapsing the named distinction substitutes one evidence or process class for another. The claim-specific distinction remains held.
operation	operation-imported-fitted-or-missing	replicated-within-subject-protocol	An imported, fitted or absent operation can manufacture a familiar theory of mind. Only the generated Fold relation and admitted dependencies act.

Axis	Eliminated form	Forced form	Exact elimination / retention basis
record	favourable-output-without-complete-trace	participant-stimulus-response-method	An output without its source, transition, missing and adverse records is not auditable. The complete required record is retained.
evidence	report-behaviour-correlation-or-confidence-substituted	first-person-with-third-person-custody	A report, behaviour, neural correlation or model confidence cannot silently become phenomenal occurrence. The exact evidence class and any bridge are named.
provenance	prior-consensus-target-or-application-selected	root-bound-forward-forcing	Prior SFT, consensus, targets or applications may register questions or test consequences but cannot select the law. Every result traces through admitted dependencies to There Is No Nothing.
generality	one-favourable-instance-with-erased-alternatives	positive-finite-successor-with-all-alternatives	One favourable instance cannot close a generated class while adverse and uncertain alternatives are erased. The base and every supplied positive finite successor preserve all favourable, adverse, absent and unresolved distinctions.
extension	free-parameter-exception-or-opaque-oracle	no-extra-rule	A fitted parameter, exception or opaque oracle can force a preferred answer. No rule beyond the admitted Fold dependencies and generated preservation conditions is added.

Unique survivor and depth independence. Sole survivor: participant-specific-observation-record__private-identity/public-generalization__replicated-within-subject-protocol__participant-stimulus-response-method__first-person-with-third-person-custody__root-bound-forward-forcing__positive-finite-successor-with-all-alternatives__no-extra-rule.

Base: The least complete carrier retains the claimed roles, transition, record, evidence class and root provenance without an extra rule.

Successor: Adding one lawful finite state, participant, content, observation or realizing component preserves every earlier identity, transition, adverse row and boundary and appends the new trace.

Exact structural value: **unique complete preserving relation in the declared 256-form grammar.** Closure scope: depth_independent; minimality and named-shape uniqueness both pass.

Operational witnesses.

- claim-carrier: the claim retains a nonempty named carrier; passed true.
- distinction-operation: the retained distinction is not collapsed into its operation; passed true.
- evidence-boundary: the evidence class does not silently become the carrier; passed true.
- exact-fold-witness: the family's exact Fold construction reproduces; passed true.

Scientific meaning. Public science can establish repeatable discrimination, reidentification, stimulus control and report custody. It does not claim literal third-person possession of another participant's red; that restraint is part of the closed result, not an omitted experiment.

The red-of-red is the specific self-observed reidentification of the red qualitative carrier. Physical red stimulus, public token, participant quality and report about that quality remain separate. The exact formal carrier is the least registered three-position recurrence: one-seventh, two-sevenths and four-sevenths return in period three and compose exactly to the One.

Adverse controls.

- false_premise: passed; expected reject the generated form lacking the complete physical carrier; observed reject the generated form lacking the complete physical carrier; receipt
sha256:9e5a7cf7ab27741a8f5fe3b6e3d16484dd6733c025d88560801e14dbdd5a912b.
- tampered_source: passed; expected reject a changed registered source identity; observed reject a changed registered source identity; receipt
sha256:29857dad740fcf3225932fb2489924ab6eeb3b519650e37373f6bc0e495cbfeb.
- tampered_artifact: passed; expected reject a missing, duplicate or additional survivor; observed reject a missing, duplicate or additional survivor; receipt
sha256:c6f728708c8566a8f473db53588f56058b79ddd97d3d65d1d4cf37e28369e2d0.
- boundary: passed; expected reject target access, forbidden proof values, imported laws and free parameters; observed reject target access, forbidden proof values, imported laws and free parameters; receipt

sha256:b8ddeb5f59399b08be8648022b1accdf7249dfc72e35a82b33ad7e30c1f56b537.

Independent reconstruction. A separately executed implementation regenerated the literal product, candidate order, every decision, the sole survivor, depth-independent closure and all four control classes. Implementation

sha256:d7bef9c4e2ba37cf00a6c84c0ee8bcbf025ea76eef152a912401ebe6158e74ec; independent certificate sha256:d6ae210f69bd3a9fe97b1b92bc0130fa2caa4fa7d6544202d7d0a19bd4d95498; external-validation identity sha256:faaa5e5c4d60bf2a98fb3b2a352f6c1c11ea03e6db32bce6b874e0e462b98a03.

Post-seal empirical-boundary comparison. Directness: participant_bound_empirical_constraint. Disposition: direct_participant_bound_and_boundary_constraint. CIE stimulus records, participant consistency, synesthetic-colour evidence and learned-category controls jointly test reidentification while direct third-person possession of another subject's red remains unclaimed.

The external target was opened only after the claim derivation seal. Predicted consequence:

physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence.

Source-derived consequence:

physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence. Exact correspondence: true. Phenomenal occurrence claimed as directly third-person possessed: false. Formal structure relabelled as empirical phenomenal fact: false.

Sources and complete feature accounting:

- CONSC-CIE-1931-CMF - International Commission on Illumination;
<https://cie.co.at/datatable/cie-1931-colour-matching-functions-2-degree-observer>; class: official physical colour-measurement standard; usable captures: 2; registered features present: colour-matching functions, standard colorimetric observer, one nm wavelength steps; absent and preserved: none; adverse/boundary role: none.
- CONSC-SYNESTHESIA-BATTERY-2007 - Eagleman et al., Journal of Neuroscience Methods;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC4118597/>; class: primary within-subject qualitative consistency protocol; usable captures: 2; registered features present: internal consistency task, three times, color, controls; absent and preserved: none; adverse/boundary role: none.
- CONSC-SYNESTHETIC-COLOUR-MATCH-2008 - Hong and Blake, Vision Research;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2423348/>; class: primary participant-specific colour matching experiment; usable captures: 1; registered features present: synesthetic color; absent and preserved: asymmetric matching, spectroradiometer, observers; adverse/boundary role: none.
- CONSC-COLOUR-CATEGORY-LEARNING-2010 - Zhou et al., PNAS;
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2890491/>; class: primary language-category and colour-discrimination control; usable captures: 1; registered features present: learned categories, perceptual discrimination, control group; absent and preserved: none; adverse/boundary role: Preserves the adverse fact that public colour categories can alter discrimination and therefore cannot be equated with private qualitative identity..

Comparison record:

- sft-consc-red-empirical-boundary-001-external-consequence: predicted physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence; source-derived physical-red-and-private-red-quality-nonidentity-with-reidentification-evidence; exact match True
- evidence disposition: direct_participant_bound_and_boundary_constraint; directness: participant_bound_empirical_constraint
- all registered source features, adverse boundaries, absences and transport failures preserved
- phenomenal occurrence was not claimed as directly third-person observed
- deliberately changed consequence rejected

Falsification boundary: Reject if another generated form preserves every registered dimension, if a required structural witness fails, if purpose-matched evidence contradicts the sealed consequence, or if report, behaviour, biology, computation, physical measurement or public colour naming is substituted for phenomenal occurrence without a separately derived bridge.

Explicit exclusions.

- semantic numerical zero
- negative proof quantity
- irrational or imaginary proof value
- completed infinity
- ungenerated continuum
- free or fitted parameter
- target-selected survivor
- prior answer as premise
- report substituted for experience
- behaviour substituted for experience
- neural correlation substituted for experience
- language fluency or confidence substituted for experience
- opaque predictor
- erased adverse, absent or unresolved row

Immutable evidence identities. Complete pre-source seal

experiments/sealed_predictions/consciousness_foundation_complete_pre_source.json; source manifest sha256:eb5be7f140c2ecf8baf1ec9ee9c8978eaa955631f25dd5a65ec2a99fc6097765; derivation seal sha256:cf97efc6aed6de83bb5dc0deca3fd9dde1c04a225ab06163cc3ed2c2d6d1207f; engine receipt sha256:cdb9dec823a5d95b78a89e3e8605d369e1f26ea20908aee781247bc85a5d21b4 at receipts/engine/model_admitted/SFT-CONSC-RED-EMPIRICAL-BOUNDARY-001-cdb9dec823a5d95b.json; empirical validation sha256:85c19a166d8c21db483c89be12a25bf845ad929f5fffca4bda58122ab46a4835; measurement receipt sha256:5701cf3636a8af68ff3f34f8d4c8d5171470659da9913da10cc226af4bc8da22; isolation sha256:bc84b6d8d55813431999425e12d8c4d1b0cb5ae92c14b614109d391719fa38a8; custody sha256:8b871d77d612f563e1f661fa373dd3571a2a7a41afd0fa133c51ef194484eeee.

27. Complete result and verification summary

The foundation contains exactly 72 admitted laws and 18,432 generated candidates. Exactly 72 forms survive. The 288 structural controls pass. Seventy-two separate implementations reconstruct the candidate order and survivor. Seventy-two post-seal empirical-boundary comparisons pass at their declared directness. All 46 inherited atomic questions are reconciled. All dependency paths reach There Is No Nothing. The engine seal is

sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a; the verification-authority seal is

sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8. Neither protected surface was modified.

The branch is current-evidence closed and extension-open, not permanently locked. A future criticism or discovery may extend or invalidate a claim only by entering the same public process: exact statement, frozen candidate grammar, complete enumeration, unique survivor or preserved halt, independent reconstruction, adverse controls, purpose-matched evidence and unchanged-engine receipt.

28. Full-field roadmap

The next editions extend the foundation through complete perceptual modalities; sound, pain, affect, embodiment and multimodal phenomenology; temporal experience; imagination and metacognition; language and reasoning; development and comparative consciousness; sleep, dream, anesthesia, altered state, cessation and recovery; nonhuman and artificial realization programmes; collective-mind boundaries; clinical-disorder handoffs; and ethical consequences stated separately from empirical admission. Applications may test these laws but never select them.

29. Limitations

- The 72-law surface is foundational, current-evidence closed and extension-open; it is not the complete Layer Two field reconstruction.
- Formal closure, first-person report, behavioral observation, neural correlation, computation and physical realization are different evidential statuses. The paper never promotes one into another without a bridge.
- The three-position red carrier is an exact structural recurrence, not a universal wavelength or intensity and not direct third-person possession of another subject's red.
- Existing sources constrain the derived boundaries; they do not prove that every system satisfying a partial correlate is conscious.
- Fifteen source identities and sixty-one features are substantial but not exhaustive of consciousness research. Three absent features and eighteen failed transport/content rows remain visible.
- Artificial realization remains an auditable test programme. No current application, including Unison AI, was used to select or certify the laws.
- Ethical implications require their own normative derivation and cannot be smuggled into empirical admission.

30. Conclusion

Foundational Consciousness and Cognitive Science is current-evidence closed and extension-open: 72 laws, 18,432 exact candidates, 72 sole survivors, 288 passing controls, 72 independent reconstructions, 72 post-seal empirical-boundary comparisons and 46 of 46 inherited atomic obligations reconciled.

The branch derives a precise solution to the structural hard problem: exterior observation and interior participation share one Fold law but retain different information. It derives binding without erasure, unity without aggregation, a subject without imported substance, finite self-reference without infinite regress, and deterministic self-opacity without ontological chance. It gives substrate independence a causal rather than rhetorical criterion. It reconstructs qualia resonance and the red-of-red as exact label-bearing recurrence while preserving the empirical distinction between physical stimulus, public report and participant quality.

That is the gravity of the result. Consciousness is not awarded to a theory because an institution favors it, to a scan because it is expensive, to an AI because it speaks persuasively or to a report because no better record was collected. Every claim must retain its carrier, alternatives, evidence class, adverse results and derivation to the One. Open criticism is unrestricted. Scientific admission is earned.

31. Repository and publication status

- Canonical repository: <https://github.com/MettaMazza/ernos-labs-sft-platform>
- Zenodo DOI: <https://doi.org/10.5281/zenodo.21636397>
- Author: Maria Smith, Ernos Labs
- Contact: Maria.Smith.Sftoe@gmail.com
- Submissions: <https://discord.gg/ucwGryVxGr>
- Current state: published open access

32. References

Primary and official external evidence:

- International Commission on Illumination. <https://cie.co.at/datatable/cie-1931-colour-matching-functions-2-degree-observer>. Evidence class: official physical colour-measurement standard.
- Kronemer et al., Nature Communications. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9707162/>. Evidence class: primary human report/no-report experiment.

- Duman et al., *Frontiers in Human Neuroscience*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC9130851/>. Evidence class: registered methodological adverse evidence.
- de Haan et al., *Brain Sciences*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10667445/>. Evidence class: primary split-brain integration experiment.
- de Haan et al., *Neuropsychologia* review. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7305066/>. Evidence class: registered unity-evidence boundary.
- Sloman et al., *PLOS ONE*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC5308842/>. Evidence class: primary introspection and confabulation experiment.
- Siclari et al., *Scientific Data*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12350935/>. Evidence class: primary open dream EEG and report database.
- Mashour et al., *eLife*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8163502/>. Evidence class: primary controlled consciousness-recovery experiment.
- Asplund et al., *Psychological Science*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3954951/>. Evidence class: primary attention-awareness experiment.
- Eagleman et al., *Journal of Neuroscience Methods*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4118597/>. Evidence class: primary within-subject qualitative consistency protocol.
- Root et al., *Behavior Research Methods*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC12960370/>. Evidence class: open synesthesia colour dataset and protocol.
- Hong and Blake, *Vision Research*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC2423348/>. Evidence class: primary participant-specific colour matching experiment.
- Zhou et al., *PNAS*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC2890491/>. Evidence class: primary language-category and colour-discrimination control.
- COGITATE Consortium. https://cogitate-consortium.github.io/cogitate-data/02_overview/. Evidence class: open preregistered multimodal adversarial dataset.
- Ferrante et al., *Nature*. <https://www.nature.com/articles/s41586-025-08888-1>. Evidence class: primary preregistered adversarial theory comparison.

Smithian Fold Theory branch dependencies:

- Smith, Maria. *From Nothing to Fold*. doi:10.5281/zenodo.21515629.
- Smith, Maria. *From Fold to Mathematics*. doi:10.5281/zenodo.21516146.
- Smith, Maria. *From Distinction to Information*. doi:10.5281/zenodo.21516916.
- Smith, Maria. *From Fold to Physics*. doi:10.5281/zenodo.21520881.
- Smith, Maria. *From Fold to Life*. doi:10.5281/zenodo.21630203.
- Smith, Maria. *From Fold to Medicine*. doi:10.5281/zenodo.21630785.

Open-science evidence supporting the institutional argument:

- Lundh A et al. *Conflicts of interest at the European Medicines Agency: a policy analysis*. 2018. <https://doi.org/10.1007/s00134-018-5293-7>.
- Fabbri A et al. *The influence of industry sponsorship on the research agenda*. 2018. <https://pubmed.ncbi.nlm.nih.gov/30252531/>.
- Gallo SA et al. *Reliability and fairness in peer review of research funding*. 2023. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10553257/>.
- Demicheli V and Di Pietrantonj C. *Peer review for improving the quality of grant applications*. Cochrane review and archive. <https://pmc.ncbi.nlm.nih.gov/articles/PMC8973940/>.
- UNESCO. *Recommendation on Open Science*. 2021. <https://www.unesco.org/en/legal-affairs/recommendation-open-science>.

- NIST. *Artificial Intelligence Risk Management Framework 1.0*.
<https://airc.nist.gov/airmf-resources/airmf/3-sec-characteristics/>.

Complete current-census successor supplement

The authoritative current scope of this paper is 77 live model-admitted claims (`consciousness_cognitive_science` 77). Counts retained inside the preceding-version narrative describe the dated freeze at which those passages were written; this successor table and the machine census control the current total.

The preceding manuscript did not contain 5 later-admitted claim section(s). They are included below in dependency order so the successor covers the complete current branch surface.

Directional synaesthetic Fold lock

Claim ID: SFT-CONSC-SYNAESTHESIA-DIRECTIONAL-LOCK-002

Exact statement: The two exact modality preimages share one binding image and may carry a stable directed cross-link without forcing the reverse link.

Dependency route: SFT-CONSC-CROSS-MODAL-QUALIA-001 -> SFT-CONSC-QUALIA-COMPOSITION-001 -> SFT-FOUNDATION-HALF-ONE-001

Candidate generator: Generate the complete eight-axis product for SFT-CONSC-SYNAESTHESIA-DIRECTIONAL-LOCK-002 and reconstruct its exact operational witness independently.

Grammar boundary: The antipodal quarter/three-quarter modality fibre with every directed-edge alternative and repeated-route witness.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: `exact-positive-fold-parts-and-held-labels__stable-directed-cross-link-over-common-half-one-image__generated-support-and-retained-record-boundary__report-preserved-ontology-separate__complete-declared-product__derivation-seal-before-target__favorable-adverse-absent-heterogeneous-unresolved__no-extra-rule`

Exact result: Quarter-One and three-quarter-One are distinct modality carriers, both Fold to half-One and together complete One. A retained trigger-to-concurrent route is repeatable and need not contain its reverse.

Closure: `depth_independent`; `minimality True`; `named-shape uniqueness True`

External status: `independently_replicated`

Engine receipt: `sha256:827a629893b47f98576e3dac63d66528a3838db90389f274c042fe9ed186a01d at receipts/engine/model_admitted/SFT-CONSC-SYNAESTHESIA-DIRECTIONAL-LOCK-002-827a629893b47f98.json`

Control	Passed	Expected	Observed
false_promise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Three-quality held orbit and nonordinary boundary

Claim ID: SFT-CONSC-NONORDINARY-THREE-QUALITY-ORBIT-002

Exact statement: The least generated three-quality experience carrier is a closed period-three Fold orbit whose states partition One; nonordinary reports remain observations without importing information outside generated support.

Dependency route: SFT-CONSC-SYNAESTHESIA-DIRECTIONAL-LOCK-002 -> SFT-CONSC-ALTERED-STATE-REPORT-BOUNDARY-001 -> SFT-CONSC-QUALIA-RECURRENCE-001 -> SFT-CONSC-RED-OF-RED-001

Candidate generator: Generate the complete eight-axis product for SFT-CONSC-NONORDINARY-THREE-QUALITY-ORBIT-002 and reconstruct its exact operational witness independently.

Grammar boundary: The exact positive unit-part Fold orbit with periods through three, support closure and report/ontology controls.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: exact-positive-fold-parts-and-held-labels__one-seventh-two-sevenths-four-sevenths-closed-orbit__generated-support-and-retained-record-boundary__report-preserved-ontology-separate__complete-declared-product__derivation-seal-before-target__favorable-adverse-absent-heterogeneous-unresolved__no-extra-rule

Exact result: The exact orbit $1/7 \rightarrow 2/7 \rightarrow 4/7 \rightarrow 1/7$ has period three and its three held states sum to One. Reports may bind or replay generated qualities beyond usual channel arrangements, but the law supplies no information from outside the generated network and makes no automatic ontological endorsement or dismissal.

Closure: depth_independent; minimality True; named-shape uniqueness True

External status: independently_replicated

Engine receipt: sha256:8ccf5643bbf358627986218d2b9291cd2ddde37f6ec2cc938bde54fe85c72aa7 at receipts/engine/model_admitted/SFT-CONSC-NONORDINARY-THREE-QUALITY-ORBIT-002-8ccf5643bbf35862.json

Control	Passed	Expected	Observed
false_promise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Exact sleep-dream period-two Fold cycle

Claim ID: SFT-CONSC-SLEEP-DREAM-PERIOD-TWO-002

Exact statement: The one-third and two-thirds carriers form a closed period-two Fold orbit, partition One and balance at half-One; physiological sleep labels are tested only after formal closure.

Dependency route: SFT-CONSC-NONORDINARY-THREE-QUALITY-ORBIT-002 -> SFT-CONSC-MEMORY-PERSISTENCE-001 -> SFT-CONSC-UNCONSCIOUS-PROCESS-001 -> SFT-CONSC-REPORT-001

Candidate generator: Generate the complete eight-axis product for SFT-CONSC-SLEEP-DREAM-PERIOD-TWO-002 and reconstruct its exact operational witness independently.

Grammar boundary: The exact positive period-two Fold orbit with state-label, balance, waking and duration controls.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: exact-positive-fold-parts-and-held-labels__one-third-two-thirds-closed-orbit-with-half-one-balance__generated-support-and-retained-record-boundary__report-preserved-ontology-separate__complete-declared-product__derivation-seal-before-target__favorable-adverse-absent-heterogeneous-unresolved__no-extra-rule

Exact result: The exact orbit $1/3 \rightarrow 2/3 \rightarrow 1/3$ is the least two-state held cycle. Its states sum to One, their balance is half-One and that balance Folds to One. Deep-sleep, REM, dream and waking are empirical correspondence labels; no universal clock duration is fitted into the formal law.

Closure: depth_independent; minimality True; named-shape uniqueness True

External status: independently_replicated

Engine receipt: sha256:96e900518c135df090379516e5793e174c96a8fab7e59d6030254c965fbeb1f51 at receipts/engine/model_admitted/SFT-CONSC-SLEEP-DREAM-PERIOD-TWO-002-96e900518c135df0.json

Control	Passed	Expected	Observed
false_promise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Cessation lock and invariant anchor

Claim ID: SFT-CONSC-CESSATION-LOCK-ANCHOR-002

Exact statement: Cessation releases occupation of the integrated half-One lock while component and record distinctions remain separately accounted; the complete One remains fixed without implying continued personal organization.

Dependency route: SFT-CONSC-SLEEP-DREAM-PERIOD-TWO-002 -> SFT-CONSC-CESSATION-001 -> SFT-CONSC-IDENTITY-CONTINUITY-001 -> SFT-CONSC-SUBSTRATE-INDEPENDENCE-001

Candidate generator: Generate the complete eight-axis product for SFT-CONSC-CESSATION-LOCK-ANCHOR-002 and reconstruct its exact operational witness independently.

Grammar boundary: Occupied/unoccupied lock labels, complete component records, fixed-anchor iterations and personal-persistence controls.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: exact-positive-fold-parts-and-held-labels__releasable-half-one-lock-and-fixed-one-anchor__generated-support-and-retained-record-boundary__report-preserved-ontology-separate__complete-declared-product__derivation-seal-before-target__favorable-adverse-absent-heterogeneous-unresolved__no-extra-rule

Exact result: The half-One integrated lock completes to One but its occupation is a releasable held state. Quarter-One and three-quarter-One components remain exactly accounted and complete One. One is invariant under further Fold action. Ending the personal organization neither makes its components numerical zero nor proves that the personal organization persists.

Closure: depth_independent; minimality True; named-shape uniqueness True

External status: independently_replicated

Engine receipt: sha256:8662c2274a82541f6c313b81263e388801e591ebc72ff93bc0bdf511fe15c2d7 at receipts/engine/model_admitted/SFT-CONSC-CESSATION-LOCK-ANCHOR-002-8662c2274a82541f.json

Control	Passed	Expected	Observed
false_promise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Complete post-seal Consciousness return comparison

Claim ID: SFT-CONSC-VALIDATION-NONORDINARY-COMPLETE-FAMILY-002

Exact statement: The sealed four-law family is compared with separately registered observations of directional synaesthesia, multi-quality/nonordinary reports, sleep-state cycling and cessation boundaries while preserving every result class.

Dependency route: SFT-CONSC-CESSATION-LOCK-ANCHOR-002 ->

SFT-FOUNDATION-MEASURED-VALUE-BOUNDARY-001 -> SFT-PHYS-MEAS-TARGET-CUSTODY-001 ->

SFT-PHYS-MEAS-UNCERTAINTY-001

Candidate generator: Generate the complete eight-axis product for

SFT-CONSC-VALIDATION-NONORDINARY-COMPLETE-FAMILY-002 and reconstruct its exact operational witness independently.

Grammar boundary: The sealed formal family against a distinct identity-first source set, retaining every observed result class.

Complete census: 256 candidates; 1 unique survivor; 4 controls

Survivor: exact-positive-fold-parts-and-held-labels__sealed-four-law-family-versus-distinct-registered-observations__generated-support-and-retained-record-boundary__report-preserved-ontology-separate__complete-declared-product__derivation-seal-before-target__favorable-adverse-absent-heterogeneous-unresolved__no-extra-rule

Exact result: The terminal comparison tests stable directional concurrent experience, bounded multi-quality and nonordinary reports, alternating sleep-state organization, and the distinction between lost integrated function and retained material components. It preserves null, adverse, heterogeneous and unresolved rows and never treats the exact Fold carriers as fitted effect sizes or clock times.

Closure: depth_independent; minimality True; named-shape uniqueness True

External status: empirically_tested_and_independently_replicated

Engine receipt: sha256:c5a0e4799ef79ce5ab7a4fdb9c975f5f3f3988c979804f79a090eb8a7b1ac2eeb at receipts/engine/model_admitted/SFT-CONSC-VALIDATION-NONORDINARY-COMPLETE-FAMILY-002-c5a0e4799ef79ce5.json

Control	Passed	Expected	Observed
false_promise	True	reject the form missing the first structural preservation	reject the form missing the first structural preservation
tampered_source	True	reject a changed source identity	reject a changed source identity
tampered_artifact	True	reject a missing, duplicate or additional survivor	reject a missing, duplicate or additional survivor
boundary	True	reject forbidden values, target feedback and extra laws	reject forbidden values, target feedback and extra laws

Preceding-version abstract preserved for chronology

The following abstract is reproduced from version 1.0.0. Its counts and status language describe that dated version and do not override the current abstract or census above.

This paper reports the foundational Consciousness and Cognitive Science branch of the third clean-room Smithian Fold Theory reconstruction at its current-evidence-closed, extension-open boundary. Seventy-two obligations in twelve ordered families generate 18,432 exact candidates and decisions, seventy-two unique survivors, seventy-two depth-independent certificates, 288 passing adverse controls, seventy-two implementation-distinct reconstructions and seventy-two post-seal empirical-boundary comparisons. Every dependency graph reaches the single premise-free theorem, There Is No Nothing. The branch uses zero axioms, zero free or fitted parameters, no negative proof quantities, no irrational or imaginary proof values, no target-selected rule and no opaque predictor.

The headline result is an exact structural reconstruction of the hard-problem boundary rather than a substitution of correlates for experience. An external observation is a many-to-one state image. Interior observation is forced when that image re-enters the same organised process. The complete two-preimage fibre makes phenomenal privacy mathematically intelligible: a participant may retain a predecessor label closed by the external image. Access, report, behaviour, neural correlation, computation and physical measurement therefore cannot become phenomenal presence by relabelling. Binding is joint closure with components retained; unity is complete mutual participation in one interior process; subject and perspective are the continuing source-located relation across changing content.

The exact constructions are explicit. One-quarter and three-quarters are distinct observer/observed preimages that share an image and compose to the One. A finite self-model closes one-quarter to one-half to the One in two nonidentity acts, ending the

regress at the Fold-fixed One. One-third and two-thirds supply the least period-two return witness for persistent label-bearing memory. The red-of-red is a label-bearing self-reidentification carried by the exact period-three support one-seventh, two-sevenths and four-sevenths, whose three parts compose to the One. Physical red stimulus, public word, participant quality and report about that quality remain different records. Substrate independence requires a complete bidirectional relation- and causality-preserving realization map; fluent output does not satisfy it.

All 356 V1 rows and 407 V2 steps were atomically reviewed. Forty-six Consciousness-owned questions, including the stronger separately registered qualia-resonance and red-of-red obligations, are closed by current V3 receipts. Fifteen external source identities contribute sixty-one preregistered features: fifty-eight are present; three absent full-text features and eighteen failed transport or content rows remain visibly preserved rather than counted as support.

Scope and ownership boundary

This paper owns the `consciousness_cognitive_science` surface comprising 77 current claims. Ownership identifies responsibility for derivation and falsification; it does not prevent criticism or lawful cross-branch use. Application performance, consensus, credentials and Lean acceptance cannot transfer ownership or select a law.

Mathematical, computational and empirical constitutions

The mathematical constitution retains exact generated carriers, relations, candidate grammars, decisions, survivors and closure boundaries. The computational constitution retains executable identities, complete traces, controls, independent reconstructions and receipts. The empirical constitution retains source registration, transport, pre-seal and post-seal chronology, custody, measurement or observation, uncertainty, favourable and adverse rows, missingness, unresolved records and falsification conditions. None of these evidence classes substitutes for another.

Registered source surface

Human-readable scholarly references remain in the inherited paper. Machine source IDs, custodians, releases, locators, source captures, access outcomes, hashes and transport status remain in the current claim packages and external-source registries. Source prestige is not evidential authority; each source is used only in its registered role. The Lean source-binding gate checks custody and identity, not empirical truth by reputation.

Data and code availability

The successor Markdown is `publications/successors/consciousness_cognitive_science/FROM_FOLD_TO_CONSCIOUSNESS_PAPER_001_V1_1.md` and its scientific predecessor is `publications/current/consciousness_cognitive_science/FROM_FOLD_TO_CONSCIOUSNESS.md`. The Lean sources and report are under `generated/lean4_validation/`; the report is `generated/lean4_validation/reports/whole_model_validation.json`. Current claims are indexed by `census/claims.json`; complete candidates, decisions, controls, certificates, observations and receipts remain under `claims/<claim-id>/` and `receipts/`. The paper's evidence map and draft deposit metadata sit beside the successor source. No remote action was performed.

References and source registry

1. Smith, Maria. The preceding version of this paper, preserved at the repository path identified above, 2026.
2. Smith, Maria. *There Is No Nothing*. Ernos Labs Methods Paper 00, versioned publication series, 2026.
3. de Moura, Leonardo, and Sebastian Ullrich. "The Lean 4 Theorem Prover and Programming Language." *Automated Deduction - CADE 28*, 2021. DOI: [10.1007/978-3-030-79876-5_37](https://doi.org/10.1007/978-3-030-79876-5_37).
4. Lean FRO. *The Lean Language Reference*. lean-lang.org/doc/reference/latest.
5. Ernos Labs. Current claim packages, source registries and Lean verification report, repository snapshot bound by the machine identities below.

Successor machine-identity appendix

- Preceding source:
publications/current/consciousness_cognitive_science/FROM_FOLD_TO_CONSCIOUSNESS.md
- Preceding source SHA-256:
sha256:62f473b326aa9cf0ecfab2dcaaf763aa752df468210aef52b0d12c40f3aebaf9
- Lean report: generated/lean4_validation/reports/whole_model_validation.json
- Lean report SHA-256:
sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf
- Ordered census SHA-256:
sha256:f3f540f77a9b677bf7ddb9f5c6ea1ea41f08e57bb07163e25f385fd4d1dcde71
- Execution manifest SHA-256:
sha256:517fd288f4484f43fdc0343b17fcabd06ea9e12c977a4512d4a891dd038fce41
- Protected engine seal:
sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a
- Verification-authority seal:
sha256:bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8
- Publication authorised: false
- Remote actions performed: false

Lean 4 independent verification update

Verified result

The pinned Lean toolchain `leanprover/lean4:v4.32.0` compiled the verification project and the executable verifier returned PASS. Its immutable report identity for this successor is

sha256:6b6a5a9a9a0b74687a6f97fb3b4fcc7455968e35c82395290837cda60d7501cf.

Verified surface	Result
Ordered census claims	2,777
Proof-bearing accepted claim gates	2,777
Source-bound claims	2,777
Candidate records	898,902
Decision records	898,902
Passed controls	11,108
Registered branches	17
Issues	0

This paper's registered branch surface contributes **77** of the accepted claims.

Native theorem proof and artifact verification are different layers

`SFTValidation/Root.lean` formalises the exact registered two-member operational grammar:

`unpresentedAbsence` and `presentedOccurrence`. The decision function rejects the former and accepts the latter.

Lean proves existence by the accepted constructor and uniqueness by exhaustive case analysis over the two constructors.

`#print axioms` reports no imported or user-declared axioms for the exported root theorems. This is a kernel-checked proof of the unique survivor inside the exact registered operational grammar.

`SFTValidation/Gates.lean` defines twelve Boolean acceptance obligations and can construct an inhabitant of `ClaimGate.Accepted` gate only when all twelve equal `true`. The runtime verifier then parses the complete ordered census and execution manifest, recomputes identities, cardinalities, one-to-one decision coverage, survivor count, controls, closure fields, empirical boundaries, certificate bindings and authoritative receipts, and calls that proof-bearing constructor for every current claim.

The exact scientific statements of the other claims are not all re-expressed as 2,776 separate Lean propositions in this version. They are validated as the repository's complete registered machine artifacts through Lean-executed checks. The source-manifest gate uses a read-only Python bridge solely to instantiate the already registered source factories; Lean consumes the result. This layer does not replace the protected admission engine, does not write receipts and does not substitute for source experiments or empirical replication.

Fail-closed custody event

The first whole-model run halted on six source-capture byte hashes. The discrepancy was traced to line-ending normalisation in the working tree, not to a changed scientific target or a failed theorem. The registered source bytes were restored, exact-byte Git attributes were added for those six captures, all 385 registered external source bindings were re-audited with no mismatch, and the unchanged Lean verification was rerun to `PASS`. The preserved halt demonstrates that source drift is detected rather than silently forgiven.

What the result supports

The result materially strengthens the model's case for formal coherence, exhaustive current-census coverage, unique-survivor enforcement, dependency and provenance integrity, cross-branch consistency and reproducibility by an implementation outside the admission engine. It does not by itself establish that every empirical claim is true in nature, that the registered grammar is the only conceivable grammar outside its stated boundary, or that SFT is superior to every rival theory. Those questions remain governed by the papers' declared experiments, falsification conditions and comparative tests.

Successor conclusion

Version 1.1.0 preserves the preceding paper's derivations, evidence classes, corrections, adverse and unresolved records, ownership limits and open frontier, while integrating the independent Lean 4 result and every later current claim in its declared scope. This paper contributes 77 current claim(s) within `consciousness_cognitive_science` to the total dependency spine.

The new formal result establishes that the registered two-class operational root has one Lean-proved survivor and that all 2,777 current claims satisfy the proof-bearing artifact gates. The major conflation resolved by this update is between native theorem proof and whole-repository artifact verification: both are valuable, but they are not the same evidential act. No empirical status is strengthened merely because the artifact graph passes, and no historical halt or correction is erased.

The current evidence therefore establishes formal coherence, current-census completeness, unique-survivor enforcement and intact source, dependency, certificate and receipt custody for this version. Field-specific empirical conclusions remain exactly those authorised by their current records. Lawful discovery, falsification, stronger evidence and versioned extension remain open, and publication itself remains pending Maria Smith's explicit confirmation.